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Contents

Introduction

Purpose

This document describes ongoing administration, management, and maintenance tasks for the Avaya Aura® Web Gateway. Use this document after deploying the Avaya Aura® Web Gateway. For more information about deployment, see *Deploying the Avaya Aura® Web Gateway*.

> Note:
> This document does not describe SDK developer applications.

Change history

Issue	Date	Summary of changes
Issue 2	February 2025	Removed the topics related to Transport Layer Security Configuration
Issue 1	December 2024	Initial release for R10.2

Avaya Aura® Web Gateway overview

The Avaya Aura® Web Gateway server acts as a gateway to Avaya Aura® clients and applications utilizing WebRTC signaling and media. Avaya Aura® Web Gateway also provides the push notification service, enabling clients to receive incoming call alerts and other notifications from the Apple Push Notification service (APNs).

You can deploy the Avaya Aura® Web Gateway through Amazon Web Services (AWS), VMware, or Avaya Virtualization Platform.

You can deploy the Avaya Aura® Web Gateway in the following environments:

- Avaya Aura® (Team Engagement)
- Avaya Aura® and Conferencing (Team Engagement and Conferencing)

To access conferencing tools, such as Avaya Meetings Server, your deployment must include Conferencing.

> Note:
> An Over-The-Top (OTT), or Conferencing-only, deployment option is available, but it is not described in this document. The Conferencing-only option follows a different deployment process. For more information, see *Deploying Avaya Meetings Server*.

New in this release

The following is a summary of new functionality that has been added to the Avaya Aura® Web Gateway in Release 10.2:

Support for Software-Only Deployment on Nutanix Environment

With Release 10.2, Avaya Aura® Web Gateway Software-Only application can be deployed on Nutanix 6.5 and later.

Red Hat Enterprise Linux (RHEL) 8.10 support

With Release 10.2, Avaya Aura® Web Gateway supports Red Hat Enterprise Linux Release 8.10.

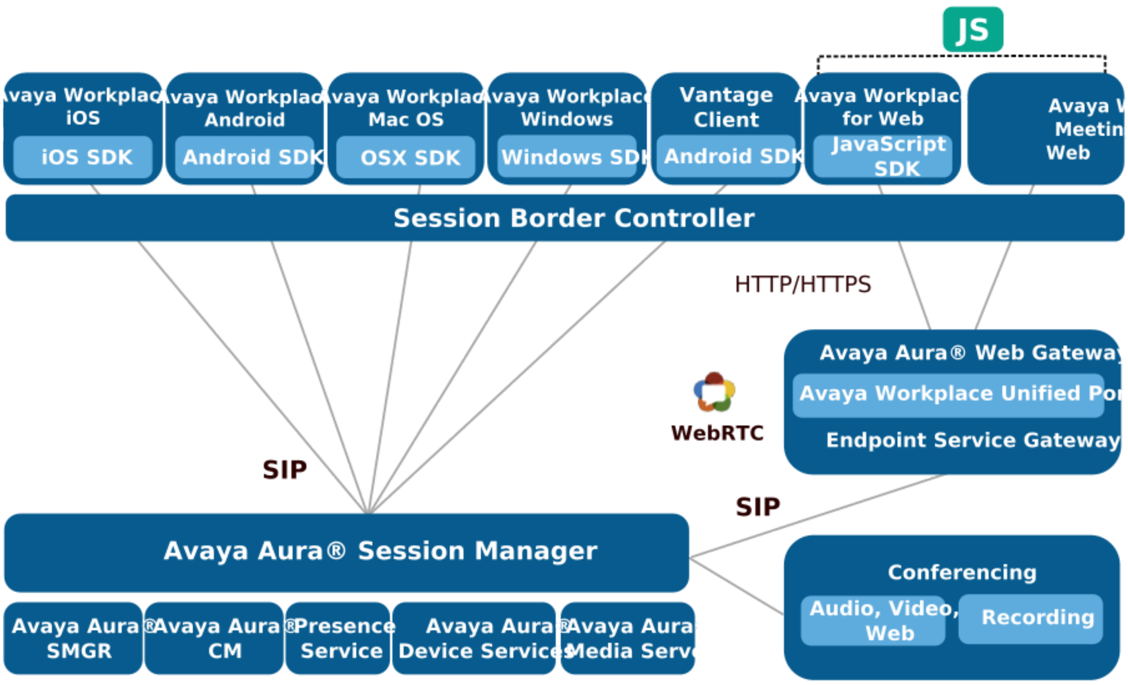
Kernel-based Virtual Machine (KVM) on RHEL 8.10 hypervisor support

With Release 10.2, Avaya Aura® Web Gateway can be deployed on Avaya-supplied KVM on RHEL Release 8.10 hypervisor (Avaya Solutions Platform 130 R6.0).

Solution architecture

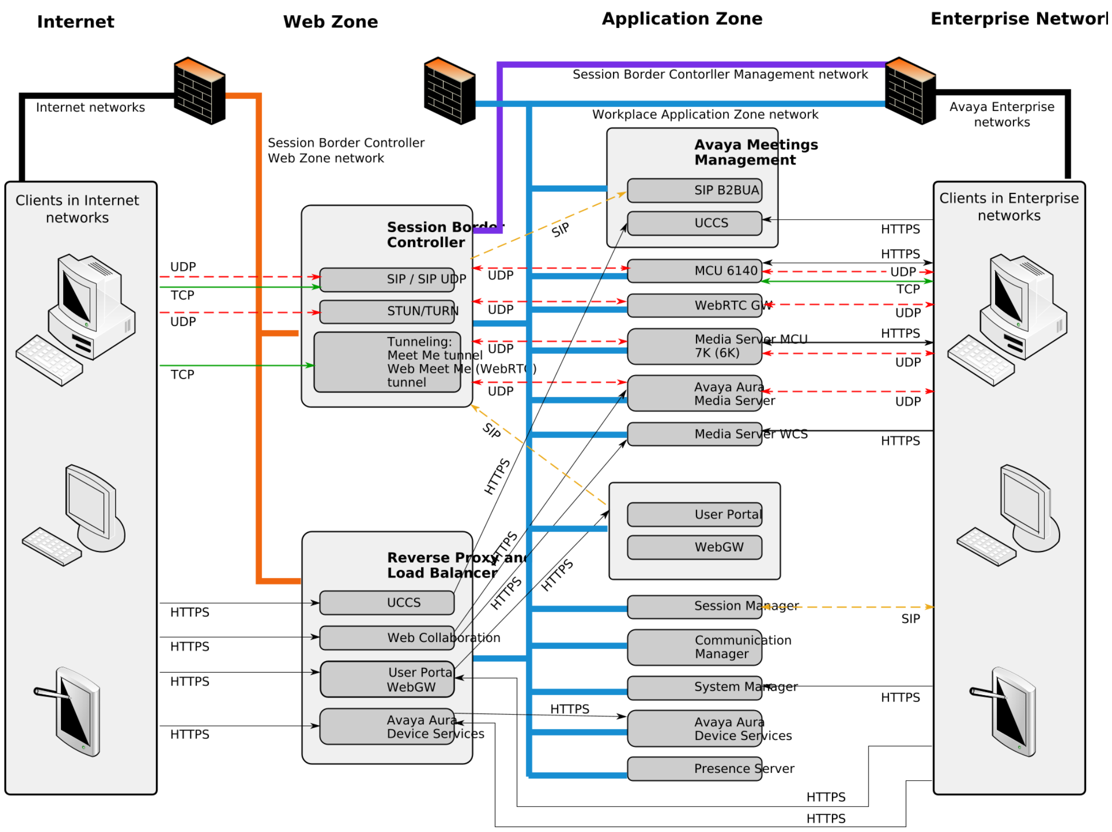
This section provides a graphical representation of the Avaya Aura® Web Gateway deployment architecture.

Figure: Avaya Workplace Client and Conferencing deployment topology



Topology diagram

This section provides a graphical representation of the Team Engagement + Conferencing deployment model topology.



For detailed information about ports, go to https://support.avaya.com/security, scroll down, and click **Avaya Product Port Matrix Documents**. Navigate to the required solution component section and then click on the appropriate Port Matrix document for the release to open it.

Geographical distribution overview

In a geographically distributed system, resources are deployed in multiple data centers to reduce media delays. For this purpose, the following components are deployed in each data center:

- Avaya Aura® Media Server
- Avaya Aura® Session Border Controller
- Avaya Aura® Web Gateway
- Web Collaboration Server
- Avaya Aura® Session Manager
- Avaya Aura® Communication Manager
- Avaya Aura® Device Services

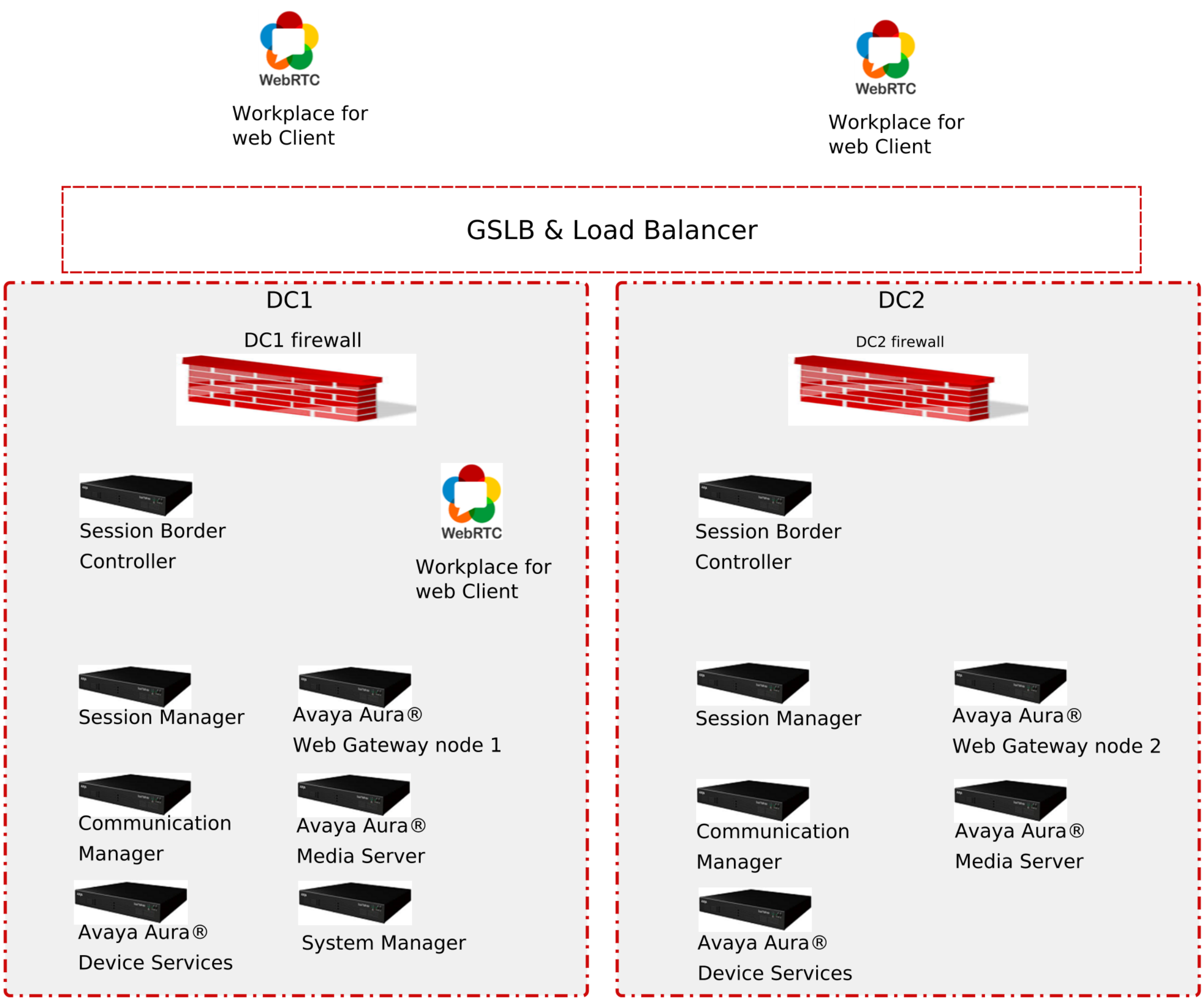
In a geographically distributed system, you must also install the following two components:

- Global Server Load Balancing (GSLB), which provides different routes and addresses based on the location of the client.
- Load balancer, which balances traffic between two or more Avaya Aura® Web Gateway nodes, which may be located in the same data center or in different data centers.

Avaya Aura® Session Manager and Avaya Aura® Communication Manager are important for call routing. To optimize media delays for point-to-point calls, deploy these components in a distributed manner across your data centers. The way in which these components are geographically distributed is outside the scope of this document. For more information about configuring Session Manager and Communication Manager, see *Administering Avaya Aura® Session Manager* and *Administering Avaya Aura® Communication Manager*.

General geographical distribution topology

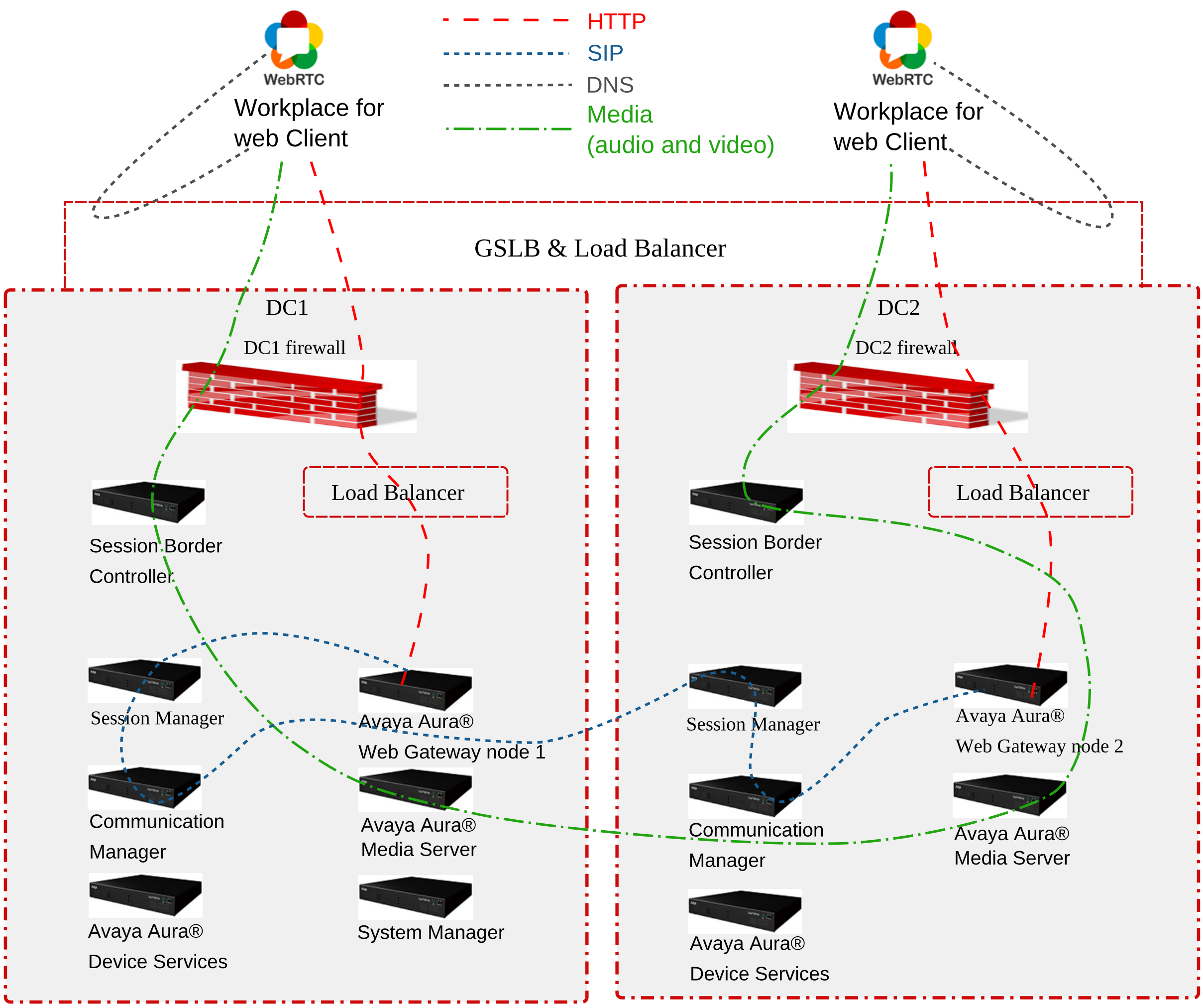
In this topology example, there are two data centers with one Avaya Aura® Web Gateway in each data center. For simplicity, System Manager is deployed in Data Center 1 (DC1), while Session Manager, Communication Manager, and Avaya Aura® Device Services are deployed in all data centers.



Signaling and media path topology when clients are located in or near different data centers

In the following topology example, there are two data centers with one Avaya Aura® Web Gateway in each data center. Clients are located in different data centers outside of the firewall and registered on the Avaya Aura® Web Gateway to receive calls. When one client makes a call to the other, the call follows the following flow:

1. Both clients log in to the corresponding Avaya Aura® Web Gateway and activate the call service.
    a. A client in data center 1 (DC1) gets the address of the DC1 load balancer and communicates with the Avaya Aura® Web Gateway deployed on that data center. The Avaya Aura® Web Gateway registers the client to the corresponding Session Manager deployed on DC1.
    b. A client in data center 2 (DC2) gets the address of the DC2 load balancer and communicates with the Avaya Aura® Web Gateway deployed on that data center. The Avaya Aura® Web Gateway registers the client to the corresponding Session Manager deployed on DC2.
2. The DC1 Avaya Aura® Web Gateway initiates the call. To route the media, the Avaya Aura® Web Gateway uses the Session Border Controller and Avaya Aura® Media Server deployed on DC1.
3. The Avaya Aura® Web Gateway sends the SIP call to Session Manager deployed on DC1.
4. Session Manager deployed on DC1 forwards the call to Session Manager deployed on DC2.
5. Session Manager deployed on DC2 forwards the SIP invite to the Avaya Aura® Web Gateway from DC2, where the second client is logged in.
6. The Avaya Aura® Web Gateway from DC2 uses the Session Border Controller and Avaya Aura® Media Server deployed on DC2 to pass the media through the firewall to the second client.



Signalling and media path topology when both clients are located in or near the same data center

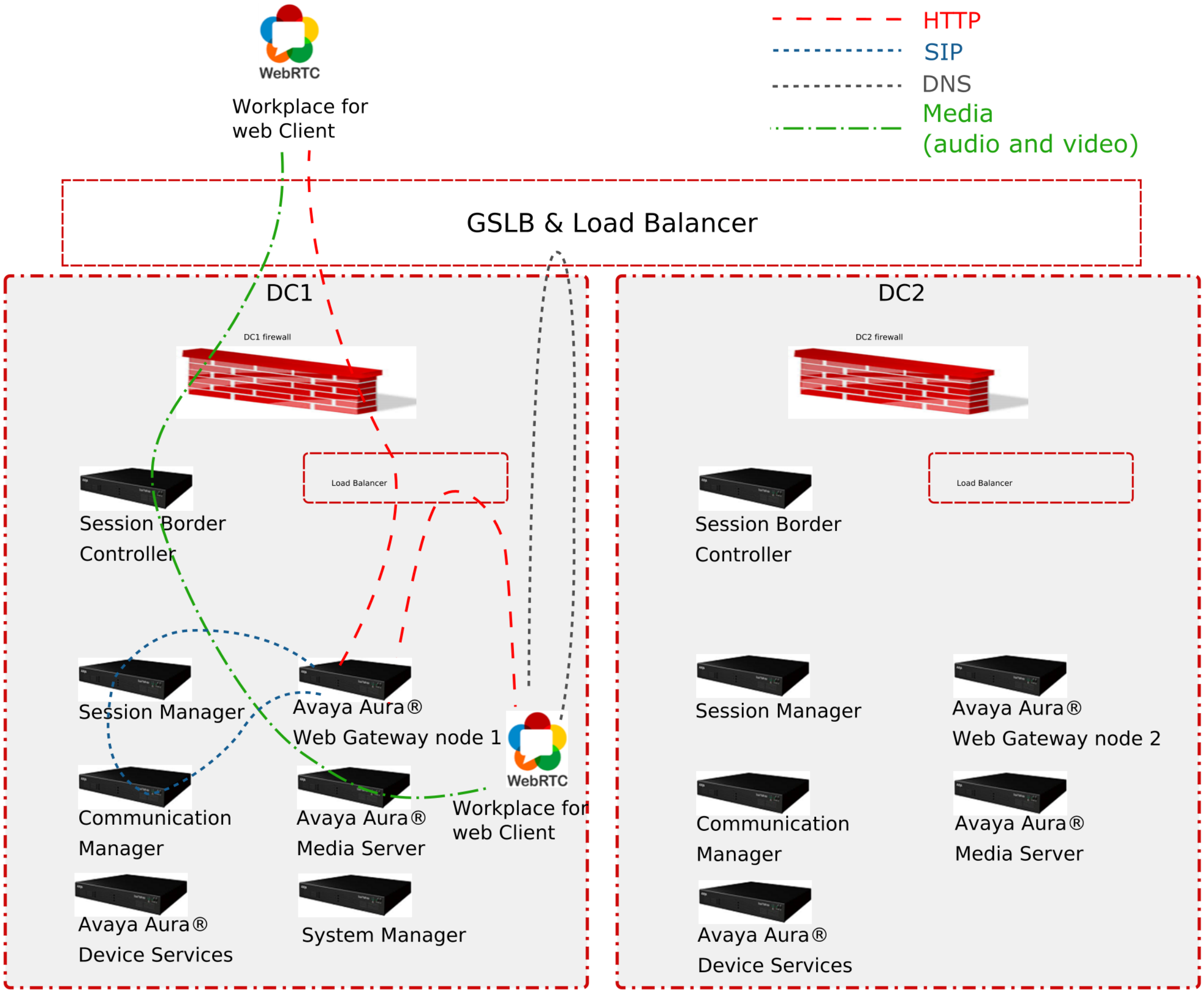
In the following topology example, there are two data centers with one Avaya Aura® Web Gateway in each data center. Two clients are located in or near the same data center (DC1):

- The first client is located outside the firewall.
- The second client is located inside the enterprise network.

Both clients are registered on the Avaya Aura® Web Gateway to receive calls. When one client makes a call to the other, the call follows the following flow:

1. Both clients log in to the Avaya Aura® Web Gateway and activate the call service.

   Both clients resolve the FQDN to the address of the load balancer deployed on DC1 and communicate with the Avaya Aura® Web Gateway deployed on DC1.
2. The external client initiates the call.
3. The Avaya Aura® Web Gateway sends the SIP call to Session Manager deployed on DC1.
4. Session Manager deployed on DC1 forwards the call to the same Avaya Aura® Web Gateway deployed on DC1, where the internal client is logged in.
5. The Avaya Aura® Web Gateway deployed on DC1 uses the Session Border Controller and Avaya Aura® Media Server deployed on DC1 to pass the media through to the internal client.



Interoperability

Product compatibility

Avaya Aura® Web Gateway interacts with the following components. For information about interoperability and supported product versions, see https://secureservices.avaya.com/compatibility-matrix/menus/product.xhtml.

Component	Description
Avaya Aura® infrastructure	The following Avaya Aura® components must be installed and configured to use TLS: • Avaya Aura® Device Services: Provides centralized contact management services using REST-based APIs. • Avaya Aura® Media Server: Supports standard media processing features. Important:

Component	Description
	Avaya Aura® Web Gateway does not support Media Server High Availability cluster configuration when using WebRTC. • System Manager: Manages Avaya Aura® components, certificates, licenses, and checks log and alarm capabilities. • Session Manager: Enables applications to perform registration and telephony functions, such as call escalation.
Conferencing	This describes the deployment for Avaya Aura® Web Gateway, but not Conferencing itself. The Avaya Meetings Server solution provides conferencing and collaboration functionality. Avaya Meetings Server is not available with the Avaya Aura® Web Gateway Avaya Workplace Client only deployment option.
Avaya SBCE	A component that provides a common element to enable secure access to the Avaya infrastructure from untrusted networks, such as the internet. In addition to SIP firewall services, this component provides the Reverse Proxy services required for HTTP signaling, media traversal, and access to other data services.

Web browser requirements

The Avaya Aura® Web Gateway administration portal supports the following web browsers:

- Internet Explorer 9, 10, or 11.
- Microsoft Edge 42 and later.
- The latest version of Mozilla Firefox or the version before it.
- Google Chrome 53 and later.

For information about supported browser versions for AWS deployments, see https://aws.amazon.com/console/faqs/#browser_support.

Data encryption

You can enable or disable data encryption when deploying the Avaya Aura® Web Gateway OVA. When data encryption is enabled, all operational data and log files are encrypted.

You can only enable data encryption on Avaya Aura® Web Gateway if you use Avaya Solutions Platform or a VMware Virtualized Environment. For software-only deployments, you must enable data encryption on the virtualization platform itself. For more information about enabling data encryption on Amazon Web Services, see How to Protect Data at Rest with Amazon EC2 Instance Store Encryption.

Once data encryption is enabled, you cannot disable it using the configuration utility or the Avaya Aura® Web Gateway administration portal. To disable data encryption, you must redeploy the Avaya Aura® Web Gateway OVA.

If you enabled data encryption and selected the **Require Encryption Pass-Phrase at Boot-Time** option, then you will need to enter the data encryption passphrase after every Avaya Aura® Web Gateway reboot. If you do not select this option, Avaya Aura® Web Gateway enables the local key store to store encryption keys, so you do not need to enter the passphrase manually. However, this is a less secure solution. Alternatively, you can set up a remote key server to store encryption keys.

Encryption of Avaya Aura® Web Gateway partitions

When you enable data encryption for Avaya Aura® Web Gateway, the following partitions are encrypted:

- sdb: */var/log/Avaya*
- sdc: */media/data*
- sdd: */media/cassandra*

The sda boot disk is always unencrypted.

Administrator responsibilities

After the Avaya Aura® Web Gateway deployment is completed, an administrator can perform the following key management tasks:

- Manage Avaya Aura® Web Gateway services using the web administration portal.
- Set up and customize optional functionality, such as Integration Windows Authentication and multisite support.
- Adjust virtual hardware resources and disk sizes.
- Schedule repairs.
- Check logs and alarms.
- Perform backup and restore operations.
- Upgrade within an Avaya Aura® Web Gateway release or migrate to a new Avaya Aura® Web Gateway release.
- Manage system layer updates.

Management tools

Setting a timeout period for an SSH session

You can extend the duration of an SSH session by using the Tmux utility. If you start an Tmux session before performing an operation, you can re-connect to that session even if the connection to the SSH terminal is reset. For more information about using the Tmux utility see See "Using the Tmux utility" on page 43.

Usage of commands and aliases

Linux alias commands

Linux aliases are defined to make frequently used commands easier to use. When an alias is available for the required operation, you can use the alias instead of typing a long path name and using `sudo`. The path name specification and `sudo` invocation are built into the aliases that Avaya provides.

Three categories of aliases with their functionality description

Alias	Description
`cdto`	Change to frequently used directories.
`app`	Perform application functions, such as install or backup.
`svc`	Manage the state of application related services.

Some of the alias commands are only available after the application has been installed.

You can type any of the aliases in a Linux shell to list the supported commands.

The following image provides an example of how the aliases are used:

```
[admin@aawg-ova ~]$ cdto

Syntax: cdto <target>

Available navigation targets:

    base            [/opt/Avaya]
    root            [/opt/Avaya/CallSignallingAgent]
    active          [/opt/Avaya/CallSignallingAgent/3.4.0.0.299]
    cas             [/opt/Avaya/CallSignallingAgent/3.4.0.0.299/CAS/3.4.0.0.299]
    misc            [/opt/Avaya/CallSignallingAgent/3.4.0.0.299/CAS/3.4.0.0.299/misc]
    bin             [/opt/Avaya/CallSignallingAgent/3.4.0.0.299/CAS/3.4.0.0.299/bin]
    config          [/opt/Avaya/CallSignallingAgent/3.4.0.0.299/CAS/3.4.0.0.299/config]
    logs            [/opt/Avaya/CallSignallingAgent/3.4.0.0.299/CAS/3.4.0.0.299/logs]
    ilogs           [/opt/Avaya/CallSignallingAgent/.CSAInstallLogs]
    tlogs           [/opt/Avaya/CallSignallingAgent/3.4.0.0.299/tomcat/8.0.24/logs]

[admin@msg-esg ~]$ app

Syntax: app <command> <arguments>

Available commands:

    install         [Run application installer for installation]
    status          [statusCSA.sh]
    configure       [configureCSA.sh]
    listnodes       [clitool-csa.sh listClusterNodes]
    collectlogs     [collectLogs.sh]
    backup          [backupCSA.sh]
    restore         [restoreCSA.sh]
    upgrade         [Run application installer for upgrade]
    rollback        [rollbackCSA.sh]
    removeinactive  [removeVersion.sh (inactive instance)]
    uninstall       [uninstallCSA.sh]
```

Each alias category displays the target command, which is in square brackets. The syntax for the command is provided in the procedures outlined in this document. Arguments that you specify after an alias are passed through to the target command.

The system can simultaneously have both an active and inactive installation of the software. For example, after an upgrade, the earlier version becomes inactive, but the new version becomes active. The alias commands operate only on the active installation unless specified.

Examples of alias commands to be used in a Linux shell

Alias example	Function provided
`cdto logs`	Changes to the log directory of the active installation on the system.
`app install`	Runs the staged application installer.
`svc telportal restart`	Restarts the telportal service.

> Note:
> The aliases must be used only from the command line in a Linux shell. Do not use them in a script. You must use the actual target command in a script.

System layer commands

The **sys** command line alias facilitates the use and discovery of system layer commands. Typing this command without arguments provides syntax help, and a list of supported system layer commands. The following is an example:

```
[admin@server-dev ~]$ sys

Execute system layer commands.

    -h, --help
        Command syntax (this help)

    -hh, --hhelp
        Verbose help

Available commands:

    secconfig       [Manage security settings]
    versions        [Query version information]
    volmgt          [Manage disk volume sizes]
    smcvemgt        [Manage Spectre/Meltdown patches]
    extension       [Manage system layer extensions]
  passwdrules     [Manage password rules]

  Command invocation syntax:
     sys <command> <arguments>

  Command syntax
    sys <command> -h

[admin@server-dev ~]$
```

Verbose help information

-hh is used for verbose help information, which provides a brief description of each available system layer command. The following is an example:

Any arguments provided after the name of the system layer command are passed through to that command.

sys secconfig command

sys secconfig provides access to the **secconfig** command, which existed in previous releases.

This command is only available for OVA-based deployments.

The following is an example of this command:

```
[admin@server4950csa ~]$ sys secconfig --hhelp

This script is used to manage run-time security settings on this
```

```
appliance.
The following command-line arguments are available:

    --help, -h
        Prints terse help (command line syntax).

    --hhelp, -hh
        Prints verbose help (this help).

    --sshCBC < --enable | --disable | --query >
    -cbc     < -e       | -d        | -q >
        Enables, disables, and queries the current state of SSH
daemon
        CBC-based ciphers.

   --fips  < --enable | --disable | --query >
   Enables, disables, and queries the current state of FIPS on the
system.

[admin@server4950csa ~]$
```

sys versions command

The **sys versions** command provides a summary of key system layer information, including the type of appliance (OVA), the version number of the system layer, the version of the current partitioning, and the OVA that was originally deployed.

```
[admin@server4889csa ~]$ sys versions

    Appliance type       : AAWG
    System layer version : 3.4.3.0.2
    Partitioning version : 2.0
    Original OVA deploy  : csa-3.5.0.0.365

[admin@server4889csa ~]$
```

sys volmgt command

Syntax help: sys volmgt --help

The **sys volmgt** command is used to query and extend disk volumes on the system.

The following provides the command line syntax for this command:

```
[admin@server4889csa ~]$ sys volmgt --help

Syntax:
    --help,                  -h
    --hhelp,                 -hh
    --version,               -v
```

```
    --status,                -st
    --summary,               -s
    --monitor  [tail|less],  -m [tail|less]
    --logs,                  -l
    --scan
    --extend <volume> [ <n>m | <n>g | <n>t --remaining ]
  --extend --all
    --reset

[admin@server4889csa ~]$
```

Verbose help: sys volmgt --hhelp

The verbose help information for the scripts provides more information about what the tool is used for.

```
[admin@server4889csa ~]$ sys volmgt --hhelp

This script provides for the ability to extend the sizes of volumes
on this
system. In order for a volume to be extended in size, the disk that
hosts
the volume must first be increased in size using the tools that are
used
to manage deployed virtual machines (VMware).

The following example illustrates how to add 20 GiB of storage to the
application log volume (/var/log/Avaya). This volume is located on
the second
disk of the system and so this example assumes that disk 2 has been
increased
in size by 20 GiB.

    sys volmgt --extend /var/log/Avaya 20g

The above example will do two things:

    1) It will extend the size of the LVM logical volume by 20 GiB.

    2) It will then extend the size of the Linux file system that is
       located inside that volume to the new size of the LVM logical
       volume.

Step (2) above may take several minutes to complete for larger
volumes. If,
for some reason, this second operation is interrupted, it can be re-
run
using the same command, but WITHOUT specifying the size argument. For
example,
the following command is used to perform step (2) only for the
application
log volume (/var/log/Avaya).

    sys volmgt --extend /var/log/Avaya
```

```
If in doubt as to whether or not all file systems have been fully
extended in
their respective volumes, step (2) can be executed across all volumes
using
a single command as follows:

    sys volmgt --extend --all

Performing step (2) on a file system that is already fully extended
in its
LVM volume is a null operation (does no harm).

Note the following general points regarding this script:

- The extending of a volume cannot be undone. Make sure the correct
volume
  is being extended, and by the correct size. To confirm any extend
  operation, the user is required to enter the response "confirm"
  (case insensitive).

- In order to avoid impacting system performance, avoid performing
extend
  operations during periods of high traffic.

- Extend operations are performed by a background process, in order
to
  avoid interference due to loss of an SSH connection. Avoid powering
down
  or rebooting a server while there is a background operation in
progress.
  The presence of a running background operation can be queried as
follows:

    sys volmgt --status

- Logical volumes on the system are referenced using their Linux file
system
  mount points, such as /var/log/Avaya and /media/data, with the
exception
  of the volume containing Linux swap, which has no mount point. The
Linux
  swap volume is referenced using "swap".

- Sizes are specified in base 2 units rather than base 10 (SI) units.
For
  example, 1g = 1 GiB = 1024 x 1024 x 1024 bytes.

- Summary information is displayed in GiB, with a resolution of two
decimal
  places. When extending the sizes of LVM volumes, units can be
specified
  in mebibytes (m), gibibytes (g), or tebibytes (t).

- Due to file system overhead allocation by the Linux kernel, the
size
  of a file system will never exactly match the size as reported by
  the LVM volume that contains that file system. To be certain that a
file
```

```
  system is fully extended to the size of the volume that contains
it,
  inspect the log file after issuing the extend operation as follows:

      sys volmgt --monitor less

  To perform such a check across all volumes:

      sys volmgt --extend --all
      sys volmgt --monitor less

The following arguments are supported by this script:

    --help, -h
        Terse help.

    --hhelp, -hh
        Verbose help (this help).

    --version, -v
        Prints the version of this script to stdout.

    --status, -st
        Prints the current status of this tool. Use this to determine
        if there is a background operation in progress, or the
results
        of the last background operation.

    --summary, -s
        Prints a summary of disks, the LVM volumes contained on each
disk,
        and the file system contained in each LVM volume. Disk
information
        includes the size of the disk and the amount of free space
        available for allocation to volumes on the disk. LVM volume
        information includes the size of the LVM volume. File system
        information includes the size of the Linux file system and
the
        current amount of space that is in use on that file system.

        Due to file system overhead allocation by the Linux kernel,
the
        size of a file system will never exactly match the size as
reported
        by the LVM volume that contains that file system. Refer to
the top of
        this help information for more information.

    --monitor [tail|less]
    -m        [tail|less]
        Browse the log file for the latest extend operation. Specify
"tail"
        to use the tail browser. Specify "less" to use the less
        browser, which allows scrolling and searching through the log
file.
        If neither is specified, the browser defaults to the tail
browser.
```

```
    --logs
        Generate a zip file in the current working directory that
contains
        all logs generated to date by this script.

    --scan
        Scan disks for newly available storage. Do this after
increasing
        the disk size of one of more disks. Once scanned, the newly
        available space appears in the "Free" column in the "--
summary"
        output, and is now available for allocation to volumes on
that disk.

        A summary is printed after the scan to show the updated
volume
        information.

    --extend <volume> [ <n>m | <n>g | <n>t --remaining ]--extend --
all
        The first form of the command operates on a single volume. If
a size
        is specified, then the LVM volume is extended by that size
(step 1),
        and the file system it contains is extended to use the new
space
        made available in that volume (step 2). If a size is not
specfied,
        then the file system contained in that volume is extended
(i.e.,
        step 2 only).

        The "--all" form of the command is used to perform step 2
across
        all volumes on the system.

        For more information, see the examples at the top of this
help.

        If "--remaining" is specified for the size, then the
specified
        volume is extended with all remaining free space on that
disk.
        If a specific increment is provided, then the volume is
extended
        by that amount, reducing the amount of free space on the disk
        by that amount. Specific sizes are in the form of a number
        (e.g., "10", "10.5", or ".5") and a unit. Units are "m" for
        mebibites, "g" for gibibytes", and "t" for tebibytes".

        The smallest increment that can be specified is 100 MiB.

        Example invocations:

            sys volmgt --extend /var/log/Avaya 10g
            sys volmgt --extend /var/log/Avaya 10.5g
            sys volmgt --extend /var/log/Avaya 0.5g
            sys volmgt --extend /var/log/Avaya .5g
```

```
             sys volmgt --extend /var/log/Avaya 500m
             sys volmgt --extend /var/log/Avaya --remaining
             sys volmgt --extend /var/log/Avaya

  --reset
        Resets internal tracking data. Use this if this script is
blocked
        on an invalid background progress indication. This condition
can
        arise if a background operation was prematurely terminated
due to,
        for example, a system reboot. Verify that no background
operations
        are in progress prior to executing this command, through
verification
        of the process id as reported by the "--status" argument.

[admin@server4889csa ~]$
```

Partitioning examples: sys volmgt --summary

The following example shows a summary of the information provided by this command for a version 2.0 partitioned system:

```
[admin@server4950csa ~]$ sys volmgt -s

                           Disk and Volume Summary

+----------- Disk ------------+------------------- Volume
--------------------+
|                             |                      LVM        File
System   |
| Num  Name     Size     Free |    Name              Size
Size     Usage |
+-----------------------------
+-----------------------------------------------+
|  1   sda      41.78    0.00 | /                    4.00
3.81      1.26 |
|                             | /home                4.00
3.81      0.05 |
|                             | /opt/Avaya          14.97
14.61      1.14 |
|                             | /tmp                 2.81
2.71      0.01 |
|                             | /var                 3.00
2.89      0.03 |
|                             | /var/log             2.00
1.91      0.00 |
|                             | /var/log/audit       3.00
2.89      0.00 |
|                             | swap                 8.00       n/
a       n/a |
+-----------------------------
+-----------------------------------------------+
|  2   sdb      60.00    0.00 | /var/log/Avaya      60.00
```

```
58.93      0.05 |
+-----------------------------
+-----------------------------------------------+
|  3   sdc      20.00    0.00 | /media/data         20.00
19.56      0.04 |
+-----------------------------
+-----------------------------------------------+
|  4   sdd      10.00    0.00 | /media/cassandra    10.00
9.71      0.02 |
+-----------------------------
+-----------------------------------------------+
```

sys smcvemgt command

The system layer **smcvemgt** command is used to manage the Linux kernel patches related to the following vulnerabilities:

- Variant #2/Spectre (CVE-2017–5715)
- Variant #3/Meltdown (CVE-2017–5754)

> Note:
> The kernel patch for the Variant #1/Spectre (CVE-2017–5754) vulnerability is permanently enabled on the system and cannot be disabled.

This command is only available for OVA-based deployments.

The choice to enable or disable these patches is a trade-off between performance and security impact:

- If the patches are enabled, the system might experience noticeable performance losses.
- If the patches are disabled, the system is not protected against the Variant #2/Spectre and Variant #3/Meltdown vulnerabilities.

By default, Linux patches for Variant #2/Spectre and Variant #3/Meltdown are enabled. The Variant #2/Spectre patch is enabled with Linux kernel defaults. In default operation mode, the Variant #2/Spectre Linux patch selects the mitigation method that is best suited for the processor architecture of the host machine.

> Note:
> To be fully functional, patches for the Variant #2/Spectre vulnerability require hardware support, which is provided by VMware and hardware vendors through microcode updates.

Changes made by the **smcvemgt** command to the Linux kernel tunalbles always cause a server reboot. The script does not manage the state of application services. To ensure that the application services are stopped before the reboot, run the `svc csa stop` command before using the **smcvemgt** command. After the reboot, manually start the application services using the `svc csa start` command.

For more information about Spectre and Meltdown kernel tunables that are affected by the **smcvemgt** command, see https://access.redhat.com/articles/3311301. For more information about the Spectre and Meltdown vulnerabilities, see https://access.redhat.com/security/vulnerabilities/speculativeexecution.

Syntax help: sys smcvemgt --help

```
[admin@server-dev ~]$ sys smcvemgt --help

Version 1.2

Syntax:
    --help,    -h
    --hhelp,   -hh
    --query,   -q
    --set,     -s  enabled
    --set,     -s  disabled
    --set,     -s  [ v2=<v2-mode> ] [ v3=<v3-mode> ]
        (v2-mode: disabled | default | kernel | user | both |
user+retp)
        (v3-mode: disabled | enabled)
    --history
```

Verbose help: sys smcvemgt --hhelp

```
[admin@srvr-dev ~]$ sys smcvemgt --hhelp

Version 1.2

This script manages the enablement status of the Linux kernel patches
for the
following Spectre and Meltdown vulnerabilities:

    Variant #2/Spectre  (CVE-2017-5715)
    Variant #3/Meltdown (CVE-2017-5754)

The kernel patch for the following related vulnerability is
permanently enabled
on the system (cannot be disabled):

    Variant #1/Spectre (CVE-2017-5753)

Note that hardware support is required for Variant #2/Spectre to be
fully
functional. CPU microcode updates must be applied in order for this
hardware
support to be provided. The "--query" argument includes an indication
as to
whether or not hardware support is provided on this server.

For more information on Spectre/Meltdown kernel tunables, refer to:

    https://access.redhat.com/articles/3311301

For additional information on the Spectre/Meltdown vulnerabilities,
refer
to:
```

```
    https://access.redhat.com/security/vulnerabilities/
speculativeexecution

Syntax:

    --help,    -h
        Provide terse help.

    --hhelp,   -hh
        Provide verbose help (this text).

    --query,   -q
        Query the configuration of the Variant #2/Spectre and Variant
#3/
        Meltdown tunables for system reboots, as well as on the
running
        system.

    --set,  -s  enabled
    --set,  -s  disabled
    --set,  -s  [ v2=<v2-mode> ] [ v3=<v3-mode> ]
        Enables and disables Variant #2/Spectre ("v2") and/or Variant
#3/
        Meltdown ("v3") patches.

        This immediately reboots the server. Applications on the
server are
        not managed by this script. Ensure that any applications are
        disabled, as required, prior to changing kernel settings with
this
        script.

        If "enabled" is specified, then both v2 and v3 are enabled,
        with v2 set to kernel default behavior. If "disabled" is
specified,
        then both v2 and v3 are disabled. Otherwise, kernel patches
        are enabled or disbled as per the specified "v2" and/or "v3"
        arguments. If a "v2" or "v3" argument is not specified, the
current
        system value for that item is retained.

        v2-mode:

            disabled
                Variant #2/Spectre is disabled.

            default
                The kernel decides how to set tunables for Variant
#2/
                Spectre, based on the processor architecture. Note
that for
                architectures prior to Skylake, the kernel selects
                retpoline ("return trampoline") over ibrs.

            kernel
                Use "ibrs" (i.e., kernel space only).

            user
```

```
                Use "ibrs_user" (i.e., userland only).

            both
                Use "ibrs_always" (i.e., kernel space and userland).

            user+retp
                Use "retpoline,ibrs_user".

        v3-mode:

            disabled
                Variant #3/Meltdown is disabled.

            enabled
                Variant #3/Meltdown is enabled.
        The following two commands are equivalent:

            sys smcvemgt enabled
            sys smcvemgt v2=default v3=enabled

        The following two commands are equivalent:

            sys smcvemgt disabled
            sys smcvemgt v2=disabled v3=disabled

    --history
        Show a history of changes made to the enablement status of
the
        Spectre and Meltdown patches.
```

sys smcvemgt usage examples

Command for querying current tunable settings

The following command queries the current tunable settings for the next boot, as well as the current runtime. This command also indicates whether there is hardware support for Variant #2/ Spectre.

```
sys smcvemgt --query
```

Command for enabling patches with default settings

The following command enables patches for Variant #2/Spectre and Variant #3/Meltdown, where Variant #2/Spectre is configured for default mode. In default mode, the kernel selects the Variant #2/Spectre mitigation mechanism based on the CPU architecture of the host machine.

```
sys smcvemgt --set enabled
```

Commands for enabling patches with specific settings

- The following command enables patches for Variant #2/Spectre and Variant #3/Meltdown, where Variant #2/Spectre is set to kernel space only.

  ```
  sys smcvemgt --set v2=kernel v3=enabled
  ```

- The following command enables patches for Variant #2/Spectre, which are configured for user space with "Retpoline", or "return trampoline". Variant#3/Meltdown retains its current settings.

  ```
  sys smcvemgt --set v2=user+retp
  ```

Command for disabling patches

The following command disables patches for Variant #2/Spectre and Variant #3/Meltdown.

```
sys smcvemgt --set disabled
```

Command for disabling patches for a specific vulnerability

The following command disables patches for Variant #3/Meltdown. Variant #2/Spectre retains its current settings.

```
sys smcvemgt --set v3=disabled
```

passwdrules command

Description

sys passwdrules allows you to review and edit complexity rules for passwords that you use to log in to the virtual machine with the Avaya Aura® Web Gateway OVA using an SSH connection.

If data encryption is enabled, these rules also apply to encryption passphrases.

Syntax

```
sys passwdrules [show] [set [options]] [set-default]
```

show

Shows the server password rule configuration, including default, minimum, and maximum values.

set

Sets server password rules. If you do not specify any options, then Avaya Aura® Web Gateway prompts you to configure each option separately. If you do not specify a certain option, Avaya Aura® Web Gateway continues to use the existing rule for this option.

set-default

Sets server password rules to their default values.

Syntax help

```
sys passwdrules [-h | --help]
```

Options

Option	Description
`diff=INTEGER`	Number of characters in the new password that must not be present in the old password.
`min-len=INTEGER`	Minimum length of the password. The default minimum length is 15 characters.
`min-digs=INTEGER`	Minimum number of digits in the password.
`min-upper=INTEGER`	Minimum number of uppercase characters in the password.
`min-lower=INTEGER`	Minimum number of lowercase characters in the password.
`min-other=INTEGER`	Minimum number of special characters in the password. Important: Avaya Aura® Web Gateway supports the following special characters: !, @, #, %, $, ^, *, ?, and _.
`min-class=INTEGER`	Minimum number of character classes that must be present in the password. The character classes are: • Digits • Uppercase characters • Lowercase characters

Option	Description
	• Special characters
`--max-repeat=INTEGER`	Maximum allowed number of consecutively repeated characters. For example, if you set this parameter to `3`, then the following password is invalid: `paaassword`, because a is repeated three times.
`--max-class-repeat=INTEGER`	Maximum allowed number of consecutively repeated characters of the same class. For example, if you set this parameter to `3`, then the `pas1sw2or3d` password is invalid, because the first three characters, which are `p`, `a`, and `s`, belong to the same character class.

Example

The following is an example of a command that sets the following password rules:

- At least 16 characters in total.
- At least two digits.
- At least one special character.
- Other settings are default.

```
sys passwdrules set --min-len=16 --min-digs=2 --min-other=1
```

Data encryption commands

The following sections contains command you can use to manage data encryption. These commands are available if you enabled data encryption during Avaya Aura® Web Gateway OVA deployment.

You cannot enable or disable data encryption using system layer commands.

> Important:
>
> - In AWS deployments, you enable data encryption on AWS itself. Therefore, you cannot use the system layer commands for disk encryption management in AWS deployments.
> - These commands are only available for OVA-based deployments.

encryptionPassphrase command

Description

You can run the **sys encryptionPassphrase** command to manage the encryption passphrase.

Syntax

```
sys encryptionPassphrase [add | change | remove | list]
```

add
: Enables you to set up an encryption passphrase.

change
: Enables you to change the existing encryption passphrase.

remove
: Removes the encryption passphrase.

list
: Displays information about passphrases and slot assignments.

encryptionStatus command

Description

The **sys encryptionStatus** command displays information about data encryption on Avaya Aura® Web Gateway , including the following:

- Whether data encryption is enabled.
- Whether the local key store is enabled.
- Whether the encryption password is used.

Syntax

```
sys encryptionStatus
```

encryptionRemoteKey command

Description

You can use the **sys encryptionRemoteKey** command to manage the remote key server. If you run the command without any parameters, Avaya Aura® Web Gateway displays help information about the command.

Syntax

```
sys encryptionRemoteKey [add <server address> [<port>] |
 remove <server address> | list]
```

add

Enables you to add a remote key server.

remove

Removes the remote key server.

list

Displays general remote key server information.

encryptionLocalKey command

Description

You can use the **disk encryptionLocalKey** command to manage the local key store. If you run this command without any parameters, Avaya Aura® Web Gateway displays help information.

Syntax

```
sys encryptionLocalKey [enable | disable]
```

enable

Enables the local key store.

disable

Disables the local key store.

Using the Tmux utility

Avaya Aura® Web Gateway resets idle SSH sessions after a certain period of time. This might affect lengthy CLI operations, such as Avaya Aura® Web Gateway installation or backup. If the connection with the SSH terminal is reset while installation or backup operations are in progress, you must restart these operations. To prevent this issue, you can start a Tmux session before performing a lengthy operation. Commands and processes that you run from a Tmux session continue to run even if the connection with the SSH terminal is reset. Therefore, you can re-connect to the Tmux session and complete the operation.

1. To start a Tmux session, run the following command from the Avaya Aura® Web Gateway CLI:
   ADDITIONAL INFORMATION:
   `tmux new-session -s <SESSION_NAME>`

   In this command, `SESSION_NAME` is a Tmux session name of your choice. For example:

   ```
   tmux new-session -s AAWG_session
   ```
2. To re-connect to the Tmux session, run the following command:
   ADDITIONAL INFORMATION:
   `tmux attach-session -t <SESSION_NAME>`

   In this command, `SESSION_NAME` is the name of the Tmux session that you created. For example:

   ```
   tmux attach-session -t AAWG_session
   ```

Avaya AuraÂ® Web Gateway management with the administration portal

Creating a client certificate

Use this procedure to create a client certificate, which can be imported into a web browser for authenticating automatic login into the Avaya Aura® Web Gateway web administration portal.

1. Open the Linux shell using your Linux administrator account credentials.
2. To generate a CSR, run the following command:
   ADDITIONAL INFORMATION:

   ```
   sudo ./createCSR.sh /tmp/AAWGportalCerts frontEndFQDN localFQDN
   organizationNameorganizationUnit locality stateOrProvince
   countryCode emailAddress
   ```

   > Important:
   > This command is a single Linux command and must be entered as a single line even if it appears as several lines in the document.

   The parameters for this script are:

   - `frontendFQDN`: For a cluster installation, this is the FQDN of the Virtual IP or external load balancer. For simple, non-clustered installations, this is the FQDN of the server where Avaya Aura® Web Gateway is installed.
   - `localFQDN`: The FQDN of the server.
   - `orgnizationName`: The name of the organization.
   - `organizationUnit`: The name of the unit or sub-organization. For example, "Design".
   - `locality`: The name of the city or town.
   - `state`: The two-digit state or province code.
   - `countryCode`: The two-digit country code.
   - `emailAddress`: The administrator email address.

   STEP RESULT:
   Avaya Aura® Web Gateway create the *oamp.csr* and *oamp.key* files in the */tmp/AAWGportalCerts* directory.
3. To generate the *.pem* file, on the System Manager web console, navigate to **Services › Security › Certificates › Authority**.
4. Click the **Add End Entity** tab and complete the following settings:
   a. Set **End Entity Profile** to **Empty**.
   b. Type your user name and password in **Username** and **Password**.

c. Type your user ID in **CN, Common name**.
   ADDITIONAL INFORMATION:
   The user ID you provide must use the same format that you used for the **UID Attribute ID** field on the LDAP Configuration tab.
d. Set **Certificate Profile** to **ENDUSER**.
e. Click **Add**.
   STEP RESULT:
   A new end entity with the specified user name is created on the System Manager web console.

5. In the left navigation pane, click the **Public Web** tab and complete the following settings:
   a. In **Username** and **Enrollment code**, type the same user name and password that you used to create an end entity.
   b. Click **Choose File** to add the *oamp.csr* file, which you generated in step See "2" on page 44.
   c. Click **OK** to generate the *.pem* file.
6. In the SSH console, run the **openssl** command to convert the *.pem* file to a *.pfx* or *.p12* file. The following is an example of the command to convert the *.pem* file to a *.p12* file:

```
openssl pkcs12 -export -out <P12_FILE_NAME> -
in <PEM_FILE_NAME>.pem -inkey oamp.key -passout pass:<PASSWORD>
```

Importing client certificates into web browsers

Use this procedure to import client certificates to the Google Chrome, Internet Explorer, or Mozilla Firefox web browser. This is an optional procedure. After you perform this procedure, you will automatically be logged in to the web administration portal. The system will bypass the Login screen.
PREREQUISITE
On the HTTP Clients tab, set **OAMP** to one of the following:

- **REQUIRED**: For certificate authentication.
- **OPTIONAL**: For certificate or password authentication.

For information on **OAMP** options, see See "Available certificate validation options" on page 91.

1. Navigate to the appropriate location in your web browser:
   - From Google Chrome, navigate to **Settings › Show advanced settings › Manage certificates**.
   - From Internet Explorer, navigate to **Tools › Internet options › Content › Certificates**.
   - From Mozzila Firefox, navigate to ≡ **› Options › Advanced › Certificates › View Certificates**.
2. Click **Import** to import the certificate to your browser.
   ADDITIONAL INFORMATION:
   When prompted for a password, enter the same password that was used in the **openssl** command to convert the *.pem* file to a *.pfx* or *.p12* file.
   ADDITIONAL INFORMATION:
   > Tip:

> To override the certificate authentication, in the SSH console, set `com.avaya.cas.common.certificateauth` to `0` as follows:
>
> ```
> CATALINA_OPTS="$CATALINA_OPTS
> -Dcom.avaya.cas.common.certificateauth=0"
> ```

Logging on to the Avaya Aura® Web Gateway administration portal

Use this procedure to log on to the Avaya Aura® Web Gateway administration portal. You can use this administration portal to configure components and update element settings anytime. You can also manage client requests from the administration portal.

The administration portal also shows the Avaya Aura® Web Gateway deployment type, which must be selected during the initial setup.

The Avaya Aura® Web Gateway administration portal supports the following web browsers:

- Internet Explorer 9, 10, or 11.
- Microsoft Edge 42 and later.
- The latest version of Mozilla Firefox or the version before it.
- Google Chrome 53 and later.

1. In your web browser, enter the URL:
   ADDITIONAL INFORMATION:
   `https://<Gateway_FQDN>:8445/admin/`
   ADDITIONAL INFORMATION:

   > Tip:
   > To obtain your FQDN, run the `hostname` command in the Avaya Aura® Web Gateway Linux shell.
   >
   > To set up an Avaya Aura® Web Gateway cluster, use the front-end FQDN for the cluster. If you are using an external load balancer, the front-end FQDN for the cluster will be the FQDN of the load balancer. If not, it will be the FQDN of the virtual IP assigned to the Avaya Aura® Web Gateway cluster.

2. Log on to the Avaya Aura® Web Gateway administration portal using one of the following:
   - Credentials of the user that was configured as an LDAP server administrator.
   - Linux administration account credentials.

System overview settings

Reviewing Avaya Aura® Web Gateway service status information

Use this procedure to view system overview information and server settings.

1. On the Avaya Aura® Web Gateway administration portal, click **System Overview**.
   ADDITIONAL INFORMATION:
   The page displays information about the deployment type, server status, and node status.
2. To modify server settings, in the Solution Servers area, click a server from the Required Server column.
   STEP RESULT:
   The appropriate Avaya Aura® Web Gateway administration portal page for the server you selected is displayed.

Avaya Aura® Media Server connectivity status indicators

On the System Overview page, Avaya Aura® Web Gateway displays the consolidated status for all Avaya Aura® Media Servers connected to Avaya Aura® Web Gateway. The following table shows the status indicators that Avaya Aura® Web Gateway can display.

Status indicator	Description
	All servers are available and can be used for calls.
	At least one server cannot accept requests. Other servers are available.
	At least one server cannot accept requests. Other servers are inactive.
	The system cannot obtain the status of any server.

Restarting services on an Avaya Aura® Web Gateway node

Use this procedure to restart services on any Avaya Aura® Web Gateway node from the administration portal.
PREREQUISITE
In a cluster environment, if you use Avaya Spaces Calling Gateway, enable Maintenance mode as described in See "Enabling Maintenance mode" on page 110. Maintenance mode prevents interoperability issues with Avaya Spaces Calling Gateway that might occur when restarting Avaya Aura® Web Gateway services.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **System Overview**.
2. In the Service Control area, select the node you want to restart from the drop-down list.
3. Click **Restart**.
   STEP RESULT:
   Avaya Aura® Web Gateway displays a pop-up window informing you that the restart is in progress.

    - If you are restarting a local node, you cannot continue working with the administration portal. You must wait until the restart is completed and then click **OK** to log in.
    - If you are restarting a remote node, press **Hide** and continue working with the administration portal. You can check whether the restart is completed and the node is back online in the Node Status area.

AFTER COMPLETING THE TASK

- In a cluster environment, if you use Avaya Spaces Calling Gateway, disable Maintenance mode as described in See "Disabling Maintenance mode" on page 111.

Identifying the seed node

In a cluster environment, you must perform certain procedures, such as upgrade, on the seed node first. Use this procedure to identify the seed node.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **System Overview**.
2. Navigate to the Cluster table.
   STEP RESULT:
   The seed node IP address and FQDN are listed in the Initial Node row.

Configuring general network settings on the Avaya Aura® Web Gateway

1. On the Avaya Aura® Web Gateway administration portal, click **General Network Settings**.
2. Click the appropriate tab.
   ADDITIONAL INFORMATION:
   The procedures below describe the tasks you can perform on each tab.

Updating System Manager settings

Use this procedure to fetch Media Server and Location lists from System Manager.

1. Click the **System Manager** tab.
2. In **FQDN**, type the System Manager FQDN address.
3. In **Port**, type the System Manager port number.
   ADDITIONAL INFORMATION:
   The default port number is 443.
4. In **Protocol**, select **https**.
5. Type the System Manager administration account login ID and password.
   ADDITIONAL INFORMATION:
   The System Manager account must have the "View access" privilege for all media server elements.
6. Click **Save**.

Configuring the Avaya Aura® Device Services host to obtain user data

1. Click the **Device Services** tab.
2. In **FQDN**, type the Avaya Aura® Device Services FQDN address.
   ADDITIONAL INFORMATION:

   > Important:
   > Ensure that the Avaya Aura® Device Services FQDN is resolvable from the Avaya Aura® Web Gateway server. Open the Linux shell using the Linux administrator account

> credentials, and type the `ping <AADS-FQDN>` command to test DNS resolution of the Avaya Aura® Device Services FQDN from the Avaya Aura® Web Gateway.

3. In **Client interface port**, type the port that is used to access Avaya Aura® Device Services.
   ADDITIONAL INFORMATION:
   The port must match the port configuration on Avaya Aura® Device Services. The default value is `443`.
4. If you are upgrading or installing the Avaya Aura® Web Gateway for the first time, type `8440` in **Server-to-server interface port**.
   ADDITIONAL INFORMATION:
   This port enables the Avaya Aura® Web Gateway to establish a connection with Avaya Aura® Device Services. The value in this field is automatically populated if the Avaya Aura® Device Services configuration already exists.

   > Tip:
   > If you need to enable port 8440 on Avaya Aura® Device Services, go to the *cdto misc* directory and run the `sudo ./dynamicconfigurations-patch.sh enable` command on every node in the cluster.
   >
   > If you need to disable this port from the firewall rule, you can use the `sudo ./dynamicconfigurations-patch.sh disable` command.

5. In **Protocol**, select **https** or **http**.
   ADDITIONAL INFORMATION:
   This selection must match the configuration on Avaya Aura® Device Services.
6. Click **Save**.
7. To clear the local device services data, in the Clear Local Device Services Data area, click **Clear**.

Managing location settings

Managing Avaya Aura® Web Gateway locations

1. Click the **Location** tab.
2. In the Web Gateway Locations area, under **Address**, verify the Avaya Aura® Web Gateway FQDN.
   ADDITIONAL INFORMATION:
   This value should be automatically populated.
3. In **Location**, select the appropriate location for each server FQDN from the list of locations.
   ADDITIONAL INFORMATION:
   The location information is retrieved from System Manager.
   a. For an Avaya Aura® Web Gateway cluster setup, set the location for each node in the cluster.
      ADDITIONAL INFORMATION:
      The FQDN for each node is added automatically.
4. Click **Save**.

Managing Avaya Aura® Media Server location and priority settings for the Avaya Aura® Web Gateway

Use this procedure to assign and prioritize locations for each Avaya Aura® Web Gateway location. In a location, the Avaya Aura® Web Gateway uses the available servers in the Assigned Locations area.

1. Click the **Location** tab.
2. In the Location Assignments and Priorities area, from **Web Gateway Location**, select an Avaya Aura® Web Gateway location.
3. To add an Avaya Aura® Media Server location to the Assigned Locations list, select the location from the Input Location list and then click **Add**.
   ADDITIONAL INFORMATION:
   You can click **Add All** to add all listed locations to the Assigned Locations list.
4. To remove an Avaya Aura® Media Server location from the Assigned Locations list, select the location and then click **Remove**.
   STEP RESULT:
   The removed Avaya Aura® Media Server locations are added to the Input Location area.
5. Drag and drop the servers in the Assigned Locations to rearrange their priority.
   ADDITIONAL INFORMATION:
   The order you set in the Assigned Locations list determines how the Avaya Aura® Media Server locations will serve the selected Web Gateway location.
6. Click **Save**.
7. Repeat the steps above for each location listed in **Web Gateway Location**.

Verifying the LDAP server configuration settings

Use this procedure to modify the LDAP configuration settings so that they are consistent with the Avaya Aura® Device Services settings.

1. Click the **LDAP Configuration** tab.
2. Modify the LDAP configuration settings as needed.
   ADDITIONAL INFORMATION:
   The settings are described in See "Enterprise LDAP Server Configuration field descriptions" on page 52.
3. Ensure that **UID Attribute ID** is set to the same value that is in Avaya Aura® Device Services.
4. Modify or update other parameters related to LDAP as required and then click **Save**.
5. Click **Test Connection** to verify the connection.
6. If the connectivity test fails, check the configured LDAP address FQDN, port, and protocol, and then run the test again.

Reviewing Avaya Aura® Media Server status

Use this procedure to review the connectivity status and availability of Avaya Aura® Media Server for a specific node in a cluster.

1. On the Avaya Aura® Web Gateway web administration portal, do one of the following:

- On the System Overview page, click **Media Services**.
- Navigate to **General Network Settings › Media Services**.

2. For a cluster environment, in the **Node Address** field, select the required node.
   STEP RESULT:
   Avaya Aura® Web Gateway displays a table with information about Media Servers for the selected node.
3. In the Status column, hover over the Avaya Aura® Media Server status icon to see performance information.
4. To edit Avaya Aura® Media Server settings, click the required Avaya Aura® Media Server host name and do one of the following:
   - If you are redirected to the System Manager web administration portal, log in using System Manager credentials with access privileges for Avaya Aura® Media Server.
   - If you are redirected to the Avaya Aura® Media Server web administration portal, log in using Avaya Aura® Media Server credentials.

LDAP server management

You must configure the enterprise LDAP server to authenticate the users and administrators of the Avaya Aura® Web Gateway. The LDAP Configuration screen on the Avaya Aura® Web Gateway web administration portal displays the enterprise LDAP server that you configured during deployment.

You cannot perform all LDAP server management tasks with the configuration utility. Use the Avaya Aura® Web Gateway web administration portal to do the following:

- Configure multiple LDAP directories.
- Specify an order in which the Avaya Aura® Web Gateway accesses LDAP directories.
- Select which LDAP directories are used for authentication.
- Configure multiple base context Distinguished Names (DNs).
- Set up LDAP synchronization.
- Configure attribute mappings.

Important:

- For secure connectivity to LDAP servers, you must import an LDAP certificate file to the Tomcat trust store. For more information, see See "Importing the secure LDAP certificate using the web administration portal" on page 89.
- If FIPS is enabled on Avaya Aura® Web Gateway, you must use the secure LDAP (LDAPS) connection to access LDAP servers.

Adding a new enterprise LDAP server

Use this procedure to add a new LDAP server to Avaya Aura® Web Gateway. After you have added a server, you cannot modify the server URL or remove the LDAP data source.
PREREQUISITE

If you are planning to use multiple authentication domains, configure the UID mapping attributes as specified in See "Configuring the UID mapping attributes when using multiple authentication domains" on page 59.

1. On the Avaya Aura® Web Gateway web administration portal, navigate to **General Network Settings › LDAP Configuration**.
   STEP RESULT:
   Avaya Aura® Web Gateway displays the Enterprise LDAP Server Configuration page.
2. Click the plus (+) icon.
   STEP RESULT:
   Avaya Aura® Web Gateway displays the New Directory tab.
3. In the **Enterprise-Directory Type** field, click the LDAP server directory that you want to add.
4. In the **Provenance Priority** field, type the priority of the enterprise LDAP server directory.
5. In the Server Address and Credentials section, specify the parameters of the enterprise LDAP server directory.
   ADDITIONAL INFORMATION:
   For more information about attributes, see See "Enterprise LDAP Server Configuration field descriptions" on page 52.
6. If you want to use the new server for authentication and authorization, do the following:
   a. Select the **Use for Authentication** check box.
   b. Configure role-related attributes, such as **Role Filter**, **Role Attribute ID**, and **Role Context DN**.
7. Click **Save**.

Enterprise LDAP Server Configuration field descriptions

Name	Description
Enterprise-Directory Type	Specifies the name of the enterprise directory. The options are: • ActiveDirectory_2012 • ActiveDirectory_2016 • ActiveDirectory_2019 • ActiveDirectory_2022 • Azure Active Directory • Domino 7.0 or 8.5.3 • LDS_2012 • Novell 8.8 • OpenLDAP 2.4.44 and 2.4.46 • OracleDirectoryServer 11.1.1

Name	Description
Provenance Priority	Specifies the provenance priority of the enterprise directory. Provenance priority is used while merging contacts. If a value is available in more than one directory, the directory value with higher provenance priority is returned. For example, if firstName is obtained from two directories, the firstName from the source with higher provenance priority is returned. You can assign a value between 2 to 10. You cannot assign Provenance priority 1 because it is always assigned to the authorization directory. Provenance priority 1 is the highest, and 10 is the lowest. Provenance priority must be different for each enterprise directory or source.

Server Address and Credentials

Name	Description
Secure LDAP	Indicates whether the LDAP server connection is secure or not. If FIPS is enabled, you must use the secure LDAP connection to access LDAP servers. If you are using a secure LDAP connection, you must also import the LDAP server trusted certificate to Avaya Aura® Web Gateway.
Import Certificate	Specifies the LDAP server trusted certificate. This field is mandatory if you are using a secure LDAP server connection. This field is only displayed when the **Secure LDAP** check box is selected. You cannot import a certificate if: • The certificate contains an unsupported critical extension. • The certificate expired. • The start date of the certificate is in the future.
Windows Authentication	Specifies whether to use Windows Authentication. The options are: • None • Negotiate If you select the Negotiate option, the system displays the **Configuration for Windows Authentication** section. Note:

Name	Description
	Windows authentication is only supported if you are using a single authentication directory. If you are using multiple authentication directories, Windows Authentication is disabled.
Address	Specifies the IP address or FQDN of the LDAP server. This field is mandatory.
Port	Specifies the port of the LDAP server. This field is mandatory.
Bind DN	Specifies the Distinguished Name (DN) of the user that has read and search permissions for the LDAP server users and roles. This is a mandatory setting. The format of the Bind DN depends on the configuration of the LDAP server. This field is mandatory. Note: Even though the parameter name is Bind DN, the format of its value is not limited to the DN format. The format can be any format that the LDAP server can support for LDAP bind. For example: for Active Directory, you can use `domain\user`, `user@domain`, as well as the actual DN of the user object.
Bind Credential	Specifies the password of the administrative user. The maximum password length depends on the LDAP server type that you use in your deployment.
Base Context DN	Specifies the complete Distinguished Name (DN) with the Organizational Unit (OU) for starting the search for users on the enterprise directory. This is the primary Base Context DN for Avaya Aura® Web Gateway. For example, `dc=domain, dc=company, dc=com`. If you are using multiple authorization domains, Avaya recommends including a domain component to the Base Context DN. For example, `dc=avaya, dc=com`. Note: Some LDAP sources, such as Domino, typically do not contain the domain component in Base Context DNs. For example, `o=MyCompany`. If Base Context DNs do not contain the domain component, Avaya Aura® Web Gateway considers them "empty" Base Context DNs when processing user login or search requests. If Avaya Aura® Web Gateway cannot find the domain specified in the request, the search continues in "empty" Base Context DNs.

Name	Description
Use additional Base Context DN	Enables Avaya Aura® Web Gateway contact search and quick search. The primary Base Context DN is used for authentication. Additional Base Context DNs are used for Avaya Aura® Web Gateway contact search and quick search, and can also be used for authentication. You can configure up to 10 additional Base Context DNs. If you select this check box, you can see the **View/Edit** button. Auto-configuration will use only the primary base context DN.
View/Edit	Enables access to the Addition Base DN Configuration page, where you can add or delete additional Base Context DNs.
UID Attribute ID	Specifies the unique attribute of the user on LDAP, which is used to search for users in the LDAP server. If you are using multiple authentication domains, you must use one of the following values: • `mail` • `userPrincipalName` • Any custom attribute that uses a domain-qualified value If you are not using multiple authentication domains, you must use one of the following values: • `sAMAccountName` • `mail` • `userPrincipalName` This field is mandatory.
Role Filter	Specifies the search filter used to search the role of the user. For example, (&(objectClass=group) (member={1})
Role Attribute ID	Specifies that the user is a member of the groups defined by that attribute. For example, objectCategory This field is mandatory.
Roles Context DN	Specifies the complete Distinguished Name (DN) to search for a user role, that is, for Role Filter. For example, dc=domain,dc=company,dc=com
Role Name Attribute	Specifies the name of the role attribute.

Name	Description
	This field is mandatory only if the **Role Name Attribute Is DN** field is set to true. For example, cn if the role is stored in a DN in the form of cn=admin, ou=Users, dc=company, dc=com.
Role Attribute is DN	Indicates whether the role attribute of the user contains DN. The default value is true.
Allow Empty Passwords	Indicates whether LDAP Server acknowledges the empty password. The default value is false.
Search Scope	Specifies the search level in the LDAP hierarchy. The options are: • Object : For searching only for the object. • One Level : For including one level in the LDAP hierarchy in the search. • Subtree : For including subtree in the LDAP hierarchy in the search. The default value is Subtree .
Role Recursion	Specifies whether role recursion is enabled. The options are: • true • false If your LDAP configuration includes nested groups, you can set the **Role Recursion** parameter to `true` so that Avaya Aura® Device ServicesAvaya Aura® Web Gateway computes role membership by searching through LDAP structures recursively. For example, the user jsmith can be in the Sales group, which can be in the AAWG users group. In this case, Role Recursion must be set to true for jsmith to be recognized as a member of the Avaya Aura® Web Gateway users group. If you set this parameter to `false`, Avaya Aura® Web Gateway does not compute the role membership information recursively. Note: Certified with 300 nested groups with 2 levels for each user.

Name	Description
Administrator Role	Specifies the administrator role in which the administrative users are assigned.
Security Administrator Role	Specifies the security administrator role in which the administrative users can manage web certificates from the web administration portal.
User Role	Specifies the user role in which the common users are assigned.
Auditor Role	Specifies the auditor role in which the users can audit the system.
Services Maintenance and Support Role	Specifies the services maintenance and support role in which users can maintain and support services.
Services Administrator Role	Specifies the services administrator role.
Language used in Directory	• Simplified Chinese (zh) • German (de) • English (en) • Spanish (es) • French (fr) • Italian (it) • Japanese (ja) • Korean (ko) • Russian (ru) • Portuguese (pt)
Active Users Search Filter	Specifies whether the user is active or inactive on LDAP Server.
Last Updated Time Attribute ID	Specifies when the user is updated on LDAP. The exact value depends on the LDAP server type that you use. Avaya recommends that you use the following values: • For Active Directory: `whenChanged`. • For OpenLDAP: `modifyTimestamp`. This field is mandatory.

Configuration for Windows Authentication

Name	Description
Service Principal Name (SPN)	Specifies the service principal name UIDAttributeID must be userPrincipalName.
Import keytab file	Imports the *tomcat.keytab* file and overwrites the existing file.
Kerberos Realm	Specifies the Kerberos realm.
DNS Domain	Specifies the DNS domain of the Domain Controller.
KDC FQDN	Specifies the FQDN of the Domain Controller.
KDC Port	Specifies the port number. The default KDC port is 88.

Button	Description
Test Connection	Tests the connection changes.
Save	Saves the changes made to the enterprise directory.
Modify Attribute Mappings	Modifies the attributes of the LDAP server.

Configuring additional base context DNs

Use this procedure to add additional base context DNs. Base context DNs are used to specify sections of the LDAP directory where LDAP searches for:

- Contacts, when performing LDAP lookups. The use of multiple base context DNs ensures maximum search performance.
- Groups, when configuring and publishing settings using the Dynamic Configuration service.

You can add up to 10 base context DNs for one LDAP source.

You can also use an additional base context DN for authentication. In this case, this additional base context DN uses the role context DN that is configured in the primary LDAP server. You can use only the primary domain or its sub-domains as an authentication domain.

1. On the Avaya Aura® Web Gateway web administration portal, navigate to **General Network Settings › LDAP Configuration**.
2. Select the appropriate LDAP server.
3. In the Server Address and Credentials section, select the **Use additional Base Context DN** check box and then click **View/Edit**.
4. In the Additional Base Context DN Configuration window, click **+**.
5. In the table, click the **Base Context DN** field and then provide the value of the base context DN.
6. If you want to use the base context DN for authentication, select the **Use for Authentication** check box.
7. Click **Save**.
8. In the Server Address and Credentials section, click **Save**.

Modifying the provenance priority

If you configured multiple LDAP servers , use this procedure to specify the order in which Avaya Aura® Web Gateway accesses LDAP directories. For example, if a givenName attribute is defined in two LDAP servers for a given user, the value that Avaya Aura® Web Gateway uses is based on the order you define.

1. On the Avaya Aura® Web Gateway web administration portal, navigate to **General Network Settings › LDAP Configuration**.
   STEP RESULT:
   Avaya Aura® Web Gateway displays the Enterprise LDAP Server Configuration page.
2. Click the **Enterprise Directory** tab that you want to use to modify the provenance priority.
3. In the **Provenance Priority** field, click **Modify**.
   STEP RESULT:
   Avaya Aura® Web Gateway displays the Source Priority Configuration pop-up window.
4. In the **Provenance Priority** column, type the priority level.
   ADDITIONAL INFORMATION:
   You can assign a value between
   2
   to
   10
   . You cannot assign Provenance priority
   1
   as it is always assigned to the authorization directory. Provenance priority
   1
   is the highest and
   10
   is the lowest. Provenance priority must be different for each enterprise directory or source.
5. Click **Save**.
6. Restart Avaya Aura® Web Gateway to apply the new priority.

Configuring the UID mapping attributes when using multiple authentication domains

If you are using multiple authentication domains, the UID mapping attribute value set for each LDAP server must be domain-qualified. Therefore, the value must include the `@domain` part. The following values are supported:

- `mail`
- `userPrincipalName`
- Any custom attribute that uses a domain-qualified value.

> Note:
> If you are not using multiple authentication domains, the Avaya Aura® Web Gateway supports only the following values for the UID mapping attribute:
>
> - `sAMAccountName`

- `mail`
- `userPrincipalName`

1. On the Avaya Aura® Web Gateway web administration portal, navigate to **General Network Settings › LDAP Configuration**.
   STEP RESULT:
   The system displays the Enterprise LDAP Server Configuration page.
2. For each LDAP server configured on the Avaya Aura® Web Gateway, in the **UID Attribute ID** field, enter the appropriate value.

Administering the LDAP server configuration

1. On the Avaya Aura® Web Gateway administration portal, navigate to **General Network Settings › LDAP Configuration**.
   STEP RESULT:
   Avaya Aura® Web Gateway displays the Enterprise LDAP Server Configuration page.
2. Select the appropriate LDAP Server.
3. Modify the details of the enterprise directory.
4. Click **Save**.

Modifying enterprise directory attribute mappings

Each LDAP directory type includes a set of pre-defined attributes mappings, which map user attributes that Avaya Aura® Web Gateway uses to LDAP attributes defined in the default LDAP schema for the respective LDAP directory type. If you use a custom LDAP schema for your LDAP directory type, you must customize the attribute mapping accordingly.

Avaya Aura® Web Gateway search results do not include attributes that are not mapped to LDAP attributes.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **General Network Settings › LDAP Configuration**.
   STEP RESULT:
   Avaya Aura® Web Gateway displays the Enterprise LDAP Server Configuration page.
2. Select the appropriate LDAP server.
3. In the Server Address and Credentials section, click **Modify Attribute Mappings**.
   STEP RESULT:
   Avaya Aura® Web Gateway displays the Enterprise Directory Mappings page. The Modify LDAP Attribute Mappings table contains the following columns:
   - Application Field Name contains user attributes that Avaya Aura® Web Gateway uses.
   - Directory Field Name contains pre-defined attributes from the standard LDAP schema.
   - Custom Field Name enables you to enter custom attribute names.
4. In the Modify LDAP Attribute Mappings section, do one of the following for the required attribute:
   - To select one of the pre-defined LDAP attributes, click the **Directory Field Name** cell and then select the required LDAP attribute from the list.

- To use a custom LDAP attribute, click the **Custom Field Name** cell and enter the name of the required custom LDAP attribute.
- To un-map the attribute, clear the **Custom Field Name** value and set **Directory Field Name** to **Choose Attribute**.

5. If you want to reset the attribute mapping to its default settings, click **Reset**.
6. Click **Save**.
   STEP RESULT:
   Avaya Aura® Web Gateway restarts to apply changes.

The following image shows an Active Directory 2012 attribute mapping example, where:

- Alias and ASCIIGivenname attributes use default mapping.
- ASCIIDisplayname attribute is mapped to the custom asciiDisplayName LDAP attribute.
- ASCIISurname is not mapped to any LDAP attribute

Modify LDAP Attribute Mappings

Application Field Name	Directory Field Name	Custom Field Name
Alias	cn	
ASCIIDisplayname	Choose Attribute	asciiDisplayName
ASCIIGivenname	givenName	
ASCIISurname	Choose Attribute	

Configuring Windows Authentication for Active Directory

Windows authentication is only supported if you are using a single authentication directory. If you are using multiple authentication directories, Windows Authentication is disabled.
PREREQUISITE
Ensure that the LDAP server you use is the Domain Controller with the appropriate Active Directory version as the server type.

1. On the Avaya Aura® Web Gateway web administration portal, navigate to **General Network Settings › LDAP Configuration**.
   STEP RESULT:
   The system displays the Enterprise LDAP Server Configuration page.
2. In the Server Address and Credentials section, do the following:
   a. In the **Windows Authentication** field, click **Negotiate**.
   b. In the Confirm Action pop-up window, click **OK**.
   c. The **UIDAttributeID** must be userPrincipalName.
   d. Ensure that the other settings on the Server Address and Credentials page are appropriate for the LDAP configuration of your Domain Controller.
3. In the Configuration for Windows Authentication section, do the following:
   ADDITIONAL INFORMATION:

   > Tip:
   > To complete the following fields, use the same values you entered when setting up the Windows Domain Controller.

   a. In **Service Principal Name**, type `HTTP` or `REST_FQDN`.
   ADDITIONAL INFORMATION:
   For example, type `HTTP` or `aads.example.com`.

b. To import the *tomcat.keytab* file transferred from the Windows Domain Controller, in **Import keytab file**, click **Import**.
ADDITIONAL INFORMATION:
In cluster deployments, the file is transferred to all nodes in the cluster. An additional option is available to send the file to specific nodes in a cluster.

You can use the following command to generate a *tomcat.keytab* file.

```
ktpass /out c:\tomcat.keytab /mapuser <Domain User
Login>@<Kerberos realm> /princ HTTP/<FRONT-END
FQDN>@<Kerberos realm> /ptype KRB5_NT_PRINCIPAL /pass
+rndPass /crypto all /kvno 0
```

In the following example, `<Domain User Login>` is `csa_user`, `<Kerberos realm>` is `EXAMPLE.COM`, and `<FRONT–END FQDN>` is `csa.example.com`.

```
ktpass /out c:\tomcat.keytab /mapuser csa_user@EXAMPLE.COM /
princ HTTP/csa.example.com@EXAMPLE.COM /ptype
KRB5_NT_PRINCIPAL /pass +rndPass /crypto all /kvno 0
```

c. In **Kerberos Realm**, type the Kerberos realm, which is usually in uppercase letters.
ADDITIONAL INFORMATION:
For example, `EXAMPLE.COM`.

d. In **DNS Domain**, type the DNS domain of the Domain Controller.
ADDITIONAL INFORMATION:
For example, `example.com`.

e. In **KDC FQDN**, type the FQDN of the Domain Controller.
ADDITIONAL INFORMATION:
This value also includes the DNS domain at the end.

For example, `ad.example.com`.

f. In **KDC Port**, do not change the default setting , which is
88
.

g. **Optional:** In a cluster deployment, click **Send Keytab File** to send the *tomcat.keytab* file to a specific node.
ADDITIONAL INFORMATION:
This option is useful if the import to a node failed or if you add a new node to your cluster.

4. Save the settings to restart the server.
ADDITIONAL INFORMATION:
The settings you specified are used to generate the files needed to configure the Tomcat JAASRealm and the corresponding Sun JAAS Login module for GSS Bind.

Configuring the internationalization parameters

The internationalization parameters specify how a user’s given name and surname are stored in Microsoft Active Directory (AD), and which language is used to store these names. Optionally, for non-Latin script languages, two of the parameters also specify how the ASCII transliteration of these names is stored.

The following procedure describes configuring the LDAP internationalization parameters when AD is used.

1. On the Avaya Aura® Web Gateway web administration portal, navigate to **General Network Settings › LDAP Configuration**.
   STEP RESULT:
   Avaya Aura® Web Gateway displays the Enterprise LDAP Server Configuration page.
2. Configure the language setting:
   ADDITIONAL INFORMATION:

Parameter	Description	Default value
Language used in Directory	The language code of one of the languages supported by the Avaya Aura® Web Gateway.	English (en)

3. Click **Save**.
4. Click **Modify Attribute Mappings**.
5. Configure the following settings:
   ADDITIONAL INFORMATION:

Parameter	Description	Default value
NativeFirstName	The attribute that stores the "given name" of the user in the language of the LDAP server.	givenName
NativeSurName	The attribute that stores the "surname" of the user in the language of the LDAP server.	sn
GivenName	This is only applicable if the language in AD is one of the non-Latin script based ones.	
SurName	This is only applicable if the language in AD is one of the non-Latin script based ones.	

   The NativeFirstName and NativeSurName parameters allow the user to identify the LDAP attributes used to store the user's native language given name and surname. These are mandatory parameters with defaults of
   givenName
   and
   sn
   .

   The GivenName and SurName parameters allows the user to identify the LDAP attributes used to store the ASCII transliteration of the user's given name and surname, respectively. These are optional parameters and only used only if the Language used in Directory parameter is set to one of the non-Latin script languages.

   The internationalization of the names must be done using the language tags specified in RFC 3866.

   To configure internationalization for Microsoft Active Directory, you must configure custom attributes for the native and the ASCII transliterations of the names, if both types of names are needed.
6. Click **Save** to apply changes and restart the Avaya Aura® Web Gateway.

Configuring Avaya Meetings Server settings on the Avaya Aura® Web Gateway

Use this procedure if your deployment includes Avaya Meetings Server.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Avaya Meetings Server › Avaya Meetings Management**.
2. In **Avaya Meetings Management IP**, type the Avaya Meetings Server Management server IP address.
   ADDITIONAL INFORMATION:
   The system automatically populates the following SIP settings after the connection with Avaya Meetings Server is established:
   - **FQDN**
   - **SIP Port**
   - **SIP Transport**
3. Click **Save**.
4. To complete the portal settings, navigate to **Avaya Meetings Server › Unified Portal Settings**.
5. In the Unified Portal Settings area, configure the settings as described in See "Unified Portal settings field descriptions" on page 65.
6. Click **Save**.

AFTER COMPLETING THE TASK
Complete the settings on the Avaya Meetings Server Management portal as described in the Avaya Meetings Server configuration section in *Deploying the Avaya Aura® Web Gateway*.

Configuring WebRTC media adaptation

WebRTC media adaptation enables the routing of WebRTC calls to the Avaya Meetings Server server through either Avaya Aura® Media Server or Avaya Meetings Media Server. Until solution 9.1.3, if Full Transcoded Video mode was used, you could disable media adaptation to route WebRTC calls directly to the Avaya Meetings Server. However, now, you should always enable media adaption by the AAMS (TE solution) or AEWG (OTT solution).

Starting from Avaya Aura® Web Gateway Release 3.7, WebRTC media adaptation is configured using Avaya Meetings Management. You must also ensure that WebRTC media adaptation is disabled on the Avaya Aura® Web Gateway administration portal.

1. On Avaya Meetings Management, configure adaptation settings as described in "Configuring Avaya Meetings Media Server for WebRTC-based calls in Team Engagement deployments" in *Administering Avaya Meetings Media Server*.
2. On the Avaya Aura® Web Gateway administration portal, navigate to **Avaya Meetings Server › Avaya Meetings Management**.
3. Clear the **Force Media Server usage for WebRTC calls** check box.

Unified Portal settings field descriptions

The following table describes the options available in **Avaya Meetings Server › Unified Portal Settings**:

Name	Description
Conference Clients	Specifies the set of clients that the Unified Portal can use to join conferences. Only the **Avaya Workplace Client and Web Client** option is supported. You cannot edit the **Conference Clients** field value.
Avaya Workplace Client Aura Video Ignore	Specifies whether the Unified Portal can ignore the Avaya Workplace Client videoCapable parameter.
Allow Recording Guest Access	Specifies whether a guest user can access recordings from the Unified Portal.
Allow Portal Guest Access	Specifies whether a guest user can access any functionality on the Unified Portal.
Allow AAWG Guest Access	Specifies whether a guest user can join meetings using the Avaya Workplace Client for Web (MeetMe) that the Unified Portal launches.
Web MeetMe WebRTC Browser Type	Specifies which web browser types can access the Unified Portal and use the following WebRTC conferencing capabilities: • Audio • Video • Presentation. For example, screen sharing. If this field does not contain a browser name and **Web MeetMe Data Only Browser Type** contains this browser name, you can access the Unified Portal and use Presentation Only conferencing capabilities. The administrator can configure forbidden and allowed browsers based on their types, engines, and versions for Avaya Workplace Client integration on the Avaya Meetings administration portal. Those settings override the browser type and version settings configured in Avaya Aura® Web Gateway for Data only and WebRTC parameters.
Web MeetMe Data Only Browser Version Exclusion	Specifies which web browsers types and versions cannot access the Unified Portal and use

Name	Description
	Presentation Only conferencing capabilities, such as screen sharing. To provide a browser version and type, use the following format: `<browser type>:<browser version>`. Use semicolons to separate entries. Use commas to separate different versions of the same browser. For example, `Chrome:64.0,65,66,67;Firefox:60.0,61.0;Safari:9` The administrator can configure forbidden and allowed browsers based on their types, engines, and versions for Avaya Workplace Client integration on the Avaya Meetings administration portal. Those settings override the browser type and version settings configured in Avaya Aura® Web Gateway for Data only and WebRTC parameters.
Web MeetMe Data Only Browser Type	Specifies which web browser types can access the Unified Portal and use Presentation Only conferencing capabilities, such as screen sharing. Note: WebRTC features, such as audio and video, are not supported in Presentation Only mode. The administrator can configure forbidden and allowed browsers based on their types, engines, and versions for Avaya Workplace Client integration on the Avaya Meetings administration portal. Those settings override the browser type and version settings configured in Avaya Aura® Web Gateway for Data only and WebRTC parameters.
Avaya Workplace Client Aura Domain	Specifies the SIP domain of enterprise users that is required for Avaya Workplace Client to join a meeting.
Web MeetMe WebRTC Browser Version Exclusion	Specifies which web browser types and versions cannot access the Unified Portal and use WebRTC conferencing capabilities. To provide a browser version and type, use the following format: `<browser type>:<browser version>`. Use semicolons to separate entries. Use commas to separate different versions of the same browser. For example: `Chrome:64.0,65,66,67;Firefox:60.0,61.0;Safari:9` The administrator can configure forbidden and allowed browsers based on their types, engines, and versions for Avaya Workplace Client integration on the Avaya Meetings administration portal. Those settings override the browser type

Name	Description
	and version settings configured in Avaya Aura® Web Gateway for Data only and WebRTC parameters.
Web MeetMe WebRTC Browser Min Version	Specifies the minimum browser version to access the Unified Portal and use WebRTC capabilities, such as audio, video, and presentation. To provide a minimum browser version, use the following format: <browser type>:<browser version>. Use semicolons to separate entries. For example, Chrome:55.0;Firefox:52.0;Edge:17. If **Web MeetMe WebRTC Browser Type** does not contain a browser name but this field contains a minimum supported version for that browser, you can use WebRTC conferencing capabilities on browser versions that satisfy the minimum version requirement. The administrator can configure forbidden and allowed browsers based on their types, engines, and versions for Avaya Workplace Client integration on the Avaya Meetings administration portal. Those settings override the browser type and version settings configured in Avaya Aura® Web Gateway for Data only and WebRTC parameters.
Web MeetMe Data Only Browser Min Version	Specifies the minimum browser version to access the Unified Portal and use Presentation Only capabilities, such as screen sharing. To provide a minimum browser version, use the following format: <browser type>:<browser version>. Use semicolons to separate entries. For example, Chrome:53.0;Firefox:52.0;IE:11;Safari:9.3.1;Edge If **Web MeetMe Data Only Browser Type** does not contain a browser name, but this field contains a minimum supported version for that browser, you can use Presentation Only conferencing capabilities on browser versions that satisfy the minimum version requirement. The administrator can configure forbidden and allowed browsers based on their types, engines, and versions for Avaya Workplace Client integration on the Avaya Meetings administration portal. Those settings override the browser type and version settings configured in Avaya Aura® Web Gateway for Data only and WebRTC parameters.

Name	Description
Avaya Workplace Client Aura Min Version	Specifies the minimum supported Avaya Workplace Client version. The default value is 3.1 .
Avaya Workplace Client MeetMe Min Version	Specifies the minimum supported Avaya Workplace Client version to join a meeting in Presentation Only mode. The default value is 3.2 .

Supported browsers and operating systems for the Avaya Meetings for Web client

Supported browsers

The Avaya Meetings for Web client supports the following browsers:

Browser	Full Video, Audio, and Data	Data Only
Google Chrome	Yes	Yes
Microsoft Edge based on Chromium	Yes	Yes
Mozilla Firefox	Yes	Yes
Apple Safari	No	Yes

The Avaya Meetings for Web client supports the latest four browser versions. For example, if the current Google Chrome version is 99, the Avaya Meetings for Web client supports Google Chrome version 96 and later.

> Note:
> The Avaya Meetings for Web client Releases older than 9.1.13 support Internet Explorer 11 for Data Only. Starting from Release 9.1.13, the Avaya Meetings for Web client does not support Internet Explorer 11.

Supported operating systems

The Avaya Meetings for Web client supports the following operating systems:

- Microsoft Windows (64-bit)
- MacOS

The Avaya Meetings for Web client does not support Android-based operating systems, such as Chrome OS.

Enabling the lockout policy for Unified Portal accounts

By default, the Unified Portal does not lock a user account if the user enters an invalid password multiple times. You can enable the lockout policy so that Unified Portal accounts become locked after multiple consecutive failed login attempts. When the lockout policy is enabled, a user has five attempts to enter a valid password by default. You can change this value as appropriate.

1. Log in to the Avaya Aura® Web Gateway CLI using an SSH connection.
2. Open the */opt/Avaya/CallSignallingAgent/<version>/nginx/1.18.0-1/conf* file in a text editor.
   ADDITIONAL INFORMATION:
   For example: `vi /opt/Avaya/CallSignallingAgent/<version>/nginx/1.18.0-1/conf`
3. In all `#limit_req zone=ups_login burst=5 nodelay;` strings in the file, remove the leading # character.
4. To change the default number of login attempts, in all `limit_req zone=ups_login burst=5 nodelay;` strings in the file, replace 5 with the appropriate number of login attempts.
   ADDITIONAL INFORMATION:
   For example, to allow three login attempts, update the strings as follows:

   ```
   limit_req zone=ups_login burst=3 nodelay;
   ```
5. Save the file.

Configuring WebRTC and presentation capabilities for the Microsoft Edge browser

To access WebRTC and presentation capabilities from the Microsoft Edge browser, you must configure the appropriate browser version on the Unified Portal Settings page. You must also enable Avaya Aura® Media Server adaptation to route WebRTC calls through Avaya Aura® Media Server.

The administrator can configure forbidden and allowed browsers based on their types, engines, and versions for Avaya Workplace Client integration on the Avaya Meetings administration portal. Those settings override the browser type and version settings configured in Avaya Aura® Web Gateway for Data only and WebRTC parameters.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Avaya Meetings Server › Avaya Meetings Management**.
2. Select the **Force Media Server usage for WebRTC calls** check box.
3. Click **Save**.
4. Navigate to **Avaya Meetings Server › Unified Portal Settings**.
5. In **Web MeetMe Data Only Browser Min Version** and **Web MeetMe Data Only Browser Min Version** fields, add the `Edge:17` entry.
   ADDITIONAL INFORMATION:
   If these fields already contain another version of Microsoft Edge, change the version to `17`.

   If you have multiple browsers, use a semicolon to separate each entry. For example, `Chrome:53.0;Firefox:52.0;Edge:17`

6. Click **Save**.

Configuring WebRTC and presentation capabilities for other browsers

To access WebRTC and presentation capabilities from your browser, you must configure the appropriate browser version on the Unified Portal Settings page. Use this procedure to update the default WebRTC and presentation configuration settings for other browsers, such as Internet Explorer, Firefox, or Chrome.

> Note:
> You cannot use Safari for audio or video calls. Safari only supports presentation capabilities, such as screen sharing.

The administrator can configure forbidden and allowed browsers based on their types, engines, and versions for Avaya Workplace Client integration on the Avaya Meetings administration portal. Those settings override the browser type and version settings configured in Avaya Aura® Web Gateway for Data only and WebRTC parameters.

This procedure does not describe the configuration for the Microsoft Edge browser. For more information about Microsoft Edge configuration, see See "Configuring WebRTC and presentation capabilities for the Microsoft Edge browser" on page 69.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Avaya Meetings Server › Unified Portal Settings**.
2. In the **Web MeetMe WebRTC Browser Min Version** field, enter the browser type and the minimum browser version for WebRTC capabilities in the `<`*`browser type`*`>:<`*`browser version`*`>` format.
   ADDITIONAL INFORMATION:
   If you have multiple browsers, use a semicolon to separate each entry. For example, `Chrome:53.0;Firefox:52.0;Edge:17`
3. In the **Web MeetMe Data Only Browser Min Version** field, enter the browser type and the minimum browser version for presentation-only capabilities in the `<`*`browser type`*`>:<`*`browser version`*`>` format.
   ADDITIONAL INFORMATION:
   If you have multiple browsers, use a semicolon to separate each entry. For example, `Chrome:53.0;Firefox:52.0;Safari:11;Edge:17`
4. Click **Save**.

External access configuration

This section describes the configuration required to access the Avaya Aura® Web Gateway from outside the enterprise network through Avaya Session Border Controller for Enterprise (Avaya SBCE), which is located at the edge of the enterprise network.

> Important:
> Ensure that the front-end port for remote access has been enabled and set as required.
>
> To configure the front-end port settings, see the "Front-end host, System Manager, and certificate configuration" section *Deploying the Avaya Aura® Web Gateway*.

Managing HTTP reverse proxy settings

Use this procedure if the Avaya Aura® Web Gateway is fronted by an HTTP reverse proxy that modifies any aspect of the Avaya Aura® Web Gateway service access URLs, such as the FQDN, HTTP protocol, port, or base path.

> Important:
> Only modify these settings if the reverse proxy serves all requests to the Avaya Aura® Web Gateway. This includes requests from clients both internal and external to the enterprise.
>
> If the reverse proxy serves only external clients, ensure that the reverse proxy's configuration accepts requests on port 443 and proxies requests to the Avaya Aura® Web Gateway on the remote access port.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **External Access › HTTP Reverse Proxy**.
2. If required, in **Front-end Port**, update the port value.
   ADDITIONAL INFORMATION:
   The default port is
   443
   .
3. In **Front-end Protocol**, select **http** or **https**.
4. If you want to use a reverse proxy FQDN instead of the front-end FQDN set up on the Avaya Aura® Web Gateway, select the **Use Front-end Host from a client request** check box.
   ADDITIONAL INFORMATION:
   If you select this check box, the certificates enabled for reverse proxy must include your FQDN in the SAN. The DNS entry for your FQDN must also point to the appropriate IP address.

   You can also select this check box if you require multiple FQDNs for the Avaya Aura® Web Gateway portal service. In this case, the reverse proxy must be configured to forward all of the IP addresses or FQDNs to the Avaya Aura® Web Gateway portal server virtual IP address. The certificate installed on the reverse proxy must include all the FQDNs that are used in the SAN. For more information about required FQDNs and certificates, see *Deploying the Avaya Aura® Web Gateway*.
5. If you have any external clients, including WebRTC, mobile, or desktop clients outside the enterprise firewall, do the following to enable a remote port:
   a. Select the **Enable port for external access** check box.
   b. In **Front-end port for external access**, enter a value between
      1024
      and
      65535
      .
      ADDITIONAL INFORMATION:
      The default port value is
      8444
      .

   ADDITIONAL INFORMATION:
   This port setting must be configured in the reverse proxy so traffic for clients outside the firewall is translated from the front-end port to this remote access port.
6. To use an external load balancer, select the **Enable use of an external load balancer** check box.
   ADDITIONAL INFORMATION:

An external load balancer is mandatory for geographically distributed systems.

7. If the FQDN of the external load balancer differs from the Avaya Aura® Web Gateway front-end FQDN, provide the FQDN of the external load balancer in the **Front-end Host** field.
   ADDITIONAL INFORMATION:

   > Important:
   > If you use an external load balancer, you must regenerate Avaya Aura® Web Gateway identity certificates so that the certificate SAN includes the external load balancer FQDN. For more information about managing identity certificates, see See "Managing identity certificates" on page 82.

8. Click **Save**.

Managing STUN server settings

Use this procedure to add and prioritize STUN servers for an Avaya Aura® Web Gateway location. You must also ensure that the STUN or TURN server details have been configured on the co-located Avaya Aura® Media Server systems.

If you use client side TURN, you must add the IP address of Avaya SBCE as a STUN server. Avaya recommends using client side TURN whenever possible.

To add a new STUN server, do the following:

1. Navigate to **External Access › STUN Servers**.
2. Click **Add**.
3. In **Address**, type the FQDN or IP address of the Avaya SBCE external B1 interface configured for STUN/TURN.
   ADDITIONAL INFORMATION:

   > Important:
   > Ensure that this IP address or FQDN is accessible to the clients. These clients might use computers that are external to the enterprise and might also require a STUN server to make a call or join a meeting.

4. In **Port**, type 3478.
5. Click **Save**.

To set the STUN priority, do the following:

6. From **Web Gateway Location**, select an Avaya Aura® Web Gateway location.
7. To add a STUN server to the Assigned STUN Servers list, select the server from the Input STUN Servers list and then click **Add**.
   ADDITIONAL INFORMATION:
   You can click **Add All** to add all listed servers to the Assigned STUN Servers list.
8. To remove a STUN server from the Assigned STUN Servers list, select the server and then click **Remove**.
9. In the Assigned STUN Servers area, drag and drop the servers to rearrange them.
   ADDITIONAL INFORMATION:
   The order you set in the Assigned STUN Servers list determines how the STUN servers will serve the selected Web Gateway location.
10. Click **Save**.
11. Repeat the steps STUN priority steps above for each location listed in **Web Gateway Location**.

Managing Avaya Session Border Controller for Enterprise connection settings

Use this procedure to add an Avaya SBCE to the Avaya Aura® Web Gateway web administration portal. Avaya SBCE is needed for external desktop clients. Avaya SBCE Release 7.2 is required for HTTP tunneling support.
PREREQUISITE
Perform the configuration on the Avaya SBCE administration portal as described in the "Avaya Session Border Controller for Enterprise configuration" section in *Deploying the Avaya Aura® Web Gateway*.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **External Access › Session Border Controller**.
2. To enable TURN usage, select the **Enable TURN in WebRTC Client** check box.
   ADDITIONAL INFORMATION:
   Avaya recommends enabling TURN in the browser client for performance reasons. For information about TURN in a WebRTC client, see See "TURN usage in a WebRTC client" on page 73.
3. Click **Add** to add a new Avaya SBCE or click **Edit** to update the information for an existing Avaya SBCE connection.
4. Complete the following settings for the connection:
   a. In **SIP Address**, type the address of the internal interface specified for signaling on Avaya SBCE.
   b. In **SIP Port**, type `5060` if you are using TCP or `5061` if you are using TLS.
   c. In **SIP Protocol**, select **TCP** or **TLS**.
   d. In **HTTP Address**, type the internal address specified for the load monitoring entry.
   e. In **HTTP Port**, type `80` if you are using HTTP protocol or `443` if you are using HTTPS protocol.
   f. In **HTTP Protocol**, select **http** or **https**.
   g. In **Location**, specify the location of the Avaya SBCE server.
      ADDITIONAL INFORMATION:
      You only need to specify the location for one node, and it is propagated to all other nodes in the cluster.
5. Click **Save**.
   STEP RESULT:
   After you add a new Avaya SBCE connection, the details for that connection are displayed on the screen. The connectivity status for the Avaya SBCE is also displayed.

TURN usage in a WebRTC client

Avaya's WebRTC solution is based on a TURN client residing on the Avaya Aura® Media Server or Scopia Elite MCU inside the enterprise. If the WebRTC browser is behind a restricted firewall that blocks UDP traffic, WebRTC media connections to Avaya Meetings Server or another WebRTC browser will not work. The solution for this issue is to use the WebRTC TURN client in the browser and send media to Avaya SBCE through UDP port 3478 or TCP/TLS port 443. The UDP port is preferable for Quality of Service (QoS) reasons. If you use TURN on TLS, media will be encoded back to UDP by the TURN server on Avaya SBCE.

> Note:

For performance reasons, Avaya recommends enabling TURN in the browser client instead of using server-side TURN.

If you use TURN from browser, disable TURN on the Avaya Aura® Media Server and Scopia Elite MCU. If you need to place a firewall between Avaya SBCE and the Avaya Aura® Media Server or Scopia Elite MCU, enable STUN on the Avaya Aura® Media Server or Scopia Elite MCU.

Avaya SBCE must be configured to support the combination of the following ports:

- STUN UDP 3478
- TURN UDP 3478
- TURN TCP/TLS 443

For more information about Avaya SBCE configuration, see *Administering Avaya Session Border Controller for Enterprise*.

Currently, the TURN feature is supported in the following browsers:

- Google Chrome
- Mozilla Firefox

Avaya SBCE connectivity status indicators

After you add an Avaya SBCE on the Avaya Aura® Web Gateway web administration portal, the connectivity status for that Avaya SBCE is also displayed. The following table describes the status indicators:

Status indicator	Status name	Description
	Active	Indicates that the server is available and can be used for calls. When you hover over this indicator, additional details, such as audio and video session, and Avaya SBCE bandwidth, are displayed.
	Overloaded	Indicates that the server is overloaded and calls cannot be made.
	Inactive	Indicates that the load monitoring cannot be retrieved for the server.
	Unknown	Indicates that the server status has not been fetched.

Reviewing the Avaya SBCE status for a specific node

Use this procedure to review the connectivity status and availability of Avaya SBCE servers for a specific node. In geographically distributed systems, Avaya SBCE from one location might be inaccessible to Avaya Aura® Web Gateway from another location, so the connectivity status might be different for different nodes.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **External Access › Session Border Controller**.

2. In a cluster environment, from the **Node address** drop-down list, select the required Avaya Aura® Web Gateway node.

Result
Avaya Aura® Web Gateway displays the connectivity status of Avaya SBCE servers for the node. For more information about the connectivity status, see See "Avaya SBCE connectivity status indicators" on page 74.

> Note:
> If the Avaya Aura® Web Gateway node is inactive, then all Avaya SBCE servers have an "Unknown" status.

Providing information about guest SIP domain and SIP proxy

You must provide information about the SIP server that Avaya Oceana® clients use for SIP communication so that the Avaya Aura® Web Gateway can send SIP requests to these clients.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **External Access › Guest SIP Proxy**.
2. To provide information about the guest SIP domain, do the following:
    a. In the **Default Guest SIP Domain** field, specify the domain configured for Avaya Oceana® clients on Session Manager.
    b. Click **Save**.
3. To provide information about the guest SIP proxy, do the following:
    a. In the Guest SIP Proxy section, click **Add**.
    b. In the new row of the table, provide the following information:
    ADDITIONAL INFORMATION:
        - **SIP Address**: The entity IP address of the Session Manager that is used by Avaya Oceana® clients.
        - **SIP Port**: The port used for SIP connection.
        - **SIP Protocol**: The protocol used for SIP connection.
        - **Location**: The appropriate location.
        - **Weight**: Enter `100`.

Log management

Changing the logging level

The logging level that you select determines which system events Avaya Aura® Web Gateway includes in log files. Avaya support personnel can use these log files for troubleshooting purposes. If the log files that you provide do not contain enough information, support personnel will ask you to select another logging level.

Unless you are told to change the logging level, use the following recommended logging levels:

- FINE

: For the
Client Application Service Logs
category.

- WARNING
: For all other log categories.

After selecting a new logging level, you must wait until the issue reoccurs before you can send updated logs to support personnel. When the issue is resolved, you must switch back to the recommended logging level.

> Important:
> Increasing logging details results in larger log files and might also decrease the system performance.

1. On the Avaya Aura® Web Gateway administration portal, click **Logs Management**.
2. In the Adjust Service Logging Level area, in **Logger**, select a log category.
3. In **Current logging level**, select the required logging level.
ADDITIONAL INFORMATION:
Unless support personnel ask you to change the logging level, use the recommended logging level for the log category.
4. Click **Save**.

Configuring log retention

Log files can contain private or sensitive information. To maximize data privacy and security, you can configure log retention and specify how long you want to keep logs. After this retention period, Avaya Aura® Web Gateway deletes log files within an hour.

The default log retention period depends on whether data encryption is enabled as follows:

- If data encryption is disabled, the default period is
1
day.
- If data encryption is enabled, the default period is
30
days.

Avaya recommends that you set the log retention period to at least
8
days, unless your organization's policies defines another retention period.

If a log file exceeds a size limit, Avaya Aura® Web Gateway deletes this log file even if the log retention period is not expired.

1. On the Avaya Aura® Web Gateway web administration portal, click **Logs Management**.
2. Select the **Use Log Retention** check box.
3. In **Retention period (days)**, set the period for which you want to keep log files on Avaya Aura® Web Gateway.
ADDITIONAL INFORMATION:
The range is
1
to
365

days.

4. Click **Save**.

Disabling log retention

When log retention is disabled, Avaya Aura® Web Gateway will only delete log files when they exceed a size limit.

1. On the Avaya Aura® Web Gateway web administration portal, click **Logs Management**.
2. Clear the **Use Log Retention** check box.
3. Click **Save**.

Downloading logs

Use this procedure to download Avaya Aura® Web Gateway logs to your computer. Support personnel can use the log files you collect for troubleshooting purposes.

You can only collect and download logs for one node at a time. You cannot download logs for an entire cluster at once.

1. On the Avaya Aura® Web Gateway administration portal, click **Logs Management**.
2. In the Collect Logs area, in **Number of rotated log files to collect (1-20)**, enter the number of logs you want to download.
   ADDITIONAL INFORMATION:
   This setting specifies the number of files from the log file history to include in the log collection. The range is
   1
   to
   20
   . If you leave this field empty, Avaya Aura® Web Gateway collects all available logs.
3. To collect logs for a node, click **Collect** in the corresponding row.
4. Wait until Avaya Aura® Web Gateway collects the logs and then click **OK**.
5. To download the logs you collected for a node, click **Download**.

Scheduling log collection

Certain log levels, such as FINEST, provide very detailed information about events on Avaya Aura® Web Gateway. However, enabling these log levels on a constant basis might negatively impact the Avaya Aura® Web Gateway performance. You can enable detailed logging temporarily for a specified time period. After the specified period expires, Avaya Aura® Web Gateway switches back to the original log level, which you configure in the Adjust Service Logging Level area.

Scheduled log collection cancels automatically if you change the original log level for any log category.

After enabling scheduled log collection for a log category, you cannot set up scheduled log collections for other log categories until the specified time period expires or you cancel scheduled log collection.

1. On the Avaya Aura® Web Gateway web administration portal, go to **Logs Management**.

2. In the Log capture scheduling area, select a log category for which you want to enable scheduled log collection.
3. In the **Planned log level** field, select a log level that you want to enable on a temporary basis for the log category.
4. If required, repeat steps 2 and 3 for other log categories.
5. In the **Schedule Start Time** field, select one of the following:
   - **Immediate**: To start collecting logs at the planned log level immediately.
   - **Specify Time**: To specify the date and time when you want to start collecting logs at the planned log level.
6. In the **Duration** field, select one of the following:
   - **Select Duration**: To specify the period during which Avaya Aura® Web Gateway collects logs at the planned log level.
   - **No Duration (Indefinite)**: To set the log level specified in the **Planned log level** field as the new default log level for the log category.

     > Note:
     > When you select **No Duration (Indefinite)**, Avaya Aura® Web Gateway does not schedule log collection at the specified log level. Instead, Avaya Aura® Web Gateway sets up the new default log level for the log category starting from the time specified in the **Schedule Start Time** field.
7. Click **Schedule**.
8. In the confirmation window, click **Save**.

Result
Avaya Aura® Web Gateway saves the configuration and displays the schedule status on the page.

Canceling scheduled log collection

You can cancel scheduled log collection at any time. If you cancel scheduled log collection when it is in progress, Avaya Aura® Web Gateway switches back to the original log level immediately.

Scheduled log collection cancels automatically if you change the original log level for any log category in the Adjust Service Logging Level area.

1. On the Avaya Aura® Web Gateway web administration portal, go to **Logs Management**.
2. In the Log capture scheduling section, click **Cancel Schedule**.
3. In the confirmation window, click **Save** and then click **OK**.

Configuring licensing

Use this procedure to verify the licenses that the Avaya Aura® Web Gateway obtains from System Manager.

1. On the Avaya Aura® Web Gateway administration portal, click **Licensing**.
   ADDITIONAL INFORMATION:
   The feature licensing values are automatically populated.
2. Ensure that the information is correct and that none of the licenses have expired.
   ADDITIONAL INFORMATION:

The values are obtained from System Manager. You can update these values on the Avaya Aura® Web Gateway administration portal by navigating to **General Network Settings › System Manager** if they are incorrect.

For more information on WebLM license configuration on System Manager, see *Administering Avaya Aura® System Manager*.

Specifying the WebLM server for the Avaya Aura® Web Gateway

Use this procedure to select a WebLM server for the Avaya Aura® Web Gateway.

By default, the Avaya Aura® Web Gateway uses the WebLM server that is hosted on System Manager. If you use the System Manager URL for the WebLM server, then the WebLM URL is automatically changed when you:

- Modify the System Manager FQDN using the configuration utility.
- Generate certificates from the Generate Identity Certificates via System Manager area of the web administration portal. For more information, see See "Generating System Manager identity certificates" on page 81.

If you use a specific WebLM URL, this URL will not be changed when the System Manager URL is changed or new identity certificates are generated.

> Note:
> Users with the Auditor role cannot edit WebLM settings.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Licensing**.
2. Do one of the following:
    - To use the WebLM server hosted on System Manager, select the **Use SMGR url for license server** check box.
    - To use a specific WebLM server, clear the **Use SMGR url for license server** check box, and then provide the FQDN of the WebLM server and the port for connecting to the WebLM server.
3. Click **Save**.

Configuring new call rejection when the audio or video port limit is exceeded

The Avaya Aura® Web Gateway license determines how many simultaneous audio or video calls Avaya Aura® Web Gateway supports. By default, if this limit is exceeded, Avaya Aura® Web Gateway raises an appropriate alarm but still allows establishing new calls. You can configure Avaya Aura® Web Gateway to reject new audio or video calls if the respective port limit is exceeded. For this purpose, Avaya Aura® Web Gateway uses the rejectUnlicensedCalls property in the *mss.start.sh* file.

When the call rejection functionality is enabled, Avaya Aura® Web Gateway displays the NOTE: Calls will be rejected if maximum ports usage is exceeded notification on the Licensing page.
PREREQUISITE
Enable EASG functionality. EASG is required to obtain the sroot privileges required to run commands in this procedure.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. To obtain sroot privileges, run the following command:
   ADDITIONAL INFORMATION:
   ```
   su - sroot
   ```
3. To navigate to the location of the *mss.start.sh* settings file, run the following commands:
   ADDITIONAL INFORMATION:
   ```
   cdto cas
   cd service.d
   ```
4. To make a backup of the *mss.start.sh* file, run the following command:
   ADDITIONAL INFORMATION:
   ```
   cp mss.start.sh <BACKUP_FILENAME>
   ```
   where `<BACKUP_FILENAME>` is a name for the backup file. You can use this backup to restore the original *mss.start.sh* file if you make any typos when editing it.
5. To open the *mss.start.sh* file in the vi text editor, run the following command:
   ADDITIONAL INFORMATION:
   ```
   vi mss.start.sh
   ```
6. Navigate to the section where CATALINA_OPTS parameters are configured.
   ADDITIONAL INFORMATION:
   These entries use the `CATALINA_OPTS="CATALINA_OPTS -D<VARIABLE>=<VALUE>"` format.
7. Configure the Avaya Aura® Web Gateway behavior when the limit for audio or video calls is exceeded as follows:
   - To reject new calls, add the following entry at the end of the section with CATALINA_OPTS variables:

     `CATALINA_OPTS="CATALINA_OPTS -DrejectUnlicensedCalls=true"`
   - To allow new calls, add the following entry at the end of the section with CATALINA_OPTS variables:

     `CATALINA_OPTS="CATALINA_OPTS -DrejectUnlicensedCalls=false"`

   ADDITIONAL INFORMATION:
   > Important:
   > If the call rejection functionality is enabled and you want to disable it, you must change the rejectUnlicensedCalls parameter value to `false`. If you remove the `CATALINA_OPTS="CATALINA_OPTS -DrejectUnlicensedCalls=true"` entry from the *mss.start.sh* file, Avaya Aura® Web Gateway continues rejecting new calls if the port limit is exceeded.
8. Save the *mss.start.sh* file.
9. To restart services and apply the changes, run the following command:
   ADDITIONAL INFORMATION:
   ```
   svc telportal restart
   ```

Security settings

Managing certificates in the Avaya Aura® Web Gateway web administration portal

You can use the Avaya Aura® Web Gateway administration portal to review and manage certificates. The management options in the administration portal do not replace the setup that you need to complete during installation. After installation is completed, use the web administration portal for management when possible. Only use the configuration utility if the administration portal is not available or for troubleshooting purposes.
PREREQUISITE

- You must have the Security Administrator role to access certificate management options.
- In a cluster environment, ensure that all nodes in the cluster are running.

1. On the web administration portal, navigate to **Security Settings › Certificate Management**.
2. Click the appropriate tab.
   ADDITIONAL INFORMATION:
   The procedures below describe the tasks you can perform on each tab.

Generating System Manager identity certificates

Use this procedure to generate System Manager security identity certificates.

Avaya Aura® Web Gateway does not accept a generated System Manager certificate if:

- The certificates contains an unsupported critical extension.
- The certificate expired.
- The start date of the certificate is in the future.

Avaya Aura® Web Gateway displays a warning message if the generated certificate cannot be imported. To fix the issue, contact the System Manager administrator.

1. Click the **SMGR Certificates** tab.
2. In the Generate Identity Certificates via System Manager area, update the following fields if they are not automatically populated:
   ADDITIONAL INFORMATION:
   - **System Manager Address**: To update this setting, navigate to **General Network Settings › System Manager**.
   - **System Manager HTTPS Port**: To update this setting, navigate to **General Network Settings › System Manager**.
   - **Common Name**: To update this setting, navigate to **External Access › HTTP Reverse Proxy**.
3. In the **Node Address** drop-down menu, do one of the following to generate a certificate:
   - Choose a node for a cluster configuration.

     If you choose the **All Cluster Nodes** option, certificates will be generated automatically for all cluster nodes.

- Keep the default setting for a standalone configuration.

4. In **System Manager Enrollment Password**, type the enrollment password as defined on the System Manager web console.
5. Click **Generate Certificates** to start requesting certificates from System Manager.
6. Restart the Avaya Aura® Web Gateway after the certificates are generated.
   STEP RESULT:
   This completes all the certificate updates. For a cluster environment, only the remote node is restarted.

Managing identity certificates

1. Click the **Identity Certificates** tab.
2. Use the following subsections to manage CSRs, keystore data, and server identity certificates.
3. After performing the required task, when prompted, restart the Avaya Aura® Web Gateway server for the changes to take effect.

Managing CSRs

1. In the Certificate Signing Requests area, do one of the following:
   - To set up a new CSR, click **Create** and then follow the steps in See "Creating CSRs" on page 82.
   - To remove an existing CSR, select it and then click **Delete**.
   - To process a signed CSR, click **Process Signing Request**. For more information, see See "Processing CA signing requests" on page 84.

Creating CSRs

This procedure provides a high-level overview of how to create CSRs.

1. If you clicked **Create** in the Certificate Signing Requests area, complete the following settings in the Create Certificate Signing dialog box and then click **Apply**.
   a. In **Alias**, type an alias using alphanumeric characters.
      ADDITIONAL INFORMATION:
      An example of an alias is `ottawacrt123`.
   b. Complete the other settings as required.
      ADDITIONAL INFORMATION:
      You can use the **Show Advanced Settings** button to view additional settings information. For more information about settings, see See "Create Certificate Signing Request screen field descriptions" on page 83.

      > Note:
      > For Open LDAP, the **Subject Alternative Name** section must contain the `localhost.localdomain` and, for IPv6, `localhost6.localdomain` records.
2. Ensure that the CSR file is successfully saved on your computer.
   ADDITIONAL INFORMATION:

The generated CSR is also added to the Certificate Signing Requests area.
STEP RESULT:
In a cluster configuration, the CSR list on all nodes is identical.

3. In a cluster environment, if the CSR is not available on a node, click **Propagate** to synchronize requests from the current node to the cluster.

AFTER COMPLETING THE TASK
Provide the CSR file to the CA for signing and apply the signed CSR as described in See "Processing CA signing requests" on page 84.
Create Certificate Signing Request screen field descriptions

Name	Description
Alias	The name of the certificate
Common Name	The FQDN of the node. For example, `amm.example.com` You cannot provide wildcard (*) characters in this field.
Use Existing FQDNs for SAN	Use this option to select a particular node in a cluster for which a certificate should be generated. You can also select all cluster nodes.
Specify SAN manually	Use this option to manually specify additional FQDNs or IP addresses for which the certificate should be generated.
Subject Alternative Name	An optional text field that can be used to further identify this certificate. You can provide multiple entries in this field. You cannot provide wildcard (*) characters in this field.
Key Bit Length	The certificate key length in bits.
Signature Algorithm	The hash algorithms to be used with the RSA signature algorithm.
Organizational Unit	The group within the company or organization creating the certificate.
Organization	The name of the company or organization creating the certificate.
City/Locality Name	The city where the certificate is being created.
State/Province Name	The state/province where the certificate is being created.
Country (ISO 3166)	The name of the country within which the certificate is being created in the ISO 3166 format. For more information about the format, see ISO 3166 country codes.

Processing CA signing requests

Use this procedure to process CA signing requests and upload signed certificates to Avaya Aura® Web Gateway.

If you want to use an identity certificate signed by a third-party CA, you must import an identity certificate chain in either PEM or PKCS12 format. This chain must contain the identity certificate signed by the third-party CA, all intermediate CA certificates, and the root CA certificate.

1. Use the appropriate CA documentation to sign the signing request with the CA.
2. In the Certificate Signing Request area, select the appropriate signing request and then click **Process Signing Requests**.
3. In the Process Signing Request dialog box, click **Choose file** to add the signed certificate and then click **Apply**.

Result
The signed certificate is removed from the Certificate Signing Requests area and added to the Keystore area.

Managing keystore data

1. In the Keystore area, do one of the following:
    - To import keystore data, click **Import** and then import keystore data.

      For more information, see See "Importing keystore data" on page 84.
    - To export a certificate in the PKCS12 format, select a keystore file and then click **Export**.

      In the Export Certificate dialog box, you can enter a password to protect your exported file.
    - To view keystore file details, click **Details**.
    - To remove an existing keystore file, select it and then click **Delete**.

Importing keystore data

Use this procedure to import third-party identity certificates to the Avaya Aura® Web Gateway Keystore.

You cannot import a certificate if:

- The certificate contains an unsupported critical extension.
- The certificate expired.
- The start date of the certificate is in the future.

Avaya Aura® Web Gateway displays a warning message if a certificate cannot be imported.

If you clicked **Import**, complete the following settings in the Import Certificate dialog box and then click **Apply**.

1. In the **Certificate Type** drop-down menu, select a format for importing the certificate.

2. From **Certificate File**, click **Choose file** to add the certificate file in the selected format.
3. If you selected the PEM format, in **Key File**, click **Choose file** to add the key file in the PEM format.
4. If you selected the PKCS12 format, in **Password**, type the password for the imported certificate.
5. In **Alias**, type an alias to be used for the imported certificate.

Managing server interface certificates

Use this procedure to manage Keystore identity certificates.

1. Navigate to the Server Interfaces area.
2. In a cluster environment, select the node to administer from the **Node Address** list.
3. Do one of the following:
    - To assign the certificate to a specific server interface, click **Assign** and then complete the settings in the Assign Certificates dialog box as described in See "Certificate assignment descriptions" on page 85.

      You cannot assign a certificate to an interface if:

        - The certificate contains an unsupported critical extension.
        - The certificate expired.
        - The start date of the certificate is in the future.

      Avaya Aura® Web Gateway displays a warning message if a certificate cannot be assigned to an interface.

      If you assign a certificate only to a selected node, then you must restart the services on that node to apply the changes. If you assign a certificate to the entire cluster, you must restart all nodes in the cluster.
    - To view details about the certificate, click **Details**.
    - To export the certificate in the PKCS12 format, click **Export**.

Certificate assignment descriptions

Name	Description
Application	Specifies the interface for REST API to the clients.
Internal	Specifies the interface being used for server-to-server component communication.
OAM	Specifies the Operations, Administration, and Maintenance (OAM) interface.

Installing third-party CA certificates

Use this procedure if you want to use certificates signed by a third-party CA instead of certificates signed by System Manager.
PREREQUISITE
Obtain a third-party CA-signed identity certificate, all intermediate CA certificates, and the root CA certificate. Do one of the following:

- Create a certificate signing request (CSR). For more information, see See "Managing CSRs" on page 82.

  > Note:
  > If the CA provides certificates as separate PEM files, you must manually generate an identity certificate chain. For more information, see See "Generating an identity certificate chain in the PEM format" on page 86.

- Obtain a certificate bundle in the PKCS12 format from the CA. The PKCS12 bundle must include a private key, the identity certificate, all intermediate CA certificates, and the root CA certificate. To view the bundle content and ensure that it includes a full certificate chain, run the `openssl pkcs12 -info -in <pkcs12_certificate_file_name>` command.

  If the CA provides the identity certificate as a separate *.p12* file, generate an identity certificate chain. For more information, see See "Generating an identity certificate chain in the PKCS12 format" on page 87.

1. Log in to the Avaya Aura® Web Gateway web administration portal.
2. Navigate to **Security Settings › Certificate Management › Identity Certificates**.
3. In the Keystore area, click **Import** and then select either the identity certificate chain in the PEM format or the PKCS12 certificate bundle.
4. In the Server Interfaces area, select the required server interface.
5. Click **Assign** and then select the certificate chain or PKCS12 bundle that you imported in step See "3" on page 86.
   STEP RESULT:
   Avaya Aura® Web Gateway restarts to apply the changes.
6. Navigate to **Security Settings › Certificate Management › Truststore**.
7. Click **Import** and then select the third-party root CA certificate and all intermediate CA certificates.

AFTER COMPLETING THE TASK
Install the third-party root CA certificate and all intermediate CA certificates on the clients.

Generating an identity certificate chain in the PEM format

If you want to use an identity certificate signed by a third-party certificate authority (CA), you must generate an identity certificate chain. An identity certificate chain must include the following certificates in order:

1. The third-party CA-signed identity certificate.
2. All intermediate CA certificates, if any.
3. The root CA certificate.

Assign this certificate chain to a specific Avaya Aura® Web Gateway server interface.

Use this procedure to generate an identity certificate chain in the PEM format.
PREREQUISITE
Ensure that you have the third-party CA-signed identity certificate, root CA certificate, and all intermediate CA certificates in the PEM format. If these files are in the PKCS12 format, you can convert these certificates to the PEM format using the OpenSSL tool.

1. Log in to the Avaya Aura® Web Gateway using your SSH credentials.
2. Navigate to the directory with the certificates.
3. Run the **cat** command as follows:
   ADDITIONAL INFORMATION:
   ```
   cat <Identity_ertificate_file_name>
   <Intermediate_CA_certificate_file_name> <Root_CA_file_name> >
   <Certificate_chain_file_name>
   ```
   If you have several intermediate CA certificates, list all intermediate CA certificate file names, separated by a space, and then list the root CA certificate file name.
   For example, if you have the identity certificate *identity.crt*, two intermediate CA certificates (*intermediateCA1.crt* and *intermediateCA2.crt*), and root CA certificate *rootCA.crt*, run the following command to generate a certificate chain with the *identityChain.crt* file name:
   ```
   cat identity.crt intermediateCA1.crt intermediateCA2.crt rootCA.crt
   > identityChain.crt
   ```

Generating an identity certificate chain in the PKCS12 format

If you want to use an identity certificate signed by a third-party certificate authority (CA), you must generate an identity certificate chain. An identity certificate chain must include the following certificates:

1. The third-party CA-signed identity certificate.
2. All intermediate CA certificates, if any.
3. The root CA certificate.

Assign this certificate chain to a specific Avaya Aura® Web Gateway server interface.

Use this procedure to generate an identity certificate chain in the PKCS12 format. You must do this if the CA does not provide a PKCS12 certificate chain that includes all required certificates.

> Tip:
> The commands in this procedure must be entered as a single line even if they appear as multiple lines in the document.

PREREQUISITE

- Ensure that the OpenSSL library is installed on Avaya Aura® Web Gateway.
- Ensure that you have the following certificate files:
    - The third-party CA-signed identity certificate in the PKCS12 format.

      Certificate files in the PKCS12 format usually have the *.p12* extension.

- ◦ A certificate chain in the PEM format that includes all intermediate and root CA certificates.

1. Log in to the Avaya Aura® Web Gateway using your SSH credentials.
2. Navigate to the directory with the certificates.
3. Run the following command to get a private key from the identity certificate file:
   ADDITIONAL INFORMATION:
   ```
   openssl pkcs12 -in <certificate_file_name> -out <private_key_file>
   -nocerts
   ```

   In this command:

   - <certificate_file_name> is the file name of the identity certificate in the PKCS12 format.
   - <private_key_file> is a file name of the private key.

   For example:
   ```
   openssl pkcs12 -in certificate.p12 -out privateKey.key
   -nocerts
   ```
4. Run the following command to get the identity certificate in the PEM format from the *.p12* certificate file:
   ADDITIONAL INFORMATION:
   ```
   openssl pkcs12 -in <certificate_file_name> -out
   <identity_certificate> -nokeys
   ```

   In this command:

   - <certificate_file_name> is the *.p12* certificate file name.
   - <identity_certificate> is a file name of the identity certificate in the PEM format.

   For example:
   ```
   openssl pkcs12 -in certificate.p12 -out certificate.pem
   -nokeys
   ```
5. Run the following command to create an identity certificate chain in the PKCS12 format:
   ADDITIONAL INFORMATION:
   ```
   openssl pkcs12 -export -out <certificate_chain> -inkey
   <private_key_file> -in <identity_certficiate> -certfile
   <CA_certificate_chain>
   ```

   In this command:

   - <certificate_chain> is the file name of the resulting identity certificate chain in the PKCS12 format.
   - <private_key_file> is the file name of the private key.
   - <identity_certificate> is the file name of the identity certificate in the PEM format.
   - <CA_certificate_chain> is the file name of a certificate chain in the PEM format that includes all intermediate and root CA certificates.

   For example:
   ```
   openssl pkcs12 –export -out certificateChain.p12 -inkey
   privateKey.key -in certificate.pem -certfile CACert.pem
   ```

Managing truststore certificates

Use this procedure to manage truststore certificates available on the Avaya Aura® Web Gateway.

1. Click the **Truststore** tab.
2. In the Truststore area, select a certificate and then click one of the following:
    - **Import**: To import a certificate in PEM or PKCS12 format.

      If the file imported into the truststore contains multiple certificates, then all CA certificates from the file will be imported. Alias names, other than the first certificate in the chain, are appended with the suffix _<X>, where <X> indicates a certificate number, such as 1, 2, or 3.

      If a new certificate is imported into the truststore or if any certificate is removed from the truststore, then you must restart the system to apply the change. In a cluster environment, all nodes in the cluster must be restarted.

      You cannot import a certificate, if the certificate contains unsupported critical extensions, if the certificate is expired, or if the start date of the certificate is in the future. If a certificate cannot be imported, Avaya Aura® Web Gateway displays a warning message.
    - **Details**: To view information about the certificate.
    - **Delete**: To delete the certificate from the truststore.

      > Important:
      > After you remove a certificate form the truststore, restart the Avaya Aura® Web Gateway to apply certificate updates. For a cluster configuration, restart the remote node.
    - **Export**: To export the certificate in the PEM format.

Importing the secure LDAP certificate using the web administration portal

For secure connectivity to LDAP servers, you must import an LDAP certificate file to the Tomcat trust store. The following procedure describes how to import the LDAP certificate using the Avaya Aura® Web Gateway web administration portal.

You cannot import an LDAP certificate if:

- The certificate contains an unsupported critical extension.
- The certificate expired.
- The start date of the certificate is in the future.

Avaya Aura® Web Gateway displays a warning message if a certificate cannot be imported.

1. On the Avaya Aura® Web Gateway web administration portal, navigate to **General Network Settings › LDAP Configuration**.
2. Select the **Secure LDAP** check box.
3. Click **Import Certificate**.

4. In the Import Certificate window, click **Choose File** and select the certificate from your computer.
5. Click **Save**.
STEP RESULT:
Avaya Aura® Web Gateway uploads the certificate to a secure LDAP Server. If a certificate is already uploaded, Avaya Aura® Web Gateway overwrites the existing certificate.

Applying third-party signed certificates to the Avaya Aura® Web Gateway

1. Open the Linux shell using your Linux administrator account credentials.
ADDITIONAL INFORMATION:
The Linux administrator account is created during the deployment process.
2. To open the Avaya Aura® Web Gateway configuration utility, run the following command:
ADDITIONAL INFORMATION:
```
app configure
```
3. Select **Front-end host, System Manager and Certificate Configuration**.
4. Select **Use System Manager for Certificates**, and then select **No**.
5. Select the following options, type the file path shown, and then select **OK** for each option:
    a. For **REST interface key file**, type `/opt/Avaya/AAWGportalCerts/frontEnd.key`.
    b. For **REST interface certificate file**, type `value /opt/Avaya/AAWGportalCerts/frontEndCerts.crt`.
    c. For **OAM interface key file**, type `value /opt/Avaya/AAWGportalCerts/frontEnd.key`.
    d. For **OAM interface certificate file**, type `value /opt/Avaya/AAWGportalCerts/frontEndCerts.crt`.
    e. For **Keystore password**, type the same keystore password that you used at the time of installation.
6. Select **Apply** and **Continue** for the specified certificates to be processed.
7. Select **Exit Configure** to exit this menu.

Configuring HTTP clients

Use this procedure to enable communication between the Avaya Aura® Web Gateway and the clients that use HTTP-based Avaya Aura® Web Gateway services over the REST and OAMP interface.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Security Settings › HTTP Clients**.
2. Set the following options:
    a. **REST** to **NONE**, which is the default value.
    STEP RESULT:
    This setting enables guest users to join conference calls.
    b. **OAMP** to **OPTIONAL**.

STEP RESULT:
This setting validates certificates presented by clients that are using Avaya Aura® Web Gateway services.

3. Click **Save**.

Configuring the certificate policy for Avaya Oceana® integration

Avaya Aura® Web Gateway validates the certificates presented by servers that generate token requests. When a third-party guest access server requests a token, Avaya Aura® Web Gateway verifies the client certificate and ensures that it is trusted.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Security Settings › HTTP Clients**.
2. In the Client-Device Certificate Policy section, set **REST** to **OPTIONAL**.
3. Click **Save**.

Available certificate validation options

Name	Description
OPTIONAL	Enables the Avaya Aura® Web Gateway to validate certificates presented by the clients to establish a secure HTTP connection with the Avaya Aura® Web Gateway.
NONE	Prevents the Avaya Aura® Web Gateway from performing any validation on certificates presented by the clients. If you select this option, the client cannot perform trusted hosts-based authentication with the Avaya Aura® Web Gateway.
OPTIONAL_NO_CA	Enables the Avaya Aura® Web Gateway to validate certificates presented by the clients. However, the Avaya Aura® Web Gateway does not require such certificates to be signed or issued by a trusted Certificate Authority (CA).
REQUIRED	Ensures that the clients present a valid certificate that is signed or issued by a trusted CA to establish a secure HTTP connection with the Avaya Aura® Web Gateway.

Adding a trusted host

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Security Settings › Trusted Hosts**.
2. Click **Add**.
3. In the new row, type the FQDN or IP address of the trusted host.
   ADDITIONAL INFORMATION:
   If you are configuring integration with Avaya Oceana®, then you must enter the FQDN of the server that requests authorization tokens.

When connecting to a cluster, add the IP address or FQDN of each node in the cluster and the virtual IP address of the cluster that you want Avaya Aura® Web Gateway to trust.

4. Click **Save**.

Configuring session security

Use this procedure to set SRTP security policies for SIP sessions.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Security Settings › Session Security**.
2. In **SRTP Policy**, select the security policy you want to apply.
   ADDITIONAL INFORMATION:
   The default option is `Best Effort`.
3. From the displayed list of encoding algorithms, select the appropriate algorithms for media encryption.
   ADDITIONAL INFORMATION:
   The selection of encoded algorithms is based on the media SRTP encoding algorithms that are supported by the back-end communications network and far-end endpoints.
4. Click **Save**.

Enabling Avaya Breeze® platform authorization

Use this procedure to enable Single Sign-On (SSO) capabilities for Avaya Aura® Web Gateway users that previously authenticated using the Avaya Breeze® platform Authorization application.

To enable authorization, you must import the Avaya Breeze® platform authorization certificate to Avaya Aura® Web Gateway. If the certificate is changed, then you must re-upload it to Avaya Aura® Web Gateway.

For more information about Authorization Service, see "Authorization Service" in *Administering Avaya Breeze® platform*.

PREREQUISITE

Obtain the Avaya Breeze® platform authorization certificate file in the *.PEM* format. For more information, see *Deploying Avaya Oceana®*.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Security Settings › Authorization**.
2. Click **Browse** and select the *.PEM* file that you exported from the Avaya Breeze® platform node.
3. Click **Save**.

OAuth configuration

OAuth is an authorization mechanism that enables you to authenticate using a combination of enterprise credentials and other factors that the enterprise has chosen, including enterprise Single Sign-On (SSO) and multi-factor authentication.

You can log in to Avaya Workplace Client using OAuth and have an SSO experience for the following features:

- Voice service.

- Services provided by Avaya Aura® Device Services, including automatic configuration, contacts, and enterprise search.
- Avaya Aura® Web Gateway for Apple Push Notifications on Avaya Workplace Client for iOS clients.
- Presence using SIP.
- Avaya Multimedia Messaging service on Avaya Aura® Presence Services.

When OAuth is used, Avaya Workplace Client accesses Avaya Aura® Web Gateway services using special OAuth access tokens. You must configure Avaya Aura® Web Gateway so that it:

- Provides Avaya Workplace Client with information where to obtain access tokens.
- Validates access tokens provided by Avaya Workplace Client.

Use the procedures listed in the sections below to set up OAuth on Avaya Aura® Web Gateway.

Before using OAuth on Avaya Aura® Web Gateway, you must configure OAuth on Avaya Aura® Device Services first. For general information about configuring OAuth on Avaya Aura® Device Services, see "OAuth configuration" in *Deploying Avaya Aura® Device Services*.

Creating client mapping

In the OAuth authorization flow, Avaya Aura® Web Gateway validates the access token received from a client to grant the client rights to use Avaya Aura® Web Gateway resources. To configure settings required for token validation, you must provide Avaya Aura® Web Gateway with the URL to discover authorization resources configured on Avaya Aura® Device Services.
PREREQUISITE
Configure OAuth on Avaya Aura® Device Services.

1. Log in to the Avaya Aura® Web Gateway web administration portal with the security administrator role.
2. Navigate to **Security Settings › OAuth 2.0 › Client ID Mapping**.
3. Click **Add**.
4. In the Create new client mapping window, complete the fields as follows:
    a. In **Client ID**, enter `Equinox`.
    ADDITIONAL INFORMATION:
    This value is case sensitive.
    b. In **OIDC Discovery URL**, enter `https://`*`<AADS front-end FQDN>`*`:`*`<AADS PORT>`*`/auth/realms/`*`<Realm>`*`/.well-known/openid-configuration`
    ADDITIONAL INFORMATION:
    In this string:
        - `<AADS front-end FQDN>` is the Avaya Aura® Device Services front-end FQDN.
        - `<AADS PORT>` is the Avaya Aura® Device Services front-end FQDN service port.
        - `<Realm>` is the Keycloak realm configured on Avaya Aura® Device Services, which is SolutionRealm by default.
5. Click **OK**.

Editing the existing client mapping

Use this procedure to review or edit identity provider mapping settings.

If you use the default OAuth configuration on Avaya Aura® Device Services, you do not need to edit the configured client mapping on Avaya Aura® Web Gateway. You only need to edit the mapping for customized configurations. An example is if access token stores the email address value in a custom attribute instead of the default `email` attribute.

1. Log in to the Avaya Aura® Web Gateway web administration portal with the security administrator role.
2. Navigate to **Security Settings › OAuth 2.0 › Client ID Mapping**.
3. Select the required client mapping and then click **Edit**.
4. Update the settings as required.
   ADDITIONAL INFORMATION:
   For more information about the settings, see See "Client mapping field descriptions" on page 94.
5. Click **OK**.

Client mapping field descriptions

The following table lists the client mapping fields.

Field	Description	Editable
Client ID	The Client ID that you provided when creating the client mapping.	✘
OIDC Discovery URL	The URL to discover the Keycloak resources you provided when creating the client mapping.	✘
Issuer	The issuer attribute that Avaya Aura® Web Gateway passes to clients in authorization tokens.	✘
Public Key	The public key that is used to sign access tokens.	✔
Access token email address attribute	The attribute in the access token that contains the email address of a user. The default value is `email`.	✔

Deleting the client mapping

After you create a client mapping, you cannot edit some mapping attributes. For example, you cannot update the discovery URL if the Avaya Aura® Device Services FQDN has changed or you want to use another Avaya Aura® Device Services system. If you need to update these attributes, you must delete the existing mapping first and then create a new mapping.

1. Log in to the Avaya Aura® Web Gateway web administration portal with the security administrator role.
2. Navigate to **Security Settings › OAuth 2.0 › Client ID Mapping**.
3. Select the required client mapping.
4. Click **Delete**.

Specifying the Avaya Aura® Device Services system used for issuing access tokens

When OAuth is configured, clients use access tokens to access Avaya Aura® Web Gateway features. Access tokens are issued by Avaya Aura® Device Services. Use this procedure to specify the Avaya Aura® Device Services system that issues access tokens for clients.
PREREQUISITE

- Ensure that OAuth is configured on Avaya Aura® Device Services. For more information, see "OAuth configuration" in *Deploying Avaya Aura® Device Services*.
- If you want to specify an Avaya Aura® Device Services URL manually, obtain the authorization endpoint URL configured on Avaya Aura® Device Services as described in See "Obtaining the authorization endpoint URL configured on Avaya Aura Device Services" on page 95.

1. Log in to the Avaya Aura® Web Gateway web administration portal with the security administrator role.
2. Navigate to **Security Settings › OAuth 2.0 › Client ID Mapping**.
3. Do one of the following:
   - Select the **Use AADS for OAuth Authorization** check box.

     If you select this check box, clients will obtain access tokens from the Avaya Aura® Device Services system configured in your deployment. This is the recommended option.
   - In **Authorization Endpoint URL**, enter the authorization endpoint URL configured on Avaya Aura® Device Services.

   > Note:
   >
   > - You only need to enter the authorization endpoint URL manually for customized configurations. If you use the default OAuth configuration on Avaya Aura® Device Services, use the **Use AADS for OAuth Authorization** check box.
   > - If the **Authorization Endpoint URL** check box is selected, you cannot modify **Authorization Endpoint URL**.
4. Click **Save**.

Obtaining the authorization endpoint URL configured on Avaya Aura® Device Services

Use this procedure if you want to specify the authorization endpoint URL configured on Avaya Aura® Device Services manually.

1. Log in to the Avaya Aura® Device Services web administration portal with the security administrator role.
2. Navigate to **Security Settings › Client ID Mapping**.
3. Select a client mapping configuration.
4. Click **Edit**.
5. Copy the **Authorize Endpoint** value.
6. Click **Cancel** to discard any changes.

Advanced settings

Configuring resource sharing

Use this procedure to enable Cross-Origin Resource Sharing (CORS) so that HTTP-based clients from different domains can access Avaya Aura® Web Gateway services.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › CORS Configuration**.
2. Select **Enable Cross-Origin Resource Sharing** and **Allow access from any origin**.
3. Click **Save**.

Managing application sessions

Use this procedure to:

- Set a timeout period for terminating inactive, idle, or unattended sessions.
- Manage concurrent HTTP sessions.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Application Management**.
2. In the Application Properties area, complete the required settings, which are described in See "Application Properties field descriptions" on page 96.
3. Click **Save**.

Application Properties field descriptions

Name	Description
Admin HTTPSession Timeout (minutes)	The timeout period for the Avaya Aura® Web Gateway web administration portal. You can enter a value between 1 and 60 minutes. The default value is 10 minutes.
Application HTTPSession Timeout (minutes)	The timeout period for application components.

Name	Description
	You can enter a value between 3 and 120 minutes. The default value is 15 minutes.
Maximum Concurrent HTTP Sessions	The maximum number of active sessions that are available for application components. If the number of sessions exceeds the configured value for any component, a 503 error, which indicates that service is unavailable, is generated by that component. You can enter a value between 100 and 1,000,000. The default value is 200,000.
Concurrent HTTP Sessions per User	The number of active sessions that are available per user. If the number of sessions exceeds the configured value for any user, a 429 error, which indicates that there are too many requests, is sent as a response to the user's request. You can enter a value between 10 and 1000. The default value is 50.

Codecs and bandwidth

Use the following procedures to add, remove, or change the priority of audio and video codecs. The Avaya Aura® Web Gateway offers these codecs to HTTP-based clients and the internal telephony infrastructure while performing call signaling negotiations.

The following codec categories are available:

- WebRTC: Offered to browser-based clients through HTTP.
- SIP: Used internally when communicating through the Avaya Aura® infrastructure or Avaya Meetings Server servers.

Configuring audio codecs and bandwidth

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Media Settings › Audio**.
2. In the SIP Audio Codecs and WebRTC Audio Codecs areas, drag and drop the codecs as required to rearrange them.
3. To add SIP or WebRTC codecs to the list, select the codec that you want to add, and click **Add**.
4. To remove SIP or WebRTC codecs from the list, select the codecs that you want to remove, and click **Remove**.
5. In **OPUS Profile**, select a bandwidth level for the Opus codec if it is included in the WebRTC Audio Codec list.
   ADDITIONAL INFORMATION:
   The Opus codec defines the audio bandwidth.
6. Click **Save**.

Configuring video codecs and bandwidth

Use this procedure to select video codecs that Avaya Aura® Web Gateway uses for SIP and WebRTC calls.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Media Settings › Video**.
2. In the SIP Video Codecs and WebRTC Video Codecs areas, drag and drop the codecs as required to rearrange them.
3. To add SIP or WebRTC codecs to the list, select the codec that you want to add, and click **Add**.
4. To remove SIP or WebRTC codecs from the list, select the codecs that you want to remove, and click **Remove**.
5. In **Call Maximum Video Bandwidth (kbps)**, do one of the following to disable video or modify the video bandwidth:
   - To disable video, select
     0
     .
   - To modify the video bandwidth, select the appropriate value.

   ADDITIONAL INFORMATION:

   > Tip:
   > For the Avaya Aura® Web Gateway to support 1020p HD video, this setting must be set to 1792 kbps or higher. You can select **Custom** to enter your own customized value.
   >
   > The bandwidth value specified will be the maximum allowed by the Avaya Aura® Web Gateway. If the bandwidth value specified in SDP or Avaya Aura® Device Services is lower than the Avaya Aura® Web Gateway value, then the system uses the lower value.
6. Click **Save**.

Configuring Avaya Aura® Media Server credentials

Use this procedure to verify and set the Avaya Aura® Media Server REST security configuration credentials.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Media Settings › AAMS Credentials**.
2. Select **Use Credentials for Authentication** to use the existing login credentials for authentication.
   STEP RESULT:
   Login and **Password** are automatically set to the same credentials that are used for the Avaya Aura® Media Server REST signaling interface. The Avaya Aura® Web Gateway uses these credentials to establish a TLS connection with the Avaya Aura® Media Server. If these credentials are updated in the Avaya Aura® Media Server, then they also need to be updated in the Avaya Aura® Web Gateway.
3. Click **Save**.

Push notification management

The push notification mechanism enables clients to receive incoming call alerts and other notifications from the Apple Push Notification service (APNs). Configuring push notifications on the Avaya Aura® Web Gateway administration portal is part of the push notification configuration procedure. Before configuring push notifications on the Avaya Aura® Web Gateway administration portal, you must create and configure an Avaya Spaces account. You also need to configure settings on Avaya Aura® Session Manager and Avaya Aura® Communication Manager. For more information about the entire configuration procedure, see the sections under Avaya push notification management.

Avaya Oceana® and Avaya Aura® Web Gateway integration checklist

Avaya Oceana® requires the Avaya Aura® Web Gateway to provide WebRTC Signaling Gateway services to video and voice-enabled SIP agents through a browser endpoint. This checklist outlines the tasks you must perform in the Avaya Aura® Web Gateway web administration portal to enable web voice and video in Avaya Oceana®.

No.	Task	Notes	Reference	✔
1	Add the FQDN of the server that requests tokens from the Avaya Aura® Web Gateway server to the trusted hosts list.	Guest clients use these tokens for authorization on the Avaya Aura® Web Gateway.	See See "Adding a trusted host" on page 91.	
2	To enable guest access calls, set the client certificate policy to **OPTIONAL**.	This is required to verify the client certificate of the third-party guest access token server.	See See "Configuring the certificate policy for Avaya Oceana integration" on page 91.	
3	Provide information about the SIP server that Avaya Oceana® clients use for SIP communication.	This information enables the Avaya Aura® Web Gateway to send SIP requests to the clients.	See See "Providing information about guest SIP domain and SIP proxy" on page 75.	

Avaya Spaces Calling and Avaya Aura® Web Gateway integration

Avaya Spaces Calling enables Avaya Spaces users to make audio calls from their Avaya Spaces accounts. The following Avaya Spaces Calling options are available:

- Avaya Cloud Calling to make calls using Avaya Aura® direct media.
- Avaya Spaces Calling browser extension.
- Avaya Workplace Client.
- Any third-party web client or softphone that supports tel URI.

Avaya Spaces Calling requires Avaya Aura® Web Gateway to provide HTTP to SIP signaling gateway services when calling through the Chrome browser extension or Avaya Cloud Calling. You must also configure other Avaya Aura® components, including Avaya Aura® System Manager and Avaya Aura® Session Manager.

This section provides Avaya Aura® Web Gateway tasks and outlines the high-level tasks required to enable audio calls through the Avaya Spaces Calling browser extension and Avaya Cloud Calling.

- To enable Avaya Spaces users to use the Avaya Spaces Calling browser extension, perform the tasks in See "Avaya Spaces Calling browser extension configuration checklist" on page 100.
- To enable Avaya Spaces users to use Avaya Cloud Calling, perform the tasks in See "Avaya Cloud Calling configuration checklist" on page 103.

For detailed information about configuring Avaya Spaces, see *Administering Avaya Spaces*.

Avaya Spaces Calling browser extension configuration checklist

The following checklist outlines the high-level tasks to perform in your deployment to support of the Avaya Spaces Calling extension in Avaya Spaces.

No.	Task	Notes	✔
1	Ensure that you have deployed each of the servers that are required for WebRTC calls.	In addition to Avaya Aura® Web Gateway, you must install and configure the following servers: • Avaya Aura® System Manager • Avaya Aura® Session Manager	

No.	Task	Notes	✔
		• Avaya Aura® Communication Manager • Avaya Aura® Device Services • Avaya Aura® Media Server • LDAP directory source, such as Active Directory or other support LDAP servers. For more information about the deployment and administration documents, see See "Documentation" on page 226.	
2	On Avaya Aura® Device Services, publish the COMM_ADDR_HANDLE_TYPE and COMM_ADDR_HANDLE_LENGTH parameters.	For more information about publishing settings on Avaya Aura® Device Services, see Chapter "Administration of the Dynamic Configuration service" in *Administering Avaya Aura® Device Services*. For more information about the parameters, see the "Avaya Aura® Device Services specific parameters" section in *Administering Avaya Aura® Device Services*.	
3	Configure Avaya Aura® Web Gateway and other servers.	Do the following: • Perform the tasks listed in the "Configuration worksheet" and the procedures listed in Chapter "System Manager, Avaya Aura® Device Services, Avaya Aura® Media Server, and Avaya Meetings Server configurations" in *Deploying the Avaya Aura® Web Gateway*. • Configure the Avaya Aura® Device Services host to obtain user data as described in See "Configuring the Avaya Aura Device Services host to obtain user data" on page 48. • Configure the Avaya Aura® Web Gateway location settings as described in See "Managing Avaya Aura Web Gateway locations" on page 49 and See "Managing Avaya Aura Media Server location and priority	

No.	Task	Notes	✔
		settings for the Avaya Aura Web Gateway" on page 50. • If the Web clients are located under a different domain than that used by Avaya Aura® Web Gateway, configure resource sharing as described in See "Configuring resource sharing" on page 96. • Configure Single Sign-On (SSO) for Avaya Spaces as described in See "Configuring SSO for an Avaya Spaces Calling browser extension" on page 104.	
4	Synchronize users from LDAP directories and Avaya Aura® to Avaya Spaces. To use the click-to-call functionality, you must also synchronize work phone numbers.	You can use one of the following options: • On Avaya Aura® Device Services, synchronize Avaya Spaces and Avaya Aura®. For more information about the synchronization process, see Chapter "Avaya Spaces integration" and the "Configuring data synchronization between Avaya Aura® Device Services and Avaya Spaces" procedure in *Administering Avaya Aura® Device Services*. • Import phone numbers of enterprise users to Avaya Spaces in .csv files. • Ask end-users to enter their work phone numbers to their Avaya Spaces user profiles directly. For more information about managing Avaya Spaces, see documents at https://resources.avayacloud.com/aspx/Avaya_Spaces.	
5	If you have clients outside your enterprise firewall, configure Avaya Session Border Controller for Enterprise.	See the procedures in the "Avaya Session Border Controller for Enterprise configuration" chapter in *Deploying the Avaya Aura® Web Gateway*.	

Avaya Cloud Calling configuration checklist

The following checklist outlines the high-level tasks to perform in your deployment to make audio calls in Avaya Spaces using Avaya Cloud Calling.

No.	Task	Notes	✔
1	Configure Avaya Aura® and other server required for enabling Avaya Cloud Calling on Avaya Aura® Web Gateway.	See See "Prerequisites for configuring Avaya Cloud Calling on Avaya Aura Web Gateway" on page 105.	
2	On Avaya Aura® Device Services, publish the COMM_ADDR_HANDLE_TYPE and COMM_ADDR_HANDLE_LENGTH parameters.	For more information about publishing settings on Avaya Aura® Device Services, see Chapter "Administration of the Dynamic Configuration service" in *Administering Avaya Aura® Device Services*. For information about the parameters, see the "Avaya Aura® Device Services specific parameters" section in *Administering Avaya Aura® Device Services*.	
3	Configure Avaya Aura® Web Gateway and other servers.	Do the following: • Perform the tasks listed in the "Configuration worksheet" and the procedures listed in Chapter "System Manager, Avaya Aura® Device Services, and Avaya Meetings Server configurations" in *Deploying the Avaya Aura® Web Gateway*. • Configure the Avaya Aura® Device Services host to obtain user data as described in See "Configuring the Avaya Aura Device Services host to obtain user data" on page 48.	
4	Configure Avaya Spaces Calling settings on Avaya Aura® Web Gateway.	See See "Configuring Avaya Spaces Calling settings on Avaya Aura Web Gateway" on page 106.	

No.	Task	Notes	✔
5	Configure Avaya Spaces Calling settings on Avaya Spaces.	Provide Avaya Aura® Web Gateway registration data to Avaya Spaces as described in See "Configuring Avaya Cloud Calling settings on Avaya Spaces" on page 107.	
6	Enable integration between Avaya Aura® Web Gateway and Avaya Spaces Calling.	See See "Enabling integration between Avaya Aura Web Gateway and Avaya Spaces Calling" on page 108.	
7	Register Avaya Spaces users on Avaya Aura® Web Gateway.	Register Avaya Spaces users on Avaya Aura® Web Gateway and enable Avaya Cloud Calling functionality for these users as described in See "Assigning Avaya Spaces users to an Avaya Aura Web Gateway system" on page 108.	

Configuring SSO for an Avaya Spaces Calling browser extension

With the Avaya Spaces SSO functionality, Avaya Spaces users can log in to the Avaya Spaces Calling extension using their Avaya Spaces login. The Avaya Spaces Calling browser extension can use the Avaya Spaces Accounts token to log in to Avaya Aura® Web Gateway without asking the user to enter their username and password.

This procedure applies to Avaya-provided web clients, such as Avaya Spaces Calling Chrome extension, and to your own custom web clients that register on Avaya Aura® Web Gateway using Avaya Spaces SSO.
PREREQUISITE
Perform the steps listed in See "Avaya Spaces Calling browser extension configuration checklist" on page 100.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Spaces Configuration › Spaces SSO**.
2. Select the **Enable SSO** check box.
3. In **Domain**, type the name of the domain configured for your company in Avaya Spaces and click **Add**.
4. If you have multiple domains configured for your company in Avaya Spaces, repeat the previous step for all these domains.
5. If you are configuring Avaya Spaces SSO for your own web client, do the following:
    a. In **Client ID**, enter the ID of your web client.
    b. Click **Add**.

    ADDITIONAL INFORMATION:
    This step is not required for the Avaya Spaces Calling browser extension because Avaya Aura® Web Gateway already contains information about its ID.

6. Click **Save**.

Prerequisites for configuring Avaya Cloud Calling on Avaya Aura® Web Gateway

Before configuring Avaya Spaces Calling integration settings on Avaya Aura® Web Gateway, ensure that the following requirements are met in your deployment:

Avaya Aura® prerequisites

Along with Avaya Aura® Web Gateway, you must install and configure the following servers:

- Avaya Aura® System Manager 8.x
- Avaya Aura® Session Manager 8.x
- Avaya Aura® Communication Manager 8.x
- Avaya Aura® Device Services 8.x
- LDAP directory source, such as Active Directory.

MDA and SIP entity link configuration:

- On System Manager, increase the currently configured Multi-Device Access (MDA) limit by 1. This is required to prevent the forced logout of Avaya Aura® Web Gateway.

  For example, if the MDA limit is currently set to 5, increase it to 6.

  To configure the MDA limit, see information about the Multi Device Access feature in *Administering Avaya Aura® System Manager*.

- Create trusted SIP entity links from each Avaya Aura® Web Gateway node to each Session Manager in your deployment.

  For more information, see "Configuring SIP Trunks for the Avaya Aura® Web Gateway on System Manager" in *Deploying the Avaya Aura® Web Gateway*.

Avaya Spaces prerequisites:

- Configure a company and a domain.
- Purchase Avaya Spaces Business or Power licenses and assign them to users Avaya Spaces that require Avaya Cloud Calling.
- Optionally, synchronize work and mobile phone numbers configured in your enterprise LDAP directories and in Avaya Spaces. You can use the following options:
  - Synchronize user data between Avaya Aura® Device Services and Avaya Spaces. For information about user data and phone number synchronization, see Chapter "Avaya Spaces integration" and the procedures "Configuring data synchronization between Avaya Aura® Device Services and Avaya Spaces" and "Configuring phone number synchronization" in *Administering Avaya Aura® Device Services*.

- Import phone numbers of enterprise users to Avaya Spaces using *.csv* files.

> Note:
> Avaya Spaces users can manually add work and mobile phone numbers to their accounts if the phone number synchronization is not enabled on Avaya Aura® Device Services.

Enterprise firewall prerequisites:

Add the following IPv4 Classless Inter-Domain Routing (CIDR) blocks to the enterprise firewall rules of your Avaya Aura® deployment:

- 155.184.0.0/20
- 155.184.16.0/22

These rules are required for the SRTP and UDP traffic from your enterprise network to the Avaya Cloud MPaaS server.

The destination ports for Avaya Cloud MPaaS are UPD ports 3000 to 4999 inclusively.

Configuring Avaya Spaces Calling settings on Avaya Aura® Web Gateway

Provide Avaya Spaces Calling Gateway settings to Avaya Aura® Web Gateway. Avaya Aura® Web Gateway uses these settings to generate authorization data, which you must export to Avaya Spaces to complete the integration process.

1. Log in to the Avaya Aura® Web Gateway administration portal as an administrator.
2. Navigate to **System Overview**.
3. In the System Settings area, in **System Name**, type a meaningful name that allows to uniquely identify your Avaya Aura® Web Gateway system.
   ADDITIONAL INFORMATION:
   Avaya Spaces displays this name to users on the User Preferences screen. Avaya Spaces also uses this name when you configure integration settings in the Avaya Spaces Admin Area.

   > Note:
   > You can change this name at any time, and it will not affect the existing integration with Avaya Spaces.

4. Navigate to **Spaces Configuration › Spaces Calling**.
5. From **Country** and **Region**, select the geographical location of your Avaya Aura® Web Gateway cluster.
6. Ensure that the **Gateway Connection Manager Address** value is `communication-services.avayacloud.com`.
7. Ensure that the **Gateway Connection Manager Port** value is `443`.
8. Click **Generate Key**.
   STEP RESULT:
   Avaya Aura® Web Gateway generates a public key, which is required to authorize Avaya Aura® Web Gateway on Avaya Spaces.
9. Ensure that the **Enable Spaces Calling Gateway Cluster** check box is not selected.
   ADDITIONAL INFORMATION:

You will select this check box after configuring Avaya Aura® Web Gateway settings on Avaya Spaces.

10. Click **Save**.
11. Click **Export key** and copy the authorization data.
ADDITIONAL INFORMATION:
You must use this data when configuring Avaya Aura® Web Gateway settings on Avaya Spaces.

    Avaya Aura® Web Gateway uses JSON format to store the authorization data. The following is an example of the authorization data:


```
{
    "systemId": "9bf8f4ab-99b1-452b-9b7f-
e75aacf31d19.mycompany.com",
    "publicKey": "-----BEGIN PUBLIC KEY-----
\nMFkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDQgAE9rtz4fuYhGm2JlvnI6lZmate8e
EX\na4wvmklSdHGYZHos7y8xNBNCEj9wc3klayOKHYIVIeL0ryVFgM16Ud5FDQ==
\n-----END PUBLIC KEY-----",
    "alg": "ES256",
    "description": "Avaya Aura Web Gateway Services"
}
```


AFTER COMPLETING THE TASK
Configure integration on the Avaya Cloud administration portal.

Configuring Avaya Cloud Calling settings on Avaya Spaces

To enable the Avaya Cloud Calling option for Avaya Spaces calling, add the Avaya Aura® Web Gateway system to the Avaya Spaces Admin Area.

For more information about configuring Avaya Cloud Calling settings on Avaya Spaces, see *Administering Avaya Spaces*.
PREREQUISITE
Configure integration settings on Avaya Aura® Web Gateway and obtain the authorization data in JSON format.

1. Log in to the Avaya Spaces Admin Area.
2. In the left navigation pane, click **Settings** for your company.
3. In the Avaya Spaces Calling area, select the **Avaya Cloud Calling** check box.
4. Click **Add new Aura System**.
STEP RESULT:
Avaya Spaces displays the Add / Edit an Aura system window.
5. In the text box, enter the authorization data that you obtained from Avaya Aura® Web Gateway.
ADDITIONAL INFORMATION:
Ensure that the authorization data you enter is a valid JSON string.
6. To automatically assign users to this Avaya Aura® Web Gateway system, select the **Auto assign all unassigned users to this Aura system** check box.
STEP RESULT:

Users that are not already assigned to another Avaya Aura® system are assigned to this system. You must opt in users before they can make calls with Avaya Cloud Calling.

7. Click **Save**.

AFTER COMPLETING THE TASK
Enable integration between Avaya Spaces Calling and Avaya Aura® Web Gateway on the Avaya Aura® Web Gateway administration portal.

Enabling integration between Avaya Aura® Web Gateway and Avaya Spaces Calling

After configuring integration settings in the Avaya Spaces Admin Area, enable integration on the Avaya Aura® Web Gateway administration portal.

1. Log in to the Avaya Aura® Web Gateway administration portal as an administrator.
2. Navigate to **Spaces Configuration › Spaces Calling**.
3. Select the **Enable Spaces Calling Gateway Cluster** check box.
4. Click **Save**.
5. In the Spaces Calling Gateway Connection Status area, ensure that the connection status for Avaya Spaces Calling Gateway is "Connected."
6. If the connection status is "Not Connected," click **Repair Connection**.
7. Log in to the Avaya Spaces Admin Area.
8. In the left navigation pane, click **Settings** of your company.
9. In the Avaya Spaces Calling area, ensure that the connection status for your Avaya Aura® Web Gateway server is "Connected."
10. To view detailed information about the connection, click **Detailed status**.

AFTER COMPLETING THE TASK
Assign Avaya Spaces users to the Avaya Aura® Web Gateway system to make calls with Avaya Cloud Calling.

Assigning Avaya Spaces users to an Avaya Aura® Web Gateway system

From the Avaya Spaces Admin Area, you must assign users to an Avaya Aura® Web Gateway system to make audio calls with Avaya Cloud Calling.
PREREQUISITE

- Complete the Avaya Aura® Web Gateway configuration.
- Add the Avaya Aura® Web Gateway system to Avaya Spaces.

1. Log in to the Avaya Spaces Admin Area.
2. In the left navigation pane, click **Settings** of your company.
3. In the Avaya Spaces Calling section, click **Assign Users to Aura Systems**.
4. Select users who require the ability to make calls using Avaya Cloud Calling.

5. In **Available Aura Systems**, select the Avaya Aura® Web Gateway system you integrated with Avaya Spaces Calling.
6. Click **Assign**.
7. For each user that you selected in step 6, click **opt-in**.
   ADDITIONAL INFORMATION:
   The **opt-in** option is only available if the user's Feature Enablement Status is "Enabled."

Disabling integration between Avaya Aura® Web Gateway and Avaya Spaces Calling

Use this procedure to disable integration between Avaya Aura® Web Gateway and Avaya Spaces Calling. When integration is disabled, the Avaya Cloud Calling option is unavailable for Avaya Spaces users. After you remove your Avaya Aura® Web Gateway system from Avaya Spaces, Avaya Spaces automatically unassigns all users assigned to that Avaya Aura® Web Gateway system and also "opts out" these users.

1. Log in to the Avaya Spaces Admin Area.
2. In the left navigation pane, click **Settings** for your company.
3. In the Avaya Spaces Calling section, click **Delete** next to the Avaya Aura® Web Gateway system that you want to disable.
4. Log in to the Avaya Aura® Web Gateway administration portal.
5. Navigate to **Spaces Configuration › Spaces Calling**.
6. Clear the **Enable Spaces Calling Gateway Cluster** check box.

Maintenance mode overview

Avaya Spaces Calling Gateway Maintenance mode is intended for use when you perform procedures that involve shutting down or restarting Avaya Aura® Web Gateway nodes, such as upgrading Avaya Aura® Web Gateway. When in Maintenance mode, some of the Avaya Spaces Calling Gateway features are disabled to ensure safety of the procedures.

The following features are unavailable in Maintenance mode:

- Failover and failback operations for Avaya Spaces users registered on Avaya Aura® Web Gateway.

  By default, Avaya Aura® Web Gateway moves users registered on a node to other nodes if the node is down. However, In Maintenance mode, Avaya Aura® Web Gateway does not move users to other nodes when you restart or shut down a node.

- Opting in Avaya Spaces users on Avaya Aura® Web Gateway from the Avaya Spaces Admin area.

  If the Avaya Spaces administrator tries to opt-in a user, Avaya Spaces informs the administrator that the opt-in functionality is unavailable because Avaya Spaces Calling Gateway is in Maintenance mode.

- Performing new calls using Avaya Spaces Calling.
- Updating the user status using the **Refresh status** button from the Avaya Spaces Admin area.

The following features remain available in Maintenance mode:

- Completing the ongoing calls.

  Avaya Aura® Web Gateway does not drop any ongoing calls established before you enable Maintenance mode. However, some mid-call operations might be unavailable when in Maintenance mode.

- Opting out Avaya Spaces users from the Avaya Spaces Admin area.

> Important:
> Maintenance mode is intended for cluster Avaya Aura® Web Gateway deployments. Do not use Maintenance mode if you have a standalone deployment.

Workflow

If you use Avaya Spaces Calling Gateway, a typical workflow for an operation that requires Avaya Aura® Web Gateway restart looks as follows:

1. Enable Maintenance mode.
2. Perform a procedure that involves restart of one or several Avaya Aura® Web Gateway nodes.
3. Disable Maintenance mode.

> Note:
> You cannot use Maintenance mode when migrating from Avaya Aura® Web Gateway Release 3.11 to Release 10.2.

Enabling Maintenance mode

If you use Avaya Spaces Calling Gateway, enable Maintenance mode before performing operations that involve restarting Avaya Aura® Web Gateway nodes, such as Avaya Aura® Web Gateway upgrade. When in Maintenance mode, most of the Avaya Spaces Calling Gateway features are unavailable.

Maintenance mode is intended for cluster Avaya Aura® Web Gateway deployments. Do not use Maintenance mode if you have a standalone deployment.

> Important:
> Disable Maintenance mode after performing the required operations.

1. Navigate to **Spaces Configuration › Spaces Calling**.
2. Select the **Enable Maintenance Mode** check box.
3. Click **Save**.
4. In the confirmation window, click **Proceed**.

AFTER COMPLETING THE TASK
Perform the required operations and then disable Maintenance mode.

Disabling Maintenance mode

After performing an operation that requires restart of Avaya Aura® Web Gateway nodes, disable Maintenance mode to re-enable the full Avaya Spaces Calling Gateway functionality.

1. Navigate to **Spaces Configuration › Spaces Calling**.
2. Clear the **Enable Maintenance Mode** check box.
3. Click **Save**.
4. In the confirmation window, click **Proceed**.

Re-establishing the connection to Gateway Connection Manager manually

Avaya Aura® Web Gateway must be connected to Gateway Connection Manager to enable users to perform calls between Avaya Spaces and Avaya Aura®. If the connection fails, Avaya Spaces Calling Gateway automatically tries to restore the connection. You can also try to repair the connection manually from the Avaya Aura® Web Gateway web administration portal.

> Note:
> The repair process might cause service disruption. Therefore, you must only run the repair if there is a problem with the connection.

1. Navigate to **Spaces Configuration › Spaces Calling**.
2. In the Connection status table, click **Repair connection** for each Avaya Aura® Web Gateway node whose connection status is **Not Connected**.
   ADDITIONAL INFORMATION:
   You cannot repair the connection for a node whose status is **Unknown** or **Node is inactive**.
3. In the confirmation window, click **Proceed**.
   STEP RESULT:
   If the repair process completed successfully, Avaya Aura® Web Gateway informs you that the connection is re-established.
4. If Avaya Aura® Web Gateway informs you that the connection cannot be repaired, do the following:
   a. Ensure that the Avaya Spaces Calling Gateway is properly configured on Avaya Aura® Web Gateway and Avaya Spaces.
   b. If the problem persists, contact Avaya support.

Gateway Connection Manager connectivity status indicators

The following table shows the Gateway Connection Manager connectivity status indicators for an Avaya Aura® Web Gateway node:

Status	Description	Is manual reconnection feasible
Connected	The Avaya Aura® Web Gateway node is connected to Gateway Connection Manager and can be used for calls between Avaya Aura® and Avaya Spaces.	Yes
Not Connected	The Avaya Aura® Web Gateway node is not connected to Gateway Connection Manager and cannot be used for calls between Avaya Aura® and Avaya Spaces.	Yes
Node is Inactive	The Avaya Aura® Web Gateway node is offline.	No
Unknown	Avaya Aura® Web Gateway cannot obtain the connection status of the node.	No

Avaya push notification management

Push notifications

The push notification mechanism enables clients to receive incoming call alerts and other notifications from the Apple Push Notification service (APNs). The push notification service sends notifications automatically. Therefore, an application can receive notifications even when it is suspended or in Sleep mode.

The Avaya Aura® Web Gateway can send push notifications about the following telephony-related events:

- Incoming calls.
- Incoming calls on an Avaya Aura® Communication Manager bridged line appearance.
- Incoming calls on an Avaya Aura® Communication Manager enhanced pickup group.
- Incoming calls on an Avaya Aura® Communication Manager team button.
- Voicemail status updates.

For messaging push notifications, use Avaya Multimedia Messaging or Avaya Aura® Presence Services Release 8.0.2 or later.

Push notification provider

The Avaya Aura® Web Gateway interacts with the APNs through a push notification provider. You must register your Avaya Aura® Web Gateway system with the push notification provider before activating push notifications. The default provider is the Avaya Push Notification provider, which must be used to support push notifications for Avaya Workplace Client. If you want to use push notifications for third-party SDK-based applications, you must use a third-party push notification provider.

For the Avaya Push Notification provider, you must create and configure an account at accounts.avayacloud.com. This account is used to store the data required to authorize your Avaya Aura® Web Gateway system. This account requirement does not apply if you are working with a third-party provider.

Checklist for push notification service configuration

Use this checklist to set up the push notification service for your iOS applications.

Depending on whether you are planning to work with the default Avaya Push Notification provider or a third-party push notification provider, the task you perform are slightly different.

No.	Task	Notes	✔
1	Set up an Avaya Spaces account.	See See "Avaya Cloud account configuration" on page 115. This step is not required for third-party notification providers.	
2	Configure the third-party push notification provider.	See the requirements in See "Third-party push notification provider requirements" on page 115. This step is not required if you are working with the Avaya Push Notification provider.	
3	Configure your enterprise network firewall.	See See "Firewall configuration" on page 117.	
4	If you are using the Avaya Push Notification Provider, determine if you want to use CA certificates built-in to RHEL.	See See "Configuring the use of CA certificates built-in to RHEL for the Avaya Push Notifications provider" on page 117. If you do not want to use built-in certificates, you must upload CA certificates to the truststore as described in See "Managing truststore certificates" on page 89.	
5	Configure the push notification provider on the Avaya Aura® Web Gateway.	• If you are working with the Avaya Push Notification provider, see See "Configuring the Avaya Push Notification provider on the Avaya Aura Web Gateway" on page 118. • If you are working with a third-party push notification provider, see See "Configuring a third-party push notification provider on the Avaya Aura Web Gateway" on page 119.	
6	Configure push notifications for mobile applications.	See See "Configuring mobile application settings" on page 121.	
7	Configure Avaya Aura® Session Manager and Avaya Aura® Communication Manager.	See See "Avaya Aura components configuration" on page 123.	

No.	Task	Notes	✔
8	Set the parameters to enable push notifications on iOS applications.	See See "Configuration parameters for iOS applications" on page 125. To enable push notifications in iOS applications, you must configure the required parameters using Avaya Aura® Device Services or a settings file. Avaya Aura® Device Services is the recommended option if it is available in your deployment.	

Third-party push notification provider requirements

By default, the Avaya Aura® Web Gateway uses the Avaya Push Notification provider for Avaya clients, such as Avaya Workplace Client for iOS. If you are planning to use push notifications for third–party Avaya™ Client SDK-based applications, you must use a third-party provider, for example, developed by the application developer. This provider must implement the following APIs:

- Interface with the Apple Push Notification service (APNs). Implementation of this interface ensures that the APNs trusts the third-party push notification provider to send notifications to Avaya clients. For more information about implementing the interface and the APIs required by the APNs, see APNs Overview.
- Interface with Avaya Aura® servers. The third-party push notification provider must trust the Avaya Aura® Web Gateway to receive push notifications from it. Avaya Aura® Web Gateway uses the OAuth-based API for authentication. For more information about the APIs, see Avaya™ Client SDK.

  Currently, the existing third-party push notification providers do not support Avaya APIs.

Avaya Cloud account configuration

If you are planning to use the Avaya Push Notification provider, you must create and configure an Avaya Cloud account. This account stores the parameters required to authorize your Avaya Aura® Web Gateway system.

Use the following sections to create and configure your Avaya Cloud account.

Registering an account

Use this procedure to register an account using your email address.

1. In your web browser, enter https://accounts.avayacloud.com/.
2. In the **Email or Phone** field, type your email address.
3. Click **Yes, sign me up!**.

STEP RESULT:
sends a confirmation email to the email address you specified.

4. In your mailbox, open the confirmation email and then click the **Confirm** button.
   STEP RESULT:
   You are redirected to the My Account page.
5. Provide your first name, last name, password, and, optionally, a photo.
6. Click **Create an account**.

Setting up a company and domain in

You must provide the company domain associated with your Avaya Aura® Web Gateway system.

1. Log in to https://accounts.avayacloud.com/ as an administrator.
2. If you have not set up your company or want to configure a new company, do the following:
   a. Click on your user name in the top-right area of the screen and then click **Add Company**.
   b. Type a name and description for your company.
   c. Click **Save**.
3. Click **Manage Companies** and click the existing company name.
4. Click the **Domains** tab.
5. Click **Add Domain**.
6. Enter the domain name and then click **OK**.
7. To verify ownership of the domain, next to the domain name, click **Verify**.
8. Do one of the following:
   - Follow the on-screen instructions to add the verification code to your domain account and then click **Verify**.
   - At the bottom-right area of the Verify Domain window, click **Manual Verify** to have Avaya personnel verify the ownership of the domain.

Adding the Avaya Mobile Push Notification Service to a company profile on your Avaya Cloud account

To activate the push notification service for the Avaya Aura® Web Gateway , you must add the Avaya Mobile Push Notification Service application to your company profile on the Avaya Cloud account.

1. Log in to https://accounts.avayacloud.com/ as an administrator.
2. From the dashboard, on the left area, click **Manage Companies**.
3. Select the required company and then click the **Apps** tab.
4. Click **Configure New App**.
5. From the **Product** drop-down list, select **Avaya Mobile Push Notification Service**.
6. Click **Save**.
   STEP RESULT:
   Avaya Cloud creates a new Avaya Mobile Push Notification Service application.

AFTER COMPLETING THE TASK
When you configure notifications, you provide authorization data to the Public Settings section of your new Avaya Mobile Push Notification Service application.

Firewall configuration

Avaya Aura® Web Gateway communicates with the Avaya Push Notification provider cloud service using HTTPS. To use the Avaya Push Notification provider, your enterprise network firewall and the HTTP proxy server must allow outbound HTTPS connections to the Avaya Push Notification provider address, which is *pnp.avaya.com:443*.

> Important:
> If you configured push notifications for Release 3.7, Avaya Aura® Web Gateway still uses the old *apnp.avaya.com* Avaya Push Notification provider address. Avaya recommends that you change this address to *pnp.avaya.com* and update your enterprise network firewall and HTTP proxy server settings to *pnp.avaya.com:443*

For applications other than Avaya Workplace Client for iOS, you must use a third-party provider. The firewall and the HTTP proxy of your enterprise must allow outbound HTTPS connections to this push notification provider.

For iOS devices running in your enterprise network, Apple push notifications rely on connectivity between iOS devices and the Apple Push Notification service network. For more information about port configuration required to receive Apple push notifications, see https://support.apple.com/en-us/ht203609.

Configuring the use of CA certificates built-in to RHEL for the Avaya Push Notifications provider

By default, Avaya Aura® Web Gateway uses built-in CA certificates of RHEL to connect to the Avaya Push Notification provider and use the APNs. In this case, you do not need to upload CA certificates to the truststore manually. If you do not want to trust the built-in CA certificates, you can disable the use of these certificates for push notifications.

If you do not use built-in CA certificates of RHEL, you must upload trusted third-party CA certificates for Avaya Push Notification provider to the Avaya Aura® Web Gateway truststore.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Security Settings › Certificate Management › Truststore**.
2. Do one of the following:
    - If you want to use built-in CA certificates of RHEL, select the **Use Operating System Trusted Root CAs for Push Notification Provider** check box.
    - If you want to use third-party CA certificates only, clear the **Use Operating System Trusted Root CAs for Push Notification Provider** check box.
3. Click **Save**.
   ADDITIONAL INFORMATION:
   Avaya Aura® Web Gateway restarts to apply changes.

Configuring the Avaya Push Notification provider on the Avaya Aura® Web Gateway

To use the Avaya Push Notification provider, you must generate authorization data and export this data to your Avaya Cloud account. You cannot edit other provider data, such as the name or address, and you cannot delete the Avaya Push Notification provider from the Avaya Aura® Web Gateway.

For a cluster, perform this procedure once, regardless of the number of nodes in the cluster.
PREREQUISITE

- Create and configure an Avaya Cloud account.
- If you do not use CA certificates built-in to RHEL, install the public certificate of the push notification provider on the Avaya Aura® Web Gateway. For more information, see See "Managing truststore certificates" on page 89.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Push Notification Settings › Provider Settings**.
2. From **Push Notification Provider**, select **Avaya Provider**.
3. In **Enter Company Domain**, enter your company domain.
   ADDITIONAL INFORMATION:
   You must enter the same company domain that you provided in your Avaya Cloud account.
4. Click **Generate Key**.
   STEP RESULT:
   The Avaya Aura® Web Gateway generates a public and private key pair and an identifier. This data is required to authorize Avaya Aura® Web Gateway on the push notification provider. The Avaya Aura® Web Gateway also updates the following values:
   - **System Id**: A unique identifier for your system.
   - **Public Key**: A public key.
5. Click **Export**.
   ADDITIONAL INFORMATION:
   Avaya Aura® Web Gateway displays a pop-up window with the authorization information in JSON format. You must provide this data to your Avaya Cloud account.

   The following is an example of the authorization data:


```
{
    "systemId": "9bf8f4ab-99b1-452b-9b7f-
e75aacf31d19.mycompany.com",
    "description": "Avaya Aura Web Gateway Services",
    "publicKey": "-----BEGIN PUBLIC KEY-----
\nMFkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDQgAE9rtz4fuYhGm2JlvnI6lZmate8e
EX\na4wvmklSdHGYZHos7y8xNBNCEj9wc3klayOKHYIVIeL0ryVFgM16Ud5FDQ==
\n-----END PUBLIC KEY-----",
    "alg": "ES256"
}
```


   Important:

> - Do not close the Provider Settings page on the Avaya Aura® Web Gateway administration portal until you finish exporting authorization data to your Avaya Spaces account.
> - Do not save the data you provided on the Provider Settings page of the Avaya Aura® Web Gateway administration portal until you export authorization data to your Avaya Cloud account. Otherwise, the push notification service might not work as expected.

6. Copy the authorization data, including the surrounding curly brackets, and save it in a file on your computer.
7. In a new browser tab, log in to your Avaya Cloud account and navigate to **Manage Companies**.
8. Select the required company and navigate to the **Apps** tab.
9. From **App**, select **Avaya Mobile Push Notification Service**.
10. In Data Configuration, select the **JSON** check box.
11. Replace the content in **Public Settings** with the content of the file that you created in step 6 and then save your changes.
    ADDITIONAL INFORMATION:
    You must ensure that the authorization data you enter in **Public Settings** is a valid JSON string. If the data uses an invalid format, the Avaya Cloud account displays a warning message, but still allows you to save changes.

    The Avaya Cloud account stores authorization data internally. **Public Settings** only displays the authorization data that you entered last. You will not lose any previously entered authorization data if you overwrite existing content with the new authorization data.
12. Return to the Provider Settings page on the Avaya Aura® Web Gateway administration portal.
13. Click **Test** to verify that your system can connect and authenticate with the push notification provider.
14. Do one of the following:
    - If the verification completed successfully, click **Save**.
    - If the verification failed, fix the issue as described in See "Avaya Aura Web Gateway cannot connect to a push notification provider" on page 217 and then and re-run the connectivity test.

      If the problem persists, contact Avaya support personnel.

Configuring a third-party push notification provider on the Avaya Aura® Web Gateway

To use push notifications on third-party iOS applications, you require a third-party push notification provider. An Avaya Breeze Client SDK application developer can implement a third-party push notification provider. Use this procedure to set up or update a third-party push notification provider on the Avaya Aura® Web Gateway.
PREREQUISITE
A third-party Avaya Breeze Client SDK application developer must implement the third-party push notification provider. For more information about push notification provider requirements, see See "Third-party push notification provider requirements" on page 115.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Push Notification Settings › Provider Settings**.
2. Do one of the following:
    - To add a new provider configuration, click **Add**.
    - To edit an existing provider, select the required provider from **Push Notification Provider** and click **Edit**.
3. In **Enter Company Domain**, enter the domain where your Avaya Aura® Web Gateway is deployed.
   ADDITIONAL INFORMATION:
   For example: `mycompany.com`
4. In **Push Notification Provider Name**, enter a name for the provider.
   ADDITIONAL INFORMATION:
   For example: `MyPushNotificationProvider`.

   This name is used in the Avaya Aura® Web Gateway administration portal for display purposes only.
5. In **Push Notification Provider Address**, enter the FQDN where your push notification provider is deployed.
   ADDITIONAL INFORMATION:
   For example: `mypushnotifications.mycompany.com`
6. In **Push Notification Provider Port**, enter the port number for the push notification provider.
   ADDITIONAL INFORMATION:
   The default port is 443.
7. Click **Generate Key**.
   STEP RESULT:
   The Avaya Aura® Web Gateway generates a public and private key pair and an identifier. This data is required to authorize Avaya Aura® Web Gateway on the push notification provider. The Avaya Aura® Web Gateway also updates the following values:
    - **System Id**: A unique identifier for your system.
    - **Public Key**: A public key.
8. Click **Export**.
   ADDITIONAL INFORMATION:
   Avaya Aura® Web Gateway displays a pop-up window with the authorization data in JSON format.

   The following is an example of the authorization data:


```
{
    "systemId": "9bf8f4ab-99b1-452b-9b7f-
e75aacf31d19.mycompany.com",
    "description": "Avaya Aura Web Gateway Services",
    "publicKey": "-----BEGIN PUBLIC KEY-----
\nMFkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDQgAE9rtz4fuYhGm2JlvnI6lZmate8e
EX\na4wvmklSdHGYZHos7y8xNBNCEj9wc3klayOKHYIVIeL0ryVFgM16Ud5FDQ==
\n-----END PUBLIC KEY-----",
    "alg": "ES256"
}
```


Important:

> - Do not close the Provider Settings page on the Avaya Aura® Web Gateway administration portal while you are exporting authorization data to your push notification provider.
> - Do not save the data you provided on the Provider Settings page of the Avaya Aura® Web Gateway administration portal until you export authorization data to your push notification portal. Otherwise, the push notification service might not work as expected.

9. Copy the system ID and public key data and provide it to your push notification provider.
   ADDITIONAL INFORMATION:
   For example:

   ```
   "systemId": "9bf8f4ab-99b1-452b-9b7f-e75aacf31d19.mycompany.com"
   "publicKey": "-----BEGIN PUBLIC KEY-----
   \nMFkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDQgAE9rtz4fuYhGm2JlvnI6lZmate8e
   EX\na4wvmklSdHGYZHos7y8xNBNCEj9wc3klayOKHYIVIeL0ryVFgM16Ud5FDQ==
   \n-----END PUBLIC KEY-----"
   ```

   The steps you must perform to provide authorization data are provider-specific. For more information, contact your third-party push notification provider vendor.
10. Return to the Provider Settings page on the Avaya Aura® Web Gateway administration portal.
11. Click **Test** to verify that your system can connect and authenticate with the push notification provider.
12. Do one of the following:
    - If the verification completed successfully, click **Save**.
    - If the verification failed, fix the issue as described in See "Avaya Aura Web Gateway cannot connect to a push notification provider" on page 217 and then and re-run the connectivity test.

      If the problem persists, contact Avaya support personnel.

Removing a third-party push notification provider

Use this procedure to remove the configuration for a third-party push notification provider from the Avaya Aura® Web Gateway. You can only remove a provider that is not used by any mobile applications. You cannot remove the default, built-in Avaya Push Notification provider.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Push Notification Settings › Provider Settings**.
2. In **Push Notification Provider**, select the required provider.
3. Click **Remove** and click **Yes** in the confirmation window.

Configuring mobile application settings

If you plan to use push notifications on a third-party CSDK-based iOS application, use this procedure to provide information about this application on Avaya Aura® Web Gateway. You can then select the required third-party push notification provider for the application.

Avaya Workplace Client for iOS configuration is pre-defined. Therefore, this procedure is not required for Avaya Workplace Client for iOS. You only need to modify the Avaya Workplace Client for iOS configuration if the default Avaya Push Notification provider configuration uses the old *apnp.avaya.com* address. For more information, see See "Reconfiguring the Avaya Push Notification provider after upgrade or migration" on page 210.

> Note:
>
> - The name of the pre-defined Avaya Workplace Client for iOS configuration is `Avaya Workplace`.
> - If you use Avaya Workplace Client for iOS, do not use other third-party iOS applications.

PREREQUISITE
Configure a push notification provider on Avaya Aura® Web Gateway. For more information, see See "Configuring the Avaya Push Notification provider on the Avaya Aura Web Gateway" on page 118 and See "Configuring a third-party push notification provider on the Avaya Aura Web Gateway" on page 119.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Push Notification Settings › Mobile Application Settings**.
2. Do one of the following:
   - To create a new mobile application configuration, click **Add**.
   - To edit existing configuration settings, from **Application Name**, select the required application and click **Edit**.
3. In **Application Name**, provide an appropriate application name.
4. In **Application Id**, provide the iOS application bundle identification string.
   ADDITIONAL INFORMATION:
   Each application must have a unique identification string.
5. In **Push Notification Provider**, select the required push notification provider.
6. To verify that the Avaya Aura® Web Gateway can send notifications to the mobile application, click **Test**.
   STEP RESULT:
7. Do one of the following:
   - If the verification is completed successfully, click **Save**.
   - If the verification failed, fix the issue as described in See "iOS application cannot connect to a push notification provider" on page 219 and re-run the connection test.

     If the problem persists, contact Avaya support personnel.

Disabling specific push notifications

Use this procedure to disable push notifications for a specific third-party application. You cannot delete the pre-defined `Avaya Workplace` configuration.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Push Notification Settings › Mobile Application Settings**.
2. In the **Application** field, select the required mobile application.
3. Click **Remove** and click **Yes** to confirm.

Disabling all push notifications

Use this procedure to disable push notifications for all third-party applications, including the default applications, such as `com.avaya.AvayaCommunicator`. If you want to restore a default application, see See "Configuring a third-party push notification provider on the Avaya Aura Web Gateway" on page 119.

PREREQUISITE

These steps apply to the current release of Avaya Aura® Web Gateway. If you have an earlier release of Avaya Aura® Web Gateway, contact Avaya personnel to obtain the appropriate script, then copy the *disable_push.sh* script to your Avaya Aura® Web Gateway server and complete step See "2" on page 123.

1. Locate the *disable_push.sh* script on the Avaya Aura® Web Gateway server.
    a. Log in to the Avaya Aura® Web Gateway using your SSH credentials.
    b. Type *cdto CAS* to navigate to the CAS folder.
    c. Type *cd push_notifications/* to navigate to the push notifications folder.
    d. Type *cp disable_push.sh ~/disable_push.sh* to copy the script to your home directory.
    e. Type *cd ~* to navigate to your home directory.
2. Run the following command to execute the script:
   ADDITIONAL INFORMATION:
   ```
   sh disable_push.sh
   ```
   Avaya Aura® Web Gateway stops providing the push notification service to all currently registered mobile devices. Mobile devices can no longer register for a push notification service on Avaya Aura® Web Gateway.
3. Restart services on all nodes of the cluster.
   ADDITIONAL INFORMATION:
   For more information, see See "Restarting services on an Avaya Aura Web Gateway node" on page 47.

Avaya Aura® components configuration

The following sections outline the settings you can modify on Avaya Aura® Session Manager and Avaya Aura® Communication Manager to improve the performance of the push notification service.

Avaya Aura® Session Manager configuration

Session Manager allows users to have multiple SIP devices associated with their accounts. The Avaya Aura® push notification solution requires that Session Manager user accounts must be configured to allow the use of multiple devices. Avaya Aura® Web Gateway counts as one additional device that registers on Session Manager to receive incoming notifications. The multiple device option is called Multiple Device Access (MDA) and is configured through a user configuration template. For more information about configuring user provisioning rules, see *Administering Avaya Aura® Session Manager*.

Avaya Aura® Communication Manager configuration

Communication Manager controls how long an incoming call alerts at a VoIP endpoint that uses SIP for call signaling. The time required to establish a call depends on various conditions, including the following:

- Call setup latency through the push notification network.
- Additional network connection steps required by iOS clients to present the incoming call notification to end users.

If you use the Apple Push Notification service in your deployment, you must change the default values of the following Communication Manager timers to provide Avaya Workplace Client for iOS users with sufficient time to answer the call. Otherwise, Communication Manager might drop the call or send it to coverage before the called party has a chance to answer the call.

Alternate route timer

By default, Communication Manager has a six second alternate router timer that expires when no 180 Ringing response is received from the called party. When the timer expires, Communication Manager drops the call. If you enable push notifications, increase this value of the **Alternate Route Timer(sec)** parameter to 10 seconds. This parameter is located in a Communication Manager SIP signaling group that is associated with the Session Manager instance used by mobile clients that are subscribed for push notifications.

Redirect on OPTIM Failure timer

The timer controls how long Communication Manager waits for a SIP 18x provisional response before canceling the call that was intended for an iOS client. The default value is 5,000 milliseconds. If you enable push notifications, increase this value to 10,000 milliseconds. The value is configured in the **Redirect on OPTIM Failure** field in a Communication Manager SIP trunk-group.

Coverage path number of ring cycles

This timer controls the number of ring cycles to alert the originally called party before directing them to the first coverage point in the coverage path. The exact duration of the wait time depends on the ring cycle length, which can be different in different countries. By default, Communication Manager waits for two ring cycles. In the **Don't Answer** field under Coverage Criteria, increase the Number of Rings to four ring cycles for the Coverage Paths that apply to users with Avaya Workplace Client running on an Apple device.

Note:

- To avoid changing coverage behavior for users who do not use Avaya Workplace Client for iOS, you can create a new coverage path with the number of rings set to four cycles for Avaya Workplace Client for iOS users. For other users, keep the existing coverage path with the number of rings set to the default two cycles.

- If an Avaya Workplace Client for iOS user already has a coverage path with the number of rings set to four cycles or higher, you do not need to edit this coverage path.
- If a user has both a deskphone and Avaya Workplace Client for iOS, the higher number of rings applies to both endpoints. That is, an incoming call rings for four cycles on the deskphone even if Avaya Workplace Client for iOS is not logged in.

For more information about Communication Manager settings, see *Administering Avaya Aura® Communication Manager*.

Configuration parameters for iOS applications

To enable the use of push notifications on applications, you must provide the following configuration parameters to iOS applications:

- TELEPHONY_PUSH_NOTIFICATION_ENABLED. This parameter is used to enable or disable push notifications. To enable push notifications, set this parameter to `1`. To disable push notifications, set this parameter to `0`.
- TELEPHONY_PUSH_NOTIFICATION_SERVICE_URL. This parameter is used to provide the Avaya Aura® Web Gateway address to applications. An example is *https://avaya-aura-web-gateway-hostname.company-domain.com*. If you are using a custom HTTPS port, the parameter must include this port number. For example: *https://avaya-aura-web-gateway-hostname.company-domain.com:8443*.

You can use one of the following methods to configure these parameters:

- Import the parameters to the Dynamic Configuration service on Avaya Aura® Device Services using the *dynamicConfigUpload.txt* file. For more information, see "Administration of the Dynamic Configuration service" in *Administering Avaya Aura® Device Services*.
- Use the *46xxsettings.txt* configuration file to push the parameters to clients.

The Dynamic Configuration service is the recommended method. Use this option if Avaya Aura® Device Services is available in your deployment.

Integrated Windows Authentication administration and management

Integrated Windows Authentication (IWA) enables you to log in to different services with the same credentials. To support IWA, some Avaya Aura® Web Gateway server administration is required. Users must be able to authenticate to the Avaya Aura® Web Gateway API using a preexisting authentication to a Windows domain. Avaya Aura® Web Gateway uses SPNEGO to negotiate authentication with the client and Kerberos to validate the authentication of the client user. User roles are retrieved normally through LDAP.

Use the following sections to complete IWA configuration on the Avaya Aura® Web Gateway and Active Directory servers. Errors in the setup might cause the authentication to fail. You can enable debug logs to assist with troubleshooting.

Avaya Aura® Web Gateway supports IWA for multiple domains. The User Principal Name (UPN) domain and the authentication domain must be the same as the root domain of the directory.

Authentication prerequisites

You must have the following to set up IWA:

- An Active Directory server.
- A DNS server for the DNS domain of Active Directory.

  For information about setting up the DNS server, see *Planning for and Administering Avaya Workplace Client for Android, iOS, Mac, and Windows*.
- A Windows client on the Active Directory domain.
- An Avaya Aura® Web Gateway server that is resolvable by the DNS.
- A domain user that is mapped to the Service Principal Name (SPN) of the Avaya Aura® Web Gateway server.
- Domain users for all individual users.
- The sAMAccountName attribute must match the user name part of the userPrincipalName attribute.

  For example, if the sAMAccountName is jdoe, then the userPrincipalName must use the following format: jdoe@<*domain.name*>.
- To log in to a computer, the user must enter the user name part of the userPrincipalName configured for that user. The domain must also match the domain part of that user userPrincipalName. The user login name format is `<domain>\<user name>`.

  For example, if the user has the "jdoe@avaya.com" userPrincipalName, where "avaya" is the domain and "jdoe" is the user name; then the user logs in to a computer using the "avaya\jdoe" account.

> Important:

- Do not change the userPrincipalName attribute configured for the user. If you change the userPrincipalName after IWA is configured, IWA will not work.
- The Active Directory, Windows client, and Avaya Aura® Web Gateway server must resolve each other FQDNs. However, they do not need to use the same DNS server or belong to the same zone.

Setting up the Windows Domain Controller

Use this procedure to add the Avaya Aura® Web Gateway SPN to a domain user on the Windows Domain Controller or the Active Directory server. The SPN must be unique across the domain. To avoid issues with duplicated SPNs, track of any SPNs assigned to users.

Avaya Aura® Web Gateway supports IWA for multiple domains. To configure IWA for multiple domains that are in different Active Directories, repeat this procedure on each Active Directory.

Important:
Enter all commands exactly as shown in this procedure, and use the following guidelines:

- The hostname used to access the Tomcat server must match the host name in the SPN exactly. Otherwise, authentication fails.
- The server must be part of the local trusted intranet for the client.
- The SPN must be formatted as `HTTP/<host name>` and must be exactly the same everywhere.
- The port number must not be included in the SPN.
- Only one SPN must be mapped to a domain user.
- The Kerberos realm is always the uppercase equivalent of the DNS domain name. For example, `EXAMPLE.COM`.
- Avaya Aura® Web Gateway supports IWA for parent and child domains. However, you cannot assign an SPN and generate a *tomcat.keytab* file for the child domain because the SPN can only be mapped to a single user in a forest. Here, you need to assign the SPN and generate a *tomcat.keytab* file for the parent domain.

1. Create a new IWA service account.
   ADDITIONAL INFORMATION:
   Do not select an account associated with an existing user.
2. Run the following command to attach the SPN to the domain name:
   ADDITIONAL INFORMATION:
   `setspn -S HTTP/<FRONT-END FQDN> <Domain user login>`

   In the following example, `<FRONT-END FQDN>` is `csa.example.com` and `<Domain user login>` is `csa.example.com`:

   ```
   setspn -S HTTP/csa.example.com csa_user
   ```

   Important:

> - If you are using Active Directory 2003, you must use `setspn -A` instead of `setspn -S`.
> - When you use `setspn -S`, the Active Directory server searches for other users with the same SPN assigned. If the server finds a duplicated SPN, see step See "3" on page 128.

3. To remove a duplicated SPN from another user, run the following command:
   ADDITIONAL INFORMATION:
   `setspn -d <SPN> <old user>`
4. Transfer the generated *tomcat.keytab* file to the Avaya Aura® Web Gateway server using the OAMP administration portal.
   ADDITIONAL INFORMATION:
   Since this is a credentials file, handle it securely and delete the original file after this file is imported into the Avaya Aura® Web Gateway server. You can generate and re-import a new *tomcat.keytab* file anytime.

Windows Domain Controller command descriptions

See "Setting up the Windows Domain Controller" on page 127 uses the following command values:

Command	Description	Example value
<FRONT—END FQDN>	The REST front host FQDN of the Avaya Aura® Web Gateway server. This is either the FQDN of the Virtual IP assigned to the cluster (if internal load balancing is used) or the FQDN of the external load balancer, if it is used.	`csa.example.com`
<Domain user login>	The Windows login ID for the domain user you created.	`csa_user`
<Kerberos realm>	The domain name for the Kerberos realm. The Kerberos realm is always the uppercase equivalent of the DNS domain name.	`EXAMPLE.COM`

Enabling encryption for the domain user

In FIPS mode, Kerberos uses Advanced Encryption Standard (AES). By default, support for Kerberos AES authentication is disabled for Active Directory users. Use this procedure to enable encryption support for the domain user that is mapped to the Avaya Aura® Web Gateway SPN.

This procedure is only required if FIPS is enabled on Avaya Aura® Web Gateway.
PREREQUISITE
Generate a keytab file as described in See "Setting up the Windows Domain Controller" on page 127.

1. In Active Directory, select the domain user that is mapped to the Avaya Aura® Web Gateway SPN.
   ADDITIONAL INFORMATION:
   For example: aads_spn_user.

2. Open the domain user properties.
3. Click the **Account** tab.
4. In the Account options area, do one of the following:
   - If you used the AES256–SHA1 encryption type when generating a keytab file, select the **This account supports Kerberos AES 256 bit encryption** check box.
   - If you used the AES128–SHA1 encryption type when generating a keytab file, select the **This account supports Kerberos AES 128 bit encryption** check box.
5. Click **OK**.

Setting up IWA on the Avaya Aura® Web Gateway administration portal

This procedure describes the changes you must perform on the Avaya Aura® Web Gateway administration portal to configure IWA.

To configure IWA for multiple domains, you must repeat this procedure for each active directory that requires the use of IWA.
PREREQUISITE
Generate a *tomcat.keytab* file for the active directory that you plan to use for IWA. If you want to configure IWA for multiple domains, you must generate a *tomcat.keytab* file for each active directory.

1. On the Avaya Aura® Web Gateway administration portal, click **LDAP Configuration**.
2. Select the required Active Directory.
3. In the Server Address and Credentials area, do the following:
   a. In the **Windows Authentication** menu, select **Negotiate**.
   b. In the Confirm Action dialog box, click **OK**.
   c. In **UID Attribute ID**, type `userPrincipalName`.
   ADDITIONAL INFORMATION:
   If this field is not set to `userPrincipalName`, you might encounter license issues and other unpredictable behavior.
   d. Ensure that the other settings are appropriate for the LDAP configuration of your Domain Controller.
   ADDITIONAL INFORMATION:
   > Important:
   > The LDAP server you use must be the domain controller with the appropriate Active Directory version as the server type.
4. In the Configuration for Windows Authentication area, complete the following information using the same values you provided when setting up the Windows Domain Controller:
   a. In **Service Principal Name (SPN)**, type `HTTP/<FRONT–END FQDN>`.
   For example, `HTTP/csa.example.com`.
   b. Click **Import** to import the *tomcat.keytab* file transferred from the Windows Domain Controller.
   STEP RESULT:
   In cluster deployments, the file is transferred to all nodes in the cluster. An additional option is available to send the file to specific nodes in a cluster.
   c. In **Kerberos Realm**, type the Kerberos realm, which is usually in all uppercase letters.
   For example, `EXAMPLE.COM`.

d. In **DNS Domain**, type the DNS domain of the Domain Controller.
   For example, `example.com`.
e. Select the **Use SRV Record** check box.
f. If **Use SRV Record** is not selected, in **KDC FQDN**, type the FQDN of the Domain Controller.
   ADDITIONAL INFORMATION:
   This value also includes the DNS domain at the end. For example, `ad.example.com`.
g. In **KDC Port**, retain the default value of 88.
   ADDITIONAL INFORMATION:
   This field is only visible if **Use SRV Record** is not selected.
h. In a cluster deployment, click **Send Keytab File** to send the *tomcat.keytab* file you imported in step See "4.b" on page 129 to a specific node.
   ADDITIONAL INFORMATION:
   This option is useful if the import to a node failed or if you add a new node to your cluster.

5. Click **Save** to retain the settings and restart the server.
   STEP RESULT:
   The updated settings are used to generate the files needed to configure the Tomcat JAASRealm and the corresponding Sun JAAS Login module for GSS Bind.

Virtual hardware adjustments

The following sections describe how to perform virtual hardware adjustments on a virtual machine.

For information about disk volume sizes, see the disk volume specifications in See "Virtual disk volume specifications" on page 131.

Virtual disk volume specifications

The following table shows the file system layout for systems deployed using Avaya-provided OVAs.

Disk Volume	Volume size can be increased	Volume Size (GiB)			
		Disk 1	Disk 2	Disk 3	Disk 4
/boot	No	0.2			
swap	Yes	8.0			
/	Yes	4.0			
/tmp	Yes	2.8			
/var	Yes	3.0			
/var/log	Yes	2.0			
/var/log/audit	Yes	3.0			
/home	Yes	4.0			
/opt/Avaya	Yes	15.0			
/var/log/Avaya	Yes		60.0		
/media/data	Yes			20.0	
/media/cassandra	Yes				10.0
Total for disk		42.0	60.0	20.0	10.0
Total disk size		132.0			

Increasing the size of a virtual disk

Each virtual disk holds one or more disk volumes. Before you can increase the size of a disk volume, you must first increase the size of the host disk to provide the required disk space.

This procedure describes how to adjust the size of a virtual disk in the Virtualization Enabled (VE) environment. The VE environment uses the standard VMware infrastructure facilities.
PREREQUISITE

- Ensure that the system layer on the virtual machine has been upgraded to the current release. You can verify this using the `sys versions` command.
- Delete all snapshots from the virtual machine. You cannot adjust disk sizes while snapshots exist.
- Determine the disk volume to be increased in size.
- Determine the disk number that hosts the disk volume. You can use the `sys volmgt --summary` command for more information.

1. If the virtual machine is installed and running, log in to the system, and shut down the operating system by running the following command:
   ADDITIONAL INFORMATION:
   ```
   sudo shutdown -h now
   ```
2. Stop your virtual machine if it is still running.
3. Click **Edit Settings**.
4. From the Hardware tab, select the hard disk to be enlarged.
5. In **Disk Provisioning**, enter a higher value for the disk size and select the appropriate unit of measure.
6. Click **OK**.
7. Power on the virtual machine.

AFTER COMPLETING THE TASK
Increase the size of the disk volume.

Increasing the virtual machine disk volume size

Use this procedure to increase the size of a disk volume. The upgrade process for an OVA from a previous release requires an increase in the size of one or more disk volumes.

In rare circumstances, Avaya support might recommend specific increments in disk volume sizes to address unexpected disk engineering issues.

> Important:
> You can only allocate free disk space to disk volumes if data encryption is disabled on Avaya Aura® Web Gateway. If data encryption is enabled, you cannot allocate free disk space.

PREREQUISITE
Increase the size of the virtual disks that host the volumes to be increased. This process makes new disk space available. For example, if the volume requires an additional 20.0 GiB of space and the host disk is currently 50.0 GiB, then you must change the size of the host disk to 70.0 GiB.

1. If the virtual machine is not running, then power it up.
2. Scan the disks on the virtual machine to detect newly available disk space by running the following command:
   ADDITIONAL INFORMATION:
   ```
   sys volmgt --scan
   ```

> Tip:
> For more information about this command, you can use the following commands:
>
> - For syntax help: `sys volmgt -h`
> - For verbose help: `sys volmgt -hh`

STEP RESULT:
After the scan is complete, an updated file system summary is displayed. The newly available disk space is reported in the Disk > Free column.

3. Allocate all of the unused space on the disk to the target volume by running the following command:
   ADDITIONAL INFORMATION:

   ```
   sys volmgt --extend <volume> --remaining
   ```

   For <volume>, specify the name of the volume as it appears in the Volume > Name column.

   All `--extend` operations are run as background tasks.

   a. To monitor the status of the operation in progress or of the last completed operation, run the following command:
   ADDITIONAL INFORMATION:

   ```
   sys volmgt --monitor less
   ```

   b. To gather all volume management logs into a zip file in the current working directory, run the following command:
   ADDITIONAL INFORMATION:

   ```
   sys volmgt --logs
   ```

   c. If a disk has multiple volumes and more than one volume is being increased in size, use one of the following commands to allocate a specific amount of unused space to a volume:
   ADDITIONAL INFORMATION:

   ```
   sys volmgt --extend <volume> <x>m
   sys volmgt --extend <volume> <x>g
   sys volmgt --extend <volume> <x>t
   ```

   In these commands, `m` means megabytes, `g` means gigabytes, `t` means terabytes, and <x> is a decimal number. For example, the following command increments the */var/log* volume by 10.5 GiB:

   ```
   sys volmgt --extend /var/log 10.5g
   ```

4. Verify that the new space has been allocated to the volume by running the following command:
   ADDITIONAL INFORMATION:

   ```
   sys volmgt --summary
   ```

   Due to disk overhead, the size of the volume reported under the Volume > LVM Size column will never exactly match the size reported under the Volume > File System > Size column.

a. If you suspect that the file system size is not correct, verify that the operation is complete by running the following command:
ADDITIONAL INFORMATION:

```
sys volmgt --status
```

b. If the status is reported as "Complete", you can correct the situation using `--extend` without an increment value:
ADDITIONAL INFORMATION:

```
sys volmgt --extend /var/log
```

This operation does not add more space to the volume that hosts the file system. Instead, it reissues the command to make full use of the current volume.

> Tip:
> Similar to using `--extend` to increase volume sizes, you can also monitor the `--extend` operation and gather logs using the following commands:
>
> ```
> sys volmgt --monitor less
> sys volmgt --logs
> ```

AWS-specific management options

If you deployed Avaya Aura® Web Gateway in an Amazon Web Services (AWS) environment, use the following sections to perform AWS-specific management operations. You can perform these management operations anytime.

Increasing the virtual disk size of AWS virtual machines

Use this procedure to increase the size of on an existing Avaya Aura® Device Services virtual machine deployed on AWS.

The following table shows the three virtual disks and the block devices of these disks, which are used by Avaya Aura® Device Services virtual machines on AWS.

Disk number	Block device	Desciption
Disk 1	*/dev/sda1*	Stores operating system and application software.
Disk 2	*/dev/xvdb*	Stores application logs.
Disk 3	*/dev/xvdc*	Stores application data.

> Note:
> You cannot reduce the disk size.

1. On the AWS console, navigate to **Services › Compute**, and then click **EC2**.
2. On the EC2 Management Console page, click **Instances**.
3. Select the instance to which you want to add storage.
   ADDITIONAL INFORMATION:
   AWS displays instance details.
4. Click the **Description** tab.
5. In the **Block devices** field, select a disk for which you want to increase the size.
   ADDITIONAL INFORMATION:
   The options are:
   - disk1
   - disk2
   - disk3

   > Note:
   > Avaya Aura® Device Services restart is required when you change the size of disk1. If you do not modify disk1, Avaya Aura® Device Services restart is not required.
6. In the **EBS ID** field, click the ID.
   ADDITIONAL INFORMATION:
   AWS displays the Volumes page with only the selected device.

7. To update the storage on the EBS disk volume, click **Actions › Modify Volume**.
8. In the Modify Volume window, in the **Size** field, enter the required disk size.
9. Click **Modify**.
10. In the Confirmation window, click **Yes**.
11. In the Status window, click **Close**.
12. If you updated disk1, do the following to restart the system for the changes to take effect:
    a. Log in to the virtual machine as an administrative user using an SSH connection.
    b. Run the `sudo reboot` command.
    ADDITIONAL INFORMATION:
    If Avaya Aura® Device Services is running when you run this command, then Avaya Aura® Device Services will be gracefully shut down.

    ADDITIONAL INFORMATION:

    > Note:
    > The process of increasing the disk size might take some time to complete. Monitor the size of the virtual disks and only proceed with any other disk-related activities after the size changes take effect.

AFTER COMPLETING THE TASK
Distribute the allocated free space between the logical volumes as described in Allocating unused disk space to logical volumes.

Block device descriptions

Block device on AWS	Disk number	Description
/dev/sda1	disk1	Stores operating system and application software.
/dev/xvdb	disk2	Stores application logs.
/dev/xvdb	disk3	Stores application data.
/dev/xvdd	disk4	Stores database commit logs.

Updating an existing stack with a new CloudFormation template

You can apply changes to an existing CloudFormation stack by updating the stack with a newer CloudFormation template. Changes to the stack can include additional nodes, new resources, and new port configuration. The system updates all the objects contained in the stack to match the new settings. Existing EBS volumes and S3 buckets are preserved.

To update an existing single-node stack you must use a new single-node stack template. To update an existing cluster you must use a new multi-node stack template. If you are expanding a cluster, you must already have a cluster with two or more nodes. You cannot expand a single AWS node into an AWS cluster
PREREQUISITE

Generate a new CloudFormation template that matches the application and profile of the existing system but includes the additional resources required. For more information, see the "Creating CloudFormation templates" section in *Deploying the Avaya Aura® Web Gateway*.

1. Sign in to the AWS console.
2. Navigate to **Services** › **Management Tools** › **CloudFormation**.
3. Select the stack to update.
4. Click **Actions** › **Update Stacks**.
5. Update the stack using the Update Stack pages.
   ADDITIONAL INFORMATION:
   You can add an additional CIDR block when going from two to three subnets. Do not change the value of any other stack parameters.

Deleting a CloudFormation stack

1. Sign in to the AWS console and click **Services** › **Management Tools** › **CloudFormation**.
2. On the CloudFormation page, select a stack to delete.
3. Click **Actions** › **Delete Stack**.
4. Click **Yes, Delete** when prompted.
   ADDITIONAL INFORMATION:
   The AMI virtual machines and other resources that are in the stack are deleted.

   > Note:
   > The stack is not completely deleted if it includes Amazon Route 53. See the next step for information about resolving this issue.

5. If deleting the stack fails due to the presence of an Amazon Route 53 DNS zone, then do the following:
   a. Navigate to **Services** › **Networking & Content Delivery** › **Route 53** › **Hosted zones**.
   b. Select the Hosted Zone that is no longer required and click **Delete Hosted Zone**.
   c. Click **Confirm**.
   d. Navigate to **Services** › **Management Tools** › **CloudFormation**.
   e. Select the stack that previously failed to delete.
   f. Click **Actions** › **Delete Stack**.
   g. Click **Yes, Delete**.

Avaya Aura® Web Gateway network settings configuration

Starting from Release 3.11, you can modify the network settings of an Avaya Aura® Web Gateway node from a single location by running the **cliChangeNet.sh** script from the node CLI. You can configure the following settings:

- IP address of the node
- Subnet mask
- Host name of the node
- Domain name of the node
- Default gateway IP address
- DNS server IP address
- Default search list
- NTP server IP or FQDN
- Time zone
- Country

The following are examples of scenarios where you can use the **cliChangeNet.sh** script:

- Re-configuring network settings for your Avaya Aura® Web Gateway standalone system or cluster.
- Updating network configuration for an Avaya Aura® Web Gateway node. For example, when moving one or several Avaya Aura® Web Gateway nodes to another data center.
- Changing your DNS or NTP servers.

cliChangeNet.sh script limitations

- You cannot use the **cliChangeNet.sh** script in AWS OVA-based deployments.
- You cannot use the **cliChangeNet.sh** script to change the following Avaya Aura® Web Gateway network settings:
  - Front-end FQDN
  - Virtual IP address

  You also cannot use **cliChangeNet.sh** to update the System Manager FQDN.

  To update these settings, use the Avaya Aura® Web Gateway configuration utility.
- If you change the IP addresses or FQDN of an Avaya Aura® Web Gateway node, you cannot restore the data using backups created before you made these network changes. This limitation applies to manual and automatic backups.

- If you perform a rollback after changing Avaya Aura® Web Gateway IP addresses or FQDNs, Avaya Aura® Web Gateway continues to use these new IP addresses or FQDNs after the rollback. The rollback process does not revert IP addresses or FQDN changes.

Avaya Aura® Web Gateway network configuration checklist

Perform the following tasks to configure Avaya Aura® Web Gateway network settings:

No.	Task	Description	✔
1	Prepare your virtual machines for configuring network settings. Back up Avaya Aura® Web Gateway.	See See "Prerequisites for configuring Avaya Aura Web Gateway network settings" on page 139.	
2	Configure network settings using the **cliChangeNet.sh** script.	See See "Configuring network settings using the cliChangeNet.sh script" on page 140. In a cluster environment, perform this procedure on the seed node first and then on all non-seed nodes.	
3	Configure network settings using the Avaya Aura® Web Gateway configuration utility.	You can configure certain Avaya Aura® Web Gateway network settings only using the configuration utility. You might also need to perform additional configuration steps depending on the changes you make in the previous step. For more information, see See "Configuring network settings using the Avaya Aura Web Gateway configuration utility" on page 141.	

Prerequisites for configuring Avaya Aura® Web Gateway network settings

There are several prerequisites for configuring Avaya Aura® Web Gateway network settings. These prerequisites are not required if you only need to update the DNS or NTP server IP addresses.

- If you plan to change the IP address or FQDN of an Avaya Aura® Web Gateway node, add a new element for this node on System Manager on the **Services › Inventory › Manage Elements** page.
- If you plan to change IP addresses or FQDNs of Avaya Aura® Web Gateway nodes and you do not use DNS servers in your deployment, add host names of all Avaya Aura® Web Gateway nodes to the */etc/hosts/* file on System Manager.

- If you plan to change IP addresses or FQDNs of Avaya Aura® Web Gateway nodes and you use third-party CA identity certificates, obtain new certificates with appropriate FQDNs from your CA.
- If you plan to change the IP address of an Avaya Aura® Web Gateway node, ensure that the new IP addresses are available in the appropriate subnet.
- If you plan to change the IP address of an Avaya Aura® Web Gateway node, ensure that the network adapter or virtual switch of the Avaya Aura® Web Gateway virtual machine is configured with the appropriate subnet for this IP address.

Configuring network settings using the cliChangeNet.sh script

Use this procedure to configure network settings of an Avaya Aura® Web Gateway node.

This procedure applies to OVA-based and software-only deployment types.

> Important:
>
> - You cannot use the **cliChangeNet.sh** script in AWS OVA-based deployments.
> - Avaya Aura® Web Gateway is unavailable when applying new network settings.

In a cluster deployment, perform this procedure on each cluster node starting from the seed node.
PREREQUISITE
Review and complete prerequisites listed in See "Prerequisites for configuring Avaya Aura Web Gateway network settings" on page 139.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Run the following command to navigate to the script location:
   ADDITIONAL INFORMATION:
   ```
   cdto misc
   ```
3. Run the following command:
   ADDITIONAL INFORMATION:
   ```
   sudo ./cliChangeNet.sh
   ```
4. Configure network settings as appropriate:
   - To change the existing value, enter a new value and press `Enter`.
   - To keep the existing value, do not enter anything and press `Enter`.
5. When prompted to apply the changes, do one of the following:
   - To confirm the changes, enter `Y`.
   - To discard any changes you made in the network configuration, enter `N`.
6. After the configuration process is complete, run the following command to restart Avaya Aura® Web Gateway:
   ADDITIONAL INFORMATION:
   ```
   svc csa restart
   ```
7. In a cluster deployment, repeat the previous steps on all remaining nodes.

AFTER COMPLETING THE TASK
Complete network configuration as described in See "Configuring network settings using the Avaya Aura Web Gateway configuration utility" on page 141.

Configuring network settings using the Avaya Aura® Web Gateway configuration utility

You must use the Avaya Aura® Web Gateway configuration utility to configure the following network settings:

- Front-end FQDN
- Virtual IP address
- System Manager FQDN
- System Manager IP addresses

You do not need to perform this procedure if you only change DNS or NTP server IP addresses.

1. Log in to the Avaya Aura® Web Gateway as an administrator.
2. Run the following command to start the configuration utility:
   ADDITIONAL INFORMATION:
   `app configure`
3. Do the following to update System Manager and front-end FQDN settings:
   a. Navigate to **Front-end host, System Manager and Certificate Configuration settings**.
   b. In **System Manager FQDN**, type the System Manager FQDN.
   c. In **System Manager Enrollment Password**, type the enrollment password of the System Manager.
   d. In **Keystore password**, type the Keystore password you used when installing Avaya Aura® Web Gateway.
   e. If you updated the front-end FQDN of the virtual IP address, in **Front-end FQDN**, type the new FQDN.
   f. If you updated the local FQDN of the node, in **Local frontend host**, type the new FQDN of the node.
   g. Click **OK** and then click **Apply**.
4. If you updated the virtual IP address of Avaya Aura® Web Gateway, do the following:
   a. Navigate to **Clustering Configuration › Virtual IP Configuration** menu.
   b. In **Virtual IP address**, type the new virtual IPv4 address.
   c. Click **OK** and click **Apply**.

   ADDITIONAL INFORMATION:

   > Note:
   > Perform this step on the seed and backup nodes only. Do not perform this step on other non-seed nodes.
5. If you use third-party identity certificates, import certificates with appropriate FQDNs to Avaya Aura® Web Gateway.
   ADDITIONAL INFORMATION:
   If you use System Manager certificates, these certificates are automatically re-generated when you perform step See "3" on page 141.

6. In **Main Menu**, click **Continue** and click **Yes**.
7. When the configuration process completes, click **Continue**.
8. In a cluster deployment, perform steps See "2" on page 141 to See "7" on page 142 on all remaining nodes of the cluster.
9. In a cluster deployment, on each cluster node, run the `app listnodes` command and ensure that all cluster nodes have the "live" status.
10. In a cluster deployment, configure RSA public and private keys for internode SSH communication.

AFTER COMPLETING THE TASK

If you have components in your deployment that use Avaya Aura® Web Gateway FQDNs or IP addresses, update these settings accordingly. For example, the following components can use Avaya Aura® Web Gateway FQDNs or IP addresses:

- Avaya Aura® Device Services uses the Avaya Aura® Web Gateway FQDN and IP address in trusted hosts.
- System Manager uses Avaya Aura® Web Gateway FQDNs on the Elements page.
- External load balancer uses the Avaya Aura® Web Gateway FQDN in its certificate.
- Avaya Session Border Controller for Enterprise uses the Avaya Aura® Web Gateway IP address and FQDN.
- Avaya Meetings Management uses the Avaya Aura® Web Gateway IP address.

Security options

Data encryption management

When you deploy the Avaya Aura® Web Gateway OVA, you can enable data encryption and configure basic settings, such as providing a passphrase or enabling the local key store. After deploying the Avaya Aura® Web Gateway OVA, you can use system layer commands to perform additional data encryption management operations, such as the following:

- Enabling a remote key server.
- Managing encryption passphrases.
- Reviewing data encryption status.

> Important:
> In AWS deployments, you enable data encryption on AWS itself. Therefore, you cannot use the system layer commands for disk encryption management in AWS deployments.

Enabling or disabling disk encryption on Avaya Aura® Web Gateway

You cannot enable or disable disk encryption by using the Avaya Aura® Web Gateway web administration portal or the configuration utility. However, you can enable or disable disk encryption when restoring Avaya Aura® Web Gateway from a backup. This procedure provides the high-level steps for enabling or disabling disk encryption.

> Important:
>
> - This procedure only applies to OVA-based deployments. You cannot enable disk encryption for software-based deployments.
> - For AWS deployments, you cannot enable disk encryption on Avaya Aura® Web Gateway. You must enable disk encryption on AWS itself. For more information, see How to Protect Data at Rest with Amazon EC2 Instance Store Encryption.

This procedure applies to cluster and standalone Avaya Aura® Web Gateway deployments.

1. Perform a full system backup:
   a. Back up each node of your Avaya Aura® Web Gateway deployment.
   b. Record configuration values from the existing Avaya Aura® Web Gateway system.
   c. Copy logs, home directory content, and backup files to an external storage.

   ADDITIONAL INFORMATION:
   For more information, see See "Backing up Avaya Aura Web Gateway" on page 179.
2. Deploy new virtual machines.
   ADDITIONAL INFORMATION:
   You can enable or disable disk encryption as needed when deploying the Avaya Aura® Web Gateway OVA. For information about the deployment procedures, see "OVA deployment" in *Deploying the Avaya Aura® Web Gateway*.

> Important:
>
> - You must use the same OVA and the same version of the Avaya Aura® Web Gateway application that was backed up.
> - You must use the original network settings, such as host names and IP addresses, from the currently installed cluster when deploying new virtual machines.

3. Restore the Avaya Aura® Web Gateway application using the backups created in step 1.
   ADDITIONAL INFORMATION:
   For information about restoring Avaya Aura® Web Gateway, see See "Restoring Avaya Aura Web Gateway in a standalone environment" on page 182 or See "Restoring an entire Avaya Aura Web Gateway cluster" on page 184.

Remote key server management

When configuring data encryption during the OVA deployment, if you choose not to enter a passhrapse after every reboot, Avaya Aura® Web Gateway stores encryption keys in a local key store. To enhance security, you can set up a remote key server. You can add multiple remote key servers.

Enabling a remote key server

Use this procedure to use a remote key sever to store encryption keys.

When adding a remote key server for the first time, you can choose either to continue using the local key store or to disable it. When both the local key store and the remote key server are enabled at the same time, Avaya Aura® Web Gateway uses the local key store to decrypt the encrypted disks at boot time.

If you already have a remote key server enabled, you can use this procedure to add another remote key server.
PREREQUISITE
Configure your remote key server. The exact configuration procedure depends on the key server you are using.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Run the following command:
   ADDITIONAL INFORMATION:
   ```
   sys encryptionRemoteKey add <server address> <port>
   ```
   In this command:
   - `<server address>` is the IP address or FQDN of the remote key server.
   - `<port>` is a port that the remote key server uses to connect to Avaya Aura® Web Gateway. This is an optional value. If you do not enter a port number, the remote key server uses port
     80
     by default.
3. When prompted, enter the existing passphrase.
4. When prompted to remove the local key store, do one of the following:
   - To disable the local key store, enter `y`.

- To continue using the local key store, enter n.

Disabling a remote key server

Use this procedure if you do not want to use a remote key server to store encryption keys. If you have multiple remote key stores, Avaya Aura® Web Gateway will continue using other remote key stores until you disable them.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Run the following command:
   ADDITIONAL INFORMATION:
   `sys encryptionRemoteKey remove <server address>`

   In this command, `<server address>` is the IP address or FQDN of the remote key server you want to disable. You must use the same IP address or FQDN that you used when enabling the remote key server.

Passphrase management

Avaya Aura® Web Gateway requires an encryption passphrase to access encrypted partitions.

When enabling data encryption during the Avaya Aura® Web Gateway OVA deployment, you set up a single passphrase. After the OVA is deployed, you can use system layer commands to:

- Configure up to seven additional passphrases.
- Change existing passphrases.
- Remove a passphrase.
- View status of passphrases.

Adding a passphrase

Use this procedure to set up additional passphrases. You can have up to seven passphrases in total. If you have multiple passphrases, you can use any of them to access encrypted disk partitions.

A passphrase you add must comply with the passphrase complexity rules.
PREREQUISITE
Ensure that you have a passphrase slot available. For more information, see See "Reviewing the passphrase status" on page 147.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Run the following command:
   ADDITIONAL INFORMATION:
   `sys encryptionPassphrase add`
3. When prompted, type one of the existing passphrases.
4. When prompted, type a new passphrase.
5. Type the new passphrase again to confirm it.

Passphrase complexity rules

All encryption passphrases you add to Avaya Aura® Web Gateway must contain at least:

- Eight characters.
- One character from each of the following character sets:
    - Uppercase letters:
      A
      to
      Z
    - Lowercase letters:
      a
      to
      z
    - Numerics:
      0
      to
      9
    - Special characters:
      !
      ,
      @
      ,
      #
      ,
      %
      ,
      $
      ,
      ^
      ,
      *
      ,
      ?
      , and
      _

Removing a passphrase

Use this task to remove an existing encryption passphrase.

Avaya Aura® Web Gateway requires at least one encryption passphrase. If you have only one passphrase, you cannot remove it.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Run the following command:
   ADDITIONAL INFORMATION:
   `sys encryptionPassphrase remove`
3. When prompted, type the encryption passphrase you want to remove.

Reviewing the passphrase status

Use this procedure to view information about the number of passphrases you have on Avaya Aura® Web Gateway.

> Tip:
> You can use this procedure before adding a new passphrase to see if Avaya Aura® Web Gateway has a free slot for a new passphrase.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Run the following command:
   ADDITIONAL INFORMATION:
   `sys encryptionPassphrase list`
   STEP RESULT:
   Avaya Aura® Web Gateway displays passphrase and slot assignment information.

```
Slot         Status      Passphrase/Remote Server
------------------------------------------------
Key Slot 0: ENABLED     Passphrase
Key Slot 1: ENABLED     Passphrase
Key Slot 2: ENABLED     Passphrase
Key Slot 3: DISABLED    empty
Key Slot 4: DISABLED    empty
Key Slot 5: DISABLED    empty
Key Slot 6: DISABLED    empty
Key Slot 7: DISABLED    empty
```

Viewing data encryption status

Use this procedure to view information about data encryption, including the following:

- Whether data encryption is enabled.
- Whether the local key store is enabled.
- Whether the encryption passphrase must be entered after every reboot.
- Information about configured remote key servers.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Run the `sys encryptionStatus` command.

Local key store management

Enabling the local key store

If you do not want to enter an encryption passphrase after every Avaya Aura® Web Gateway reboot and you do not have a remote key server, you can enable the local key store. However, the local key store is less secure than a remote key store or entering a passphrase after every reboot.

You can enable the local key store even if a remote key server is enabled. When both the local key store and the remote key server are enabled at the same time, Avaya Aura® Web Gateway uses the local key store to decrypt the encrypted disks at boot time.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Run the following command:
   ADDITIONAL INFORMATION:
   `sys encryptionLocalKey enable`
3. When prompted, enter the encryption passphrase.

Disabling the local key store

When you disable the local key store and do not have a remote key server, you will need to enter your passphrase after every Avaya Aura® Web Gateway reboot. If at least one remote key server is configured, then Avaya Aura® Web Gateway will use this remote key server to store encryption keys.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Run the following command:
   ADDITIONAL INFORMATION:
   `sys encryptionLocalKey disable`

Advanced Intrusion Detection Environment tool management

Advanced Intrusion Detection Environment (AIDE) is a tool for monitoring file system changes. AIDE creates a baseline database of system layer files and then verifies the integrity of the files by comparing the database to the actual state of the system layer files. AIDE only performs file integrity checks and does not provide other checks, such as rootkit searches.

> Important:
> Software-only deployments do not include the AIDE tool. If you want to use AIDE on Avaya Aura® Web Gateway deployed as a software-only application, you must install AIDE separately.

On Avaya Aura® Web Gateway, you can manage AIDE using the **fsit.sh** system layer script, which is located in the */opt/Avaya/bin* directory. The **fsit.sh** script is available after you deploy the Avaya Aura® Web Gateway OVA.

Avaya recommends that you run AIDE scanning at least once a week.

Creating a baseline database

Use this procedure to create a baseline database of operational system and Avaya Aura® Web Gateway files. AIDE uses this database as a reference when performing file system integrity scans.

You must create a database before you start running AIDE scans. Do not create a new database before each scan. Create a new baseline after you installed Avaya Aura® Web Gateway and then create a new baseline each time after you upgraded Avaya Aura® Web Gateway.

The database creation process can take up to 50 minutes depending on the system data.

1. Log in to the virtual machine with the deployed Avaya Aura® Web Gateway OVA as an administrator.
2. Run the following command to navigate to the **fsit.sh** script location:
   ADDITIONAL INFORMATION:
   `cd /opt/Avaya/bin`
3. Run the following command to create an AIDE database as a background process:
   ADDITIONAL INFORMATION:
   `sudo ./fsit.sh -updateDB &`

   Creating the database as a background process enables you to perform other tasks at the same time.

Result

AIDE creates the baseline database and records the results in the */var/log/aide/aide_update_status.log* file. If the database is created successfully, the log contains the Completed entry and a time stamp. Otherwise, the log contains the Failed entry. You can view the log file content using the **cat** command.

The following is an example of the **cat** command output when the baseline database is created successfully.

```
[admin@aads68 bin]$ sudo cat /var/log/aide/aide_update_status.log
Completed (Wed Dec 11 15:27:24 IST 2019)
[admin@aads68 bin]$
```

AIDE scanning

During the scanning process, AIDE compares the actual state of the file system with the reference values stored in the baseline database and informs you if there are any differences. You can either run AIDE scanning manually or configure automatic scanning. AIDE scanning can take up to 50 minutes depending on the system data. AIDE records the scanning results to the */var/log/aide/aide_scan_status.log* file.

You must run AIDE scans on a regular basis.

Excluding files and directories from AIDE scanning

By default, AIDE checks the integrity of all Avaya Aura® Web Gateway files and directories, except for the following:

- */opt/spirit/LogTail**
- */opt/spirit/logging/*
- */opt/spirit/persist/persisted_ids.properties*

Avaya Aura® Web Gateway regularly updates certain files and directories, such as */opt/Avaya/CallSignallingAgent/<version>/logs/*. AIDE includes all changed files in the scanning report, regardless of the reason for the change. This might result in false positive alarms in the scan report. You can add certain files and directories you want to exclude from scanning to the */etc/aide.conf* AIDE configuration file. AIDE will not include these files and directories in scanning reports.

> Other:
> Be cautions when excluding a file or directory from scanning because an intruder might place a malicious file in a place that AIDE does not check.

1. Log in to the virtual machine with the deployed Avaya Aura® Web Gateway OVA as an administrator.
2. Open the */etc/aide.conf* file in a text editor.
   ADDITIONAL INFORMATION:
   For example, to open the file in vi, run the `vi /etc/aide.conf` command.
3. For each file or directory you want to exclude from scanning, add an entry in the following format:
   ADDITIONAL INFORMATION:
   `!<path to file or directory>`

   In this entry, `<path to file or directory>` is the absolute path to the file or directory you want to exclude. For example, enter `!/opt/Avaya/CallSignallingAgent/3.8.0.0.001/logs/` to exclude the entire directory from the report.

   If you want to exclude a single file, add the `$` character at the end of the entry. For example: `!/opt/Avaya/CallSignallingAgent/3.8.0.0.001/logs/File.log$`

   The entry format supports regular expressions. For example, you can enter `!/opt/Avaya/CallSignallingAgent/3.8.0.0.001/logs/File*` to exclude only files and directories that have names starting with *File*.
4. Save the */etc/aide.conf* file.

Running AIDE scanning manually

Use this procedure to run AIDE scanning manually. Scanning can take up to 50 minutes, depending on the system data. Avaya recommends that you run AIDE scanning at least once a week.

Run AIDE scanning during the least busy periods to minimize the impact on Avaya Aura® Web Gateway performance.
PREREQUISITE
Ensure that you created a baseline database for your current Avaya Aura® Web Gateway release.

1. Log in to the virtual machine with the deployed Avaya Aura® Web Gateway OVA as an administrator.
2. Run the following command to navigate to the **fsit.sh** script location:
   ADDITIONAL INFORMATION:
   `cd /opt/Avaya/bin`
3. Run the following command to run AIDE scanning as a background process:
   ADDITIONAL INFORMATION:
   `sudo ./fsit.sh -scanNow &`

Result
AIDE checks the file system integrity and records the results of the scan to the */var/log/aide/aide_scan_report.log* file.
AFTER COMPLETING THE TASK
Review the scanning results.

Configuring automatic scanning

Use this procedure to configure automatic AIDE scanning. Automatic scanning is performed once a day. Scanning can take up to 50 minutes, depending on the system data.

Schedule AIDE scanning for the least busy periods to minimize the impact on Avaya Aura® Web Gateway performance.

PREREQUISITE

Ensure that you created a baseline database for your current Avaya Aura® Web Gateway release.

1. Log in to the virtual machine with the deployed Avaya Aura® Web Gateway OVA as an administrator.
2. Run the following command to navigate to the **fsit.sh** script location:
   ADDITIONAL INFORMATION:
   `cd /opt/Avaya/bin`
3. Run the following command to configure automatic scanning:
   ADDITIONAL INFORMATION:
   `sudo ./fsit.sh -enableScan <hour> <minute>`

   Replace `<hour>` and `<minute>` with the time when you want to run the scan process. Use the 24–hour time format. For 12 a.m., enter `0`.

   For example, enter `sudo ./fsit.sh -enableScan 16 00` to run scanning at 4:00 p.m.

Result

AIDE checks the file system integrity once a day and records the results of the scan to the */var/log/aide/aide_scan_report.log* file.

AFTER COMPLETING THE TASK

Review the scanning results.

Disabling automatic scanning

Use this procedure to disable automatic AIDE scanning.

1. Log in to the virtual machine with the deployed Avaya Aura® Web Gateway OVA as an administrator.
2. Run the following command:
   ADDITIONAL INFORMATION:
   `cd /opt/Avaya/bin`
3. Run the following command:
   ADDITIONAL INFORMATION:
   `sudo ./fsit.sh -disableScan`

Reviewing the AIDE scanning report

Use this procedure to review the results of the latest AIDE scan. The report contains the following:

- The baseline database status.
- Information about the automatic scanning configuration.
- The last database update and scan dates.

- The number of scanned files and directories.
- The number of added, removed, or changed files.
- Information about added, removed, or changed files.
- Detailed Information about changes.

1. Log in to the virtual machine with the deployed Avaya Aura® Web Gateway OVA as an administrator.
2. Run the following command to navigate to the **fsit.sh** script location:
   ADDITIONAL INFORMATION:
   `cd /opt/Avaya/bin`
3. Run the following command:
   ADDITIONAL INFORMATION:
   `sudo ./fsit.sh -status`
   STEP RESULT:
   The scan results are displayed in the terminal window.

The following example shows a scan report with differences between the baseline database and the actual system state found:

```
AIDE 0.15.1 found differences between database and filesystem!!
Start timestamp: 2019-12-11 15:26:52

Summary:
  Total number of files:        55724
  Added files:                  0
  Removed files:                0
  Changed files:                1


---------------------------------------------------
Changed files:
---------------------------------------------------

changed: /opt/Avaya/bin/fsit.sh

---------------------------------------------------
Detailed information about changes:
---------------------------------------------------


File: /opt/Avaya/bin/fsit.sh
 SHA512   : 7v0sOtK9erPSw8dw8GLTCacbvdHO28qj , bdxUm2J38kb/1yfDVO/KiA+55CeJ1hfX
```

The following example shows a scan report with no differences found:

```
AIDE, version 0.15.1

### All files match AIDE database. Looks okay!
```

CylancePROTECT antivirus software overview

Avaya Aura® Web Gateway supports the CylancePROTECT antivirus software. CylancePROTECT is an antivirus tool that uses mathematical models and machine learning algorithms instead of reactive signatures and databases to detect and neutralize malicious software. CylancePROTECT can identify threat on the operating system and memory layers.

To use CylancePROTECT, you must install the CylancePROTECT Agent on each Avaya Aura® Web Gateway node in your deployment. For information about installation, configuration, and usage of CylancePROTECT, see the CylancePROTECT Administration Guide.

Avaya Aura® Web Gateway supports CylancePROTECT version 2.1.1570 (Agent version 1570) or higher.

Out-of-band management

By default, Avaya Aura® Web Gateway uses a single network interface for all traffic types. For additional security, you can use the out-of-band management functionality to isolate client traffic from system management traffic by using different network interfaces for these traffic types.

When you enable out-of-band management on Avaya Aura® Web Gateway, the management access on the public interface is disabled. You can administer Avaya Aura® Web Gateway using the configured management interface. End users cannot access the management interface. This prevents unauthorized login attempts from the end user network.

You can configure out-of-band management after installing or upgrading Avaya Aura® Web Gateway.

Out-of-band management configuration checklist

No.	Task	Notes	✔
1	Add and configure a second network interface.	See See "Configuring a second network interface" on page 153.	
2	Configure out-of-band management settings on the Avaya Aura® Web Gateway administration portal.	See See "Configuring out-of-band management settings" on page 157.	
3	If you configure out-of-band management in a cluster deployment, update internode SSH communication settings.	See See "Configuring RSA public and private keys for SSH connections in a cluster" on page 186.	
4	Generate identity certificates for the new interface.	See See "Generating identity certificates for the second network interface" on page 158.	

Configuring a second network interface

You can add a second network interface by editing your virtual machine settings. Then you can configure this interface using RHEL CLI commands.

In a cluster deployment, perform this procedure on all nodes in the cluster.
PREREQUISITE

- Prepare the following data for the second network interface:

    - IP address. In a cluster deployment, you must use a unique IP address for each node.
    - Network mask
    - Gateway address

- Enable EASG functionality. This is required to obtain the sroot user privileges, which are required to perform this procedure.

1. Log in to Avaya Aura® Web Gateway using an SSH connection an administrator.
2. To obtain the sroot user privileges, run the following command:
   ADDITIONAL INFORMATION:

   ```
   su - sroot
   ```

3. Run the `ip address show` command to ensure that you have only one network interface.
   ADDITIONAL INFORMATION:
   For example:

   ```
   [admin@aawg ~]$ ip address show
   1: lo:  <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
   UNKNOWN qlen 1000
       link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
       inet 127.0.0.1/8 scope host lo
          valid_lft forever preferred_lft forever
       inet6 ::1/128 scope host
          valid_lft forever preferred_lft forever
   2: eth0:  <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq
   state UP qlen 1000
       link/ether 00:0C:29:a8:86:79 brd ff:ff:ff:ff:ff:ff
       inet 192.0.2.26/24 brd 192.0.2.255 scope global eth0
          valid_lft forever preferred_lft forever
       inet6 2001:db8::fea8:8679/64 scope link
          valid_lft forever preferred_lft forever
   ```

   In this example, `lo` is the loopback interface and `eth0` is the only network interface.

4. Do the following to enable Avaya Aura® Web Gateway to process packets from any network:
   a. Open the */etc/sysctl.conf* file in a text editor.
      ADDITIONAL INFORMATION:
      For example: `vi /etc/sysctl.conf`
   b. Find the `net.ipv4.conf.all.rp_filter` parameter and change its value to `2`.
      ADDITIONAL INFORMATION:

      ```
      net.ipv4.conf.all.rp_filter = 2
      ```

   c. Save the file.
   d. Run the `reboot` command to restart Avaya Aura® Web Gateway.
   e. Run the following command to verify the updated configuration:
      ADDITIONAL INFORMATION:
      `sysctl -a 2>/dev/null | grep "\.rp_filter"`

      The following is an example of the command output:

```
[admin@aawg ~]$ sysctl -a 2>/dev/null | grep "\.rp_filter"
net.ipv4.conf.all.rp_filter = 2
net.ipv4.conf.default.rp_filter = 1
net.ipv4.conf.eth0.rp_filter = 1
net.ipv4.conf.lo.rp_filter = 0
```

5. On the VMware virtual machine where Avaya Aura® Web Gateway is deployed, add a new network interface.
   ADDITIONAL INFORMATION:
   For information about adding network interfaces on VMware, see the documentation for the VMware version you used to deploy Avaya Aura® Web Gateway.
6. Run the `ip address show` command to ensure that you added a second network interface.
   ADDITIONAL INFORMATION:
   For example:

   ```
   [admin@aawg ~]$ ip address show
   1: lo:  <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
   UNKNOWN qlen 1000
       link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
       inet 127.0.0.1/8 scope host lo
          valid_lft forever preferred_lft forever
       inet6 ::1/128 scope host
          valid_lft forever preferred_lft forever
   2: eth0:  <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq
   state UP qlen 1000
       link/ether 00:0C:29:a8:86:79 brd ff:ff:ff:ff:ff:ff
       inet 192.0.2.26/24 brd 192.0.2.255 scope global eth0
          valid_lft forever preferred_lft forever
       inet6 2001:db8::fea8:8679/64 scope link
          valid_lft forever preferred_lft forever
   3: eth1:  <BROADCAST,MULTICAST> mtu 1500 noop state DOWN qlen
   1000
       link/ether 00:0C:29:a8:86:80 brd ff:ff:ff:ff:ff:ff
   ```

   In this example, `eth0` is the existing network interface and `eth1` is the second network interface you added in the previous step.
7. Run the following command to copy the configuration file for the second network interface from the configuration file for the existing network interface:
   ADDITIONAL INFORMATION:
   `cp /etc/sysconfig/network-scipts/ifcfg-eth0 /etc/sysconfig/network-scripts/ifcfg-eth1`

   The following is an example of the configuration file for the existing network interface:

   ```
   [admin@aawg ~]$ cat /etc/sysconfig/network-scripts/ifcfg-eth0
   DEVICE=eth0
   BOOTPRO=static
   ONBOOT=yes
   TYPE=Ethernet
   IPADDR=192.0.2.26
   NETMASK=255.255.255.0
   GATEWAY=192.0.2.1
   ```

8. Open the */etc/sysconfig/network-scripts/ifcfg-eth1* configuration file for the second network interface in a text editor.
   ADDITIONAL INFORMATION:
   For example: `vi /etc/sysconfig/network-scripts/ifcfg-eht1`
9. Modify the following parameters:
   a. DEVICE: Enter `eth1`.
   b. IPADDR: Enter the IP address of the second network interface.
   c. NETMASK: Enter the subnetwork mask of the second network interface IP address.
   d. GATEWAY: Enter the gateway address.

   ADDITIONAL INFORMATION:
   The following is an example of the updated configuration file for the second network interface:

```
[admin@aawg ~]$ cat /etc/sysconfig/network-scripts/ifcfg-eth1
DEVICE=eth1
BOOTPRO=static
ONBOOT=yes
TYPE=Ethernet
IPADDR=198.51.100.29
NETMASK=255.255.255.0
GATEWAY=198.51.100.1
```

10. Save the file.
11. Run the `sudo systemctl restart NetworkManager` command to apply changes.
12. Run the `ip address show` command to verify that the second network interface is configured.
    ADDITIONAL INFORMATION:
    For example:

```
[admin@aawg ~]$ ip address show
1: lo:  <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0:  <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq
state UP qlen 1000
    link/ether 00:0C:29:a8:86:79 brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.26/24 brd 192.0.2.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 2001:db8::fea8:8679/64 scope link
       valid_lft forever preferred_lft forever
3: eth1:  <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq
state UP qlen 1000
    link/ether 00:0C:29:a8:86:80 brd ff:ff:ff:ff:ff:ff
    inet 198.51.100.29/24 brd 198.51.100.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 2001:db8::fea8:8680/64 scope link
       valid_lft forever preferred_lft forever
```

13. From another computer, run the `ping` command to ensure that both network interfaces are accessible:
    ADDITIONAL INFORMATION:
    `ping <FIRST NETWORK INETRFACE IP ADDRESS>`

    `ping <SECOND NETWORK INTERFACE IP ADDRESS>`

    For example:

    ```
    ping 192.0.2.26
    ping 198.51.100.29
    ```

Configuring out-of-band management settings

After configuring a second network interface, you can configure out-of-band management settings on the Avaya Aura® Web Gateway administration portal. Out-of-band management enables you to isolate client traffic from system management traffic.
PREREQUISITE
Ensure that you have two network interfaces configured on the Avaya Aura® Web Gateway.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Security Settings › OOB Management**.
2. From **Portal and AAWG Services**, select the network interface that you want to use for client traffic.
3. From **OAM**, select the network interface that you want to use for system management traffic.
4. To block SSH connections to the network interface you use for client traffic, select the **Block ssh traffic on the in-band (Portal and AAWG Services) network** check box.
5. Click **Save**.
6. In the Confirm Action window, click **Yes** and then wait until the configuration process is completed.
7. Click **OK**.
8. To verify the changes, do the following:
    a. Log out from the Avaya Aura® Web Gateway administration portal.
    b. In a new browser window, try to open the Avaya Aura® Web Gateway administration portal using the IP address of the network interface you configured for client traffic.
    c. Ensure that the browser displays the Site cannot be reached message.
    d. In a new browser window, open the Avaya Aura® Web Gateway administration portal using the IP address of the network interface you configured for system management traffic.
    e. Log in to the Avaya Aura® Web Gateway administration portal.
    f. If you selected the **Block ssh traffic on the in-band (Portal and AAWG Services) network** check box, log in to the Avaya Aura® Web Gateway CLI using SSH with the IP address of the network interface you configured for client traffic.
    g. Ensure that you cannot log in to the Avaya Aura® Web Gateway CLI using SSH.

AFTER COMPLETING THE TASK
In a cluster deployment, update internode SSH communication settings as described in See "Configuring RSA public and private keys for SSH connections in a cluster" on page 186.

Generating identity certificates for the second network interface

When out-of-band management is configured, Avaya Aura® Web Gateway uses two network interfaces to access the Avaya Aura® Web Gateway administration portal. You must generate new identity certificates so your browser can trust the Avaya Aura® Web Gateway administration portal.

By default, all new generated certificates contain the IP address of the second interface in their Subject Alternative Name (SAN) property. Therefore, you do not need to add DNS records for this interface.
PREREQUISITE

- Configure out-of-band management on the Avaya Aura® Web Gateway administration portal.
- In a cluster environment, update internode SSH communication settings as described in See "Configuring RSA public and private keys for SSH connections in a cluster" on page 186.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Security Settings › Certificate Management › SMGR Certificates**.
2. If you have a cluster or a single node and you do not plan to use an FQDN to access the Avaya Aura® Web Gateway administration portal, do the following:
   a. In a cluster environment, ensure that **Node Address** is set to **All Cluster Nodes**.
   b. In **System Manager Enrollment Password**, enter the System Manager enrollment password.
   c. Click **Generate Certificates**.
   d. Accept all default settings and wait until System Manager generates new certificates.

   STEP RESULT:
   Avaya Aura® Web Gateway applies the changes by restarting its nodes. After the restart, the identity certificate of each node contains the IP address of the second network interface in the SAN field.
3. If you have a cluster or a single node and you plan to use an FQDN for accessing the Avaya Aura® Web Gateway administration portal, do the following:
   a. In **Node Address**, select the required node.
   b. Select the **Show settings** check box.
   c. In **New Subject Alternative Name**, enter the FQDN that is mapped to the IP address of the secondary interface of the node and then click **Add**.
   d. In **System Manager Enrollment Password**, enter the System Manager enrollment password.
   e. Click **Generate Certificates**.
   f. Accept all default settings and wait until System Manager generates new certificates.
   STEP RESULT:
   Avaya Aura® Web Gateway applies the changes by restarting its nodes. After the restart, the identity certificate of each node contains the following new entries in its SAN field:
      - The FQDN address that you provided.
      - The IP address, which is assigned automatically.
   g. In a cluster environment, repeat these substeps for all remaining nodes.

Restoring the default out-of-band management settings

You can restore the default out-of-band management settings if the network interface that you assigned for system management traffic is not reachable. With the default settings, services are accessible on all network interfaces.

1. Log in to an Avaya Aura® Web Gateway node using an SSH connection.
2. Run the `cdto cas` command.
3. Run the `sudo ./os/securty/firewall.pl` command.
4. Enter the administrator password and then wait until the process is complete.

firewall.pl script

The **firewall.pl** script enables you to specify which network interfaces are used for the system management and client traffic types.

If you run the command without arguments, the out-of-band management settings are reset to the default configuration, where all traffic types use all network interfaces. This can be helpful if the network interface you selected for system management traffic is not reachable.

The script is located in the */opt/Avaya/CallSignallingAgent/<version>/CAS/<version>/os/security/* directory.

Syntax

```
sudo ./firewall.pl [--oam={<OAM_interface>|all}] [--
custom={<Client_interface>|all]} [--block_ssh={y|n}] [--print_ports]
```

--oam

Sets a network interface for system management traffic. For example, `eth1`. If you want to use all interfaces for system management traffic, enter `all`.

--custom

Sets a network interface for client traffic. For example, `eth0`. If you want to use all interfaces for client traffic, enter `all`.

--block_ssh

Specifies whether Avaya Aura® Web Gateway blocks SSH traffic on the network interface that is used for client traffic. To allow SSH traffic, enter y. To block SSH traffic, enter n.

--print_ports

Prints the current firewall configuration.

Example

The following command enables client traffic on all configured network interfaces, sets the eth1 network interface for system management traffic, and allows SSH traffic on the system interface that is used for client traffic.

```
sudo ./firewall.pl --oam=eth1 --custom=all --block_ssh=n
```

Enabling additional STIG hardening

The Security Technical Implementation Guides (STIGs) are the requirements that, when implemented, enhance application security and monitoring capabilities. By default, Avaya Aura® Web Gateway enables essential STIG hardening options during the system layer installation or upgrade. These default settings are appropriate for most deployments. If your organization requires stricter STIG compliance, use this procedure to enable additional Linux STIG security hardening options.

> Important:
> You cannot enable additional STIG hardening in software-only deployments.
>
> Note:
> After you enable additional STIG hardening, ensure that the length of the automatic backup password is at least 15 characters.

1. Log in to the virtual machine with the Avaya Aura® Web Gateway OVA as an administrator.
2. Run the following command to enable additional STIG hardening options:
   ADDITIONAL INFORMATION:
   ```
   sys secconfig --stig --enable
   ```
3. When prompted, press c to apply new password rules.
4. To review the STIG hardening status, run the following command:
   ADDITIONAL INFORMATION:
   ```
   sys secconfig --stig --query
   ```

Disabling additional STIG hardening

Use this procedure to disable additional STIG hardening. The default STIG hardening options, which are enabled during the system layer installation or upgrade process, will still be available. Avaya Aura® Web Gateway also continues to use the password complexity rules that were set up when you enabled additional STIG hardening.

1. Log in to the virtual machine with the Avaya Aura® Web Gateway OVA as an administrator.
2. Run the following command to disable additional STIG hardening:
   ADDITIONAL INFORMATION:
   ```
   sys secconfig --stig --disable
   ```
3. To review the STIG hardening status, run the following command:
   ADDITIONAL INFORMATION:

```
sys secconfig --stig --query
```

Configuring UEFI Secure Boot for OVA-based virtual machines

Unified Extensible Firmware Interface (UEFI) is an interface between the operating system and the platform firmware that replaces the legacy BIOS interface. Secure Boot is an UEFI feature that secures the operating system boot-up process by preventing the loading of boot loaders and drivers that are not signed with a trusted digital signature.

By default, Avaya Aura® Web Gateway Release 10.2.x OVA-based deployments have UEFI and Secure Boot enabled. For security reasons, Avaya recommends that you use Secure Boot whenever possible.

For more information about managing Secure Boot settings on VMware virtual machines, see UEFI Secure Boot in the VMware product documentation.
PREREQUISITE
Deploy a virtual machine.

1. Log in to the vSphere Client.
2. Navigate to a Avaya Aura® Web Gateway virtual machine.
3. Right-click the virtual machine and select **Edit Settings**.
4. Click the **VM Options** tab.
5. Click **Boot Options**.
6. Ensure that **Firmware** is set to **EFI (recommended)**.
7. Do one of the following:
    - To enable Secure Boot, select the **Secure Boot** check box.
    - To disable Secure Boot, clear the **Secure Boot** check box.
8. Click **OK**.

Checking the UEFI Secure Boot status

You can check whether the UEFI Secure Boot is enabled on a virtual machine.

1. Log in to a virtual machine using an SSH connection.
2. To check the status, run the following command:
   ADDITIONAL INFORMATION:
   ```
   mokutil --sb-state
   ```
   STEP RESULT:
    - If the command output is SecureBoot enabled, UEFI Secure Boot is enabled.
    - If the command output is Failed to read SecureBoot, UEFI Secure Boot is disabled.

Characters supported for Avaya Aura® Web Gateway passwords

You can use the following characters for Avaya Aura® Web Gateway passwords:

- Uppercase letters:
  A
  to
  Z
- Lowercase letters:
  a
  to
  z
- Numerics:
  0
  to
  9
- Special characters: !, @, #, %, $, ^, *, ?, and _

These rules apply to all passwords that you enter manually to access resources and for Keystore passwords.

> Note:
> The web administration portal supports Active Directory-based authentication. Therefore, you can use other special characters, such as double quotes ("), for web administration portal passwords.

Password policy for human-user accounts

Human-user accounts are accounts that require you to enter a password manually to access Avaya Aura® Web Gateway resources. To protect data and prevent unauthorized access to the system, Avaya Aura® Web Gateway provides default complexity rules for human-user accounts. Each password you create for human-user accounts must meet these requirements.

Avaya Aura® Web Gateway enforces the default complexity rules for the following human-user accounts:

- SSH administration access to Avaya Aura® Web Gateway CLI
- Keepalived authentication
- Cassandra database administrator

If your organization requires a different password policy, you can modify the default complexity rules by using the **sys passwdrules** command.

> Note:
> - Avaya Aura® Web Gateway does not enforce complexity rules for the following passwords because they are not configured on Avaya Aura® Web Gateway:

> - Avaya Aura® Web Gateway web administration portal password.
> - System Manager enrollment password.
>
> - Avaya Aura® Web Gateway does not apply complexity rules for the Keystore password because it is not considered to be a human-user account.

Configuring password rules

You can modify the default password complexity policy for human-user accounts.

1. Log in to Avaya Aura® Web Gateway using an SSH connection.
2. Run the `sys passwdrules` command with appropriate arguments.

Avaya Aura® Web Gateway entry points

The following table shows accounts that you can use to access various Avaya Aura® Web Gateway resources.

Login name	Accessible resources	Password change method
SSH administration user	• Virtual machine resources. • Avaya Aura® Web Gateway web administration portal, if additional STIG hardening is disabled.	Using the **passwd** command in the Avaya Aura® Web Gateway CLI.
Cassandra database administration user	CLI-based Cassandra database administration.	Using the Avaya Aura® Web Gateway configuration utility.
Keepalived account	In a cluster environment, this account is used for IP takeover and to ensure High Availability.	Using the Avaya Aura® Web Gateway configuration utility.

Monitoring and maintenance options

Monitoring services

1. Run the following command to view the status of individual Avaya Aura® Web Gateway services:
   ADDITIONAL INFORMATION:
   ```
   svc csa status
   ```

Viewing performance statistics

You can review various performance metrics and system health indicators, such as CPU or memory usage, on the Avaya Aura® Web Gateway web administration portal. You can review performance metrics for specific nodes from logs currently available on Avaya Aura® Web Gateway or from an external performance log file.

> Note:
> Avaya Aura® Web Gateway stores performance logs a maximum of 30 days. You can collect logs for the future use as described in See "Downloading logs" on page 77. Performance logs have the *perf_CLF.log* file name format and they are located in the */opt/Avaya/CallSignallingAgent/<build_number>/logs* directory. Each log file contains metrics for a specific day.

1. On the Avaya Aura® Web Gateway portal, go to **System Information › Performance Metrics**.
2. View performance data for a node for a specific day as follows:
   a. Select **Choose Performance log from node by date**.
   b. In **Nodes**, select the required node.
   c. In **Date**, select the required date.
3. View performance metrics from an external performance log file as follows:
   a. Select **Choose Performance Log file**.
   b. From **File**, select a log file
4. Click **Retrieve**.
   STEP RESULT:
   Avaya Aura® Web Gateway displays performance metrics in the Performance Metrics area. By default, Avaya Aura® Web Gateway displays the Summary tab.

   If Avaya Aura® Web Gateway does not contain performance logs for the selected date, it displays a warning message.

   Retrieving logs from an external file might take several seconds depending on the log file size and network bandwidth.
5. In the Performance Metrics area, click the tab that contains performance metrics you want to review.
   ADDITIONAL INFORMATION:

For more information about performance metrics, see See "Performance charts" on page 165.

6. If you review performance metrics for the current date, click **Refresh** to update the metrics.

Performance charts

The following table lists the performance charts that you can see on the Performance Metrics page.

Chart	Description
Summary	Displays smaller versions of all available charts.
System load	Displays CPU information, such as CPUs load, time spent waiting for input or output operations, or time used by user space processes. For example, you can use this chart to monitor whether the system load average threshold is exceeded due to spikes.
Java memory usage	Displays information about the Java virtual machine (JVM) memory usage.
HTTP Session Number	Displays the number of active HTTP sessions over time.
Servers Request Latency/ Count	Displays statistics related to requests to other solution components, such as System Manager, Avaya Meetings Server, or Session Manager. • The Servers Request Count chart displays the number of requests per 30 seconds. • The Servers Request Latency chart displays the average latency for a single request.
Call Resource Request Latency/Count	Displays statistics related to requests to the Call service API, such as creating or updating a call, or getting information about a call. • The Servers Request Count chart displays the number of requests per 30 seconds. • The Servers Request Latency chart displays the average latency for a single request.
Clients Resource Request Latency/Count	Displays statistics related to requests to the Clients service API, such as keepalive messages or getting information about clients. • The Servers Request Count chart displays the number of requests per 30 seconds.

Chart	Description
	• The Servers Request Latency chart displays the average latency for a single request.
Presence Session Resource Request Latency/ Count	Displays statistics related to requests to the Presence service API, such as creating, deleting, updating, or getting information about Presence sessions. • The Servers Request Count chart displays the number of requests per 30 seconds. • The Servers Request Latency chart displays the average latency for a single request.
Service Resource Request Latency/Count	Displays statistics related to requests to activate or deactivate call requests. • The Servers Request Count chart displays the number of requests per 30 seconds. • The Servers Request Latency chart displays the average latency for a single request.
Token Resource Request Latency/Count	Displays statistics related to requests to the IView Token service. • The Servers Request Count chart displays the number of requests per 30 seconds. • The Servers Request Latency chart displays the average latency for a single request.
Root Resource Request Latency/Count	Displays statistics related to requests to the root API, such as getting information about available resources. • The Servers Request Count chart displays the number of requests per 30 seconds. • The Servers Request Latency chart displays the average latency for a single request.
Configuration Resource Request Latency/Count	Displays statistics related to requests to the Configuration API, such as getting information about the Avaya Aura® Web Gateway configuration. • The Servers Request Count chart displays the number of requests per 30 seconds. • The Servers Request Latency chart displays the average latency for a single request.
SIP Status Counter	Displays the number of failures for SIP requests. This chart displays statistics for various SIP request types, such as

Chart	Description
	INVITE, NOTIFY, BYE, REFER, REGISTER, OPTIONS. PUBLISH, and SUBSCRIBE.

Log management

Use the Avaya Aura® Web Gateway web administration portal to adjust logging levels, manage log retention, and collect logs for troubleshooting purposes. You can also use the perfLogViewer tool to view performance logs.

Important logs and alarms

Log descriptions

Code	Description
AUD_CSA-00012	The database has been manually stopped.
AUD_CSA-00014	The Tomcat server has been manually stopped.
AUD_CSA-00016	The Sip relay server has been manually stopped.
AUD_CSA-00018	Nginx has been manually stopped.
AUD_CSA-00022	Recovery Manager has been manually stopped.
AUD_CSA-00023	The SNMP IP address and product ID have been updated.
AUD_CSA-00032	The maximum rate of requests or responses within a given time period has been exceeded.
AUD_CSA-00041	Recovery Manager monitoring has been disabled.
AUD_CSA-00043	Recovery Manager Watchdog restart has been disabled.
AUD_CSA-00060	ESG Telportal has been manually stopped.
AUD_CSA-00107	The Keepalived service has been manually stopped.
AUD_CSA-00109	Overload controls have been disabled.

Alarm descriptions

Code	Description
1.3.6.1.4.1.6889.2.94.0.20	Failed to reach the LDAP server.

Code	Description
1.3.6.1.4.1.6889.2.94.0.21	
1.3.6.1.4.1.6889.2.94.0.24 1.3.6.1.4.1.6889.2.94.0.25	Some Cassandra nodes are down or unreachable.
1.3.6.1.4.1.6889.2.94.0.26 1.3.6.1.4.1.6889.2.94.0.27	Failed to reach Avaya Meetings Server Management.
1.3.6.1.4.1.6889.2.94.0.28 1.3.6.1.4.1.6889.2.94.0.29	Failed to reach Session Manager because the signalling request timed out.
1.3.6.1.4.1.6889.2.94.0.34 1.3.6.1.4.1.6889.2.94.0.35	Failed to reach Avaya Aura® Device Services.
1.3.6.1.4.1.6889.2.94.0.38 1.3.6.1.4.1.6889.2.94.0.39	Failed to reach all Avaya Media Server services.
1.3.6.1.4.1.6889.2.94.0.56 1.3.6.1.4.1.6889.2.94.0.57	Approaching full capacity of memory usage.
1.3.6.1.4.1.6889.2.94.0.62 1.3.6.1.4.1.6889.2.94.0.63	REST certificate alarm.
1.3.6.1.4.1.6889.2.94.0.64 1.3.6.1.4.1.6889.2.94.0.65	OAMP certificate alarm.
1.3.6.1.4.1.6889.2.94.0.66 1.3.6.1.4.1.6889.2.94.0.67	Tomcat certificate alarm.
1.3.6.1.4.1.6889.2.94.0.76 1.3.6.1.4.1.6889.2.94.0.77	The system is operating in License Error Mode (major).
1.3.6.1.4.1.6889.2.94.0.78 1.3.6.1.4.1.6889.2.94.0.79	The system is operating in License Restricted Mode (critical).
1.3.6.1.4.1.6889.2.94.0.80 1.3.6.1.4.1.6889.2.94.0.81	No audio licenses are available.
1.3.6.1.4.1.6889.2.94.0.82 1.3.6.1.4.1.6889.2.94.0.83	Audio licenses are at a critical threshold.
1.3.6.1.4.1.6889.2.94.0.86 1.3.6.1.4.1.6889.2.94.0.87	Time server synchronization.

Code	Description
1.3.6.1.4.1.6889.2.94.0.92	The average CPU usage has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.94	The average CPU usage has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.104	The request latency for Session Manager has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.106	The request latency for Session Manager has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.110	The request latency for Avaya Meetings Server Management has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.112	The request latency for Avaya Meetings Server Management has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.122	The request latency call resource for all calls has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.124	The request latency call resource for all calls has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.128	The request latency call resource for getting calls has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.130	The request latency call resource for getting calls has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.134	The request latency call resource for getting call capabilities has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.136	The request latency call resource for getting call capabilities has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.140	The request latency call resource for creating calls has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.142	The request latency call resource for creating calls has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.146	The request latency call resource for updating calls has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.148	The request latency call resource for updating calls has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.152	The request latency call resource for activated or deactivated calls has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.154	The request latency call resource for activated or deactivated calls has exceeded the critical threshold.

Code	Description
1.3.6.1.4.1.6889.2.94.0.158	The request latency for getting Avaya Meetings Server Management tokens has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.160	The request latency for getting Avaya Meetings Server Management tokens has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.164	The request latency for getting resources from the root resource has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.166	The request latency for getting resource from the root resource has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.170	The request latency for getting all clients from the client resource has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.172	The request latency for getting all clients from the client resource has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.176	The request latency for getting clients from the client resource has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.178	The request latency for getting clients from the client resource has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.182	The request latency for the KeepAlive server from the resource has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.184	The request latency for the KeepAlive server from the resource has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.188	The request latency for the configuration resource has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.190	The request latency for the configuration resource has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.194	The request latency for creating presence session has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.196	The request latency for creating presence session has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.200	The request latency for getting presence session has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.202	The request latency for getting presence session has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.206	The request latency for updating presence session has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.208	The request latency for updating presence session has exceeded the critical threshold.

Code	Description
1.3.6.1.4.1.6889.2.94.0.212	The request latency for deleting presence session has exceeded the major threshold.
1.3.6.1.4.1.6889.2.94.0.214	The request latency for deleting presence session has exceeded the critical threshold.
1.3.6.1.4.1.6889.2.94.0.216	Avaya Media Server is down or unreachable.
1.3.6.1.4.1.6889.2.94.0.218	Avaya Media Server is not provided.
1.3.6.1.4.1.6889.2.94.0.220	Avaya Media Server is overloaded.
1.3.6.1.4.1.6889.2.94.0.222	Connection to Avaya Media Server has failed.
1.3.6.1.4.1.6889.2.94.0.224	Avaya Media Server is not found.
1.3.6.1.4.1.6889.2.94.0.230	The video licenses are unavailable.
1.3.6.1.4.1.6889.2.94.0.232	The video licenses are in the critical state.
1.3.6.1.4.1.6889.2.94.0.234	The server node licenses are unavailable.
1.3.6.1.4.1.6889.2.94.0.236	The server node is at a critical threshold.

Viewing performance logs

Use this procedure to view performance logs using the perfLogViewer tool. This tool processes special performance logs that track system performance. Avaya Aura® Web Gateway creates performance log on a daily basis.

You can view the following data using the perfLogViewer tool:

- Calls Resource Request Latency
- Clients Resource Request Latency
- Configuration Resource Request Latency
- HTTP Session Number
- Presence Session Resource Request Latency
- Root Resource Request Latency
- Server Request Latency
- Service Resource Request Latency
- SIP Status Counter
- System Average Load
- Token Resource Request Latency

1. From the Avaya Aura® Web Gateway web administration portal, download an archive with Avaya Aura® Web Gateway logs on your computer.
   ADDITIONAL INFORMATION:

For more information, see See "Downloading logs" on page 77.

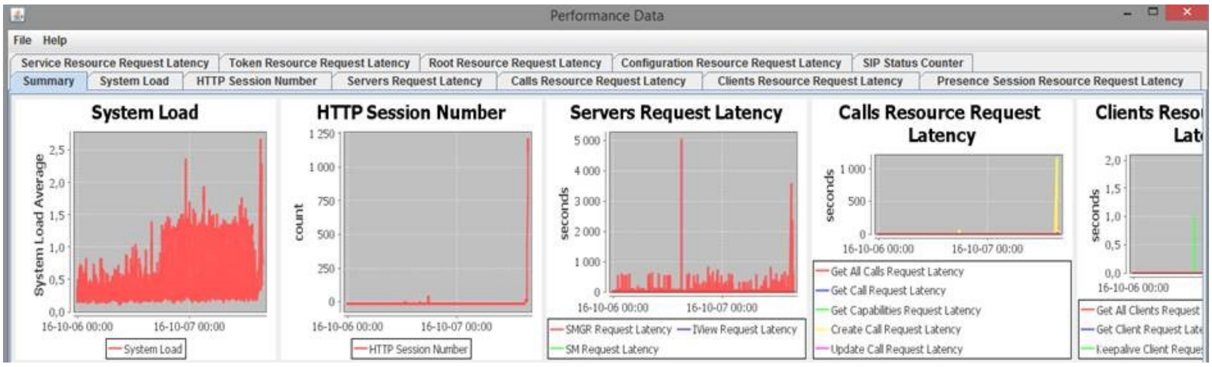
2. Extract performance logs from the archive with Avaya Aura® Web Gateway logs.
   ADDITIONAL INFORMATION:
   Performance logs have the *perf_CLF.log.<LOG_FILE_NUMBER>* file name format. In the log archive, performance logs are located in the */opt/Avaya/CallSignallingAgent/<build_number>/logs* directory.

3. Copy the perfLogViewer tool on your computer from the following location on Avaya Aura® Web Gateway:
   ADDITIONAL INFORMATION:
   /opt/Avaya/CallSignallingAgent/<build_number>/CAS/<build_number>/lib/perfLogViewer-<version_number>.jar

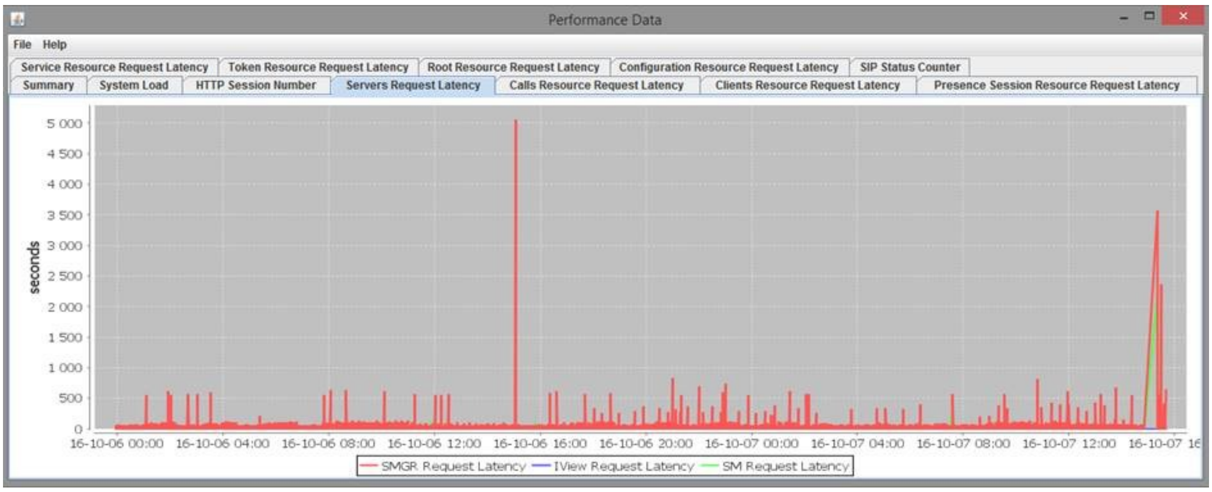
   You can use any file transfer utility, such as SFTP or SCP.

4. Double-click the *.jar* file to run the perfLogViewer tool.

5. In the perfLogViewer tool, click **File › Open** and then select the required performance file.
   ADDITIONAL INFORMATION:
   The following image provides an example of the summary page displaying some of the performance metrics:



6. To investigate a metric in detail, click the required data tab.
   ADDITIONAL INFORMATION:
   The following image provides an example of the server request latency over time:



7. To investigate an area of the graph in more detail, press and hold the left mouse button to select the area of interest.
   ADDITIONAL INFORMATION:
   The following image displays the maximized portion of a graph using the zoom in functionality:

ADDITIONAL INFORMATION:

Note:
If perfLogViewer cannot load a log because the system and log locales are different, then close the *.jar* file and run the following command in the Windows command prompt:

```
java -Duser.language.format=en -Duser.country.format=US -jar
perfLogViewer-<version_number>.jar.
```

Configuring threshold values for specific alarms

You can modify the default threshold values for the following alarms that Avaya Aura® Web Gateway raises on System Manager:

- CPU average usage exceeds a threshold.
- The number of available server node, audio, or video licenses exceeds a threshold.

PREREQUISITE
Enable EASG functionality. EASG is required to obtain the sroot privileges, which you must use to run commands in this procedure.

1. Log in to the initial node as an administrator using an SSH connection.
2. To obtain sroot privileges, run the following command:
   ADDITIONAL INFORMATION:
   ```
   su - sroot
   ```
3. To navigate to the location of the alarm settings file, run the following commands:
   ADDITIONAL INFORMATION:
   ```
   cdto cas
   cd service.d
   ```
4. Open the *mss.start.sh* file in the vi text editor:
   ADDITIONAL INFORMATION:
   ```
   vi mss.start.sh
   ```
5. Navigate to the section where CATALINA_OPTS parameters are configured.
   ADDITIONAL INFORMATION:

These entries use the `CATALINA_OPTS="CATALINA_OPTS -D<VARIABLE>=<VALUE>"` format.

6. At the end of the section, add an entry for an appropriate alarm in the following format:
   ADDITIONAL INFORMATION:
   `CATALINA_OPTS="$CATALINA_OPTS -D<PARAMETER>=<VALUE>"`

   In this entry, `PARAMETER` is the parameter name for the alarm and `VALUE` is the threshold. When this threshold is reached, Avaya Aura® Web Gateway raises the alarm on System Manager.

   For example, to raise a warning alarm when the CPU usage exceeds 60%, add the following entry:

   ```
   CATALINA_OPTS="$CATALINA_OPTS
   -Dcom.avaya.csa.audit.perf.warning.system.load.threshold=60"
   ```

   > Note:
   >
   > - The command does not contain any space characters between `-D` and the parameter name.
   > - Do not append the percent character to the threshold value.

7. Repeat the previous step to configure threshold values for other alarms.
8. Save the file.
9. To restart services and apply changes, run the following command:
   ADDITIONAL INFORMATION:

   ```
   svc telportal restart
   ```

Configurable alarms

The following table lists the Avaya Aura® Web Gateway alarms on System Manager for which you can configure threshold values.

Alarm	Parameter in *mss.start.sh*	Value range
Alarm indicating that the number of remaining available server node, audio, or video licenses is less than the specified threshold expressed in percents.	com.avaya.csa.servernode.licensing.warning.threshold.percentage	0-100 percents The default value is 10.
Alarm indicating that the average CPU usage exceeded the specified warning threshold expressed in percents.	com.avaya.csa.audit.perf.warning.system.load.threshold	0-100 percents The default value is 70.
Alarm indicating that the average CPU usage exceeded the specified major threshold expressed in percents.	com.avaya.csa.audit.perf.major.system.load.threshold	0-100 percents The default value is 80.

Alarm	Parameter in *mss.start.sh*	Value range
Alarm indicating that the average CPU usage exceeded the specified critical threshold expressed in percents.	com.avaya.csa.audit.perf.critical.system.load.threshold	0-100 percents The default value is 90.

> Note:
> For the CPU usage alarms, Avaya Aura® Web Gateway firstly checks if the average CPU usage exceeds the threshold specified for a critical alarm, then the threshold for the major alarm, and then the threshold for the warning alarm. Therefore, Avaya Aura® Web Gateway will never raise warning or major alarms if thresholds configured for these alarms are higher that the threshold for the critical alarm.

Managing Cassandra repairs

The repair process synchronizes the data between nodes to provide consistency. Use this procedure to set up the date and time for the Cassandra repair process. You can run the repair on a weekly or monthly basis. You can also disable the repair.

The Avaya Aura® Web Gateway performs the repair on one node at a time. By default, the Avaya Aura® Web Gateway performs the repair once a week on Sunday at 01:20 a.m.

> Tip:
> Set the time when the system usage is low to minimize impact on system performance.

Cassandra repair results are stored in the *CAS.log* file.

1. Log in to the seed node CLI as the root user using an SSH connection.
2. Run the following commands to open the *catalina.properties* file in the vi editor:
   ADDITIONAL INFORMATION:
   ```
   cdto active
   vi mss/<tomcat_version>/conf/catalina.properties
   ```
3. To run the repair on a weekly basis, do the following:
   a. Edit the `com.avaya.cas.cassandra.repair.time.hour=<hour>` string to set the hour when the procedure starts.
      ADDITIONAL INFORMATION:
      This parameter uses a 24-hour notation. For example, if you want to start the procedure at 4 p.m., enter `16`.
   b. Edit the `com.avaya.cas.cassandra.repair.time.minute=<minute>` string to set the minute when the procedure starts.
   c. Edit the `com.avaya.cas.cassandra.repair.time.weekday=<day_of_week>` string to set the required day of the week.
      ADDITIONAL INFORMATION:
      The week starts from Sunday. For example, for Sunday, enter `1`, for Monday, enter `2`, and so on.
   d. In the `com.avaya.cas.cassandra.repair.time.monthday=<day_of_month>` string, set `<day_of_month>` to `0`.

   For example, to run the repair procedure on Wednesdays at 8:30 p.m., edit the entries as follows:

```
com.avaya.cas.cassandra.repair.time.hour=8
com.avaya.cas.cassandra.repair.time.minute=30
com.avaya.cas.cassandra.repair.time.weekday=4
com.avaya.cas.cassandra.repair.time.monthday=0
```

4. To run the repair on a monthly basis, do the following:
    a. Edit the `com.avaya.cas.cassandra.repair.time.hour=<hour>` string to set the hour when the procedure starts.
    ADDITIONAL INFORMATION:
    This parameter uses a 24-hour notation. For example, if you want to start the procedure at 4 p.m., enter `16`.
    b. Edit the `com.avaya.cas.cassandra.repair.time.minute=<minute>` string to set the minute when the procedure starts.
    c. In the `com.avaya.cas.cassandra.repair.time.weekday=<day_of_week>` string, set `<day_of_week>` to `0`.
    d. Edit the `com.avaya.cas.cassandra.repair.time.monthday=<day_of_month>` string to set the required day of the month.

    For example, to run the repair procedure on the second day of each month at 2:40 p.m., edit the entries as follows:

```
com.avaya.cas.cassandra.repair.time.hour=14
com.avaya.cas.cassandra.repair.time.minute=40
com.avaya.cas.cassandra.repair.time.weekday=0
com.avaya.cas.cassandra.repair.time.monthday=2
```

5. To disable the repair, set both `com.avaya.cas.cassandra.repair.time.weekday` and `com.avaya.cas.cassandra.repair.time.weekday` to `0` or to any other valid non-zero value.
For example:

```
com.avaya.cas.cassandra.repair.time.weekday=0
com.avaya.cas.cassandra.repair.time.monthday=0
```

6. Save the file.
7. Run the `svc telportal restart` command to restart services.
8. In a cluster environment, repeat the previous steps on all non-seed nodes in the cluster.

Enhanced Access Security Gateway support for the Avaya AuraÂ® Web Gateway

Enabling the Enhanced Access Security Gateway from the CLI

Use this procedure to enable Enhanced Access Security Gateway (EASG) functionality in Avaya Aura® Web Gateway. Avaya support engineers can use this functionality to access your computer and resolve product issues in real time.

1. Open the SSH console as an administrator.
2. Run the following command to check the EASG status:
   ADDITIONAL INFORMATION:
   ```
   EASGStatus
   ```
   ADDITIONAL INFORMATION:
   By default, the EASG status is disabled.
3. Run the following command to enable EASG:
   ADDITIONAL INFORMATION:
   ```
   sudo /usr/sbin/EASGManage --enableEASG
   ```
4. Run the following command to verify the product certificate:
   ADDITIONAL INFORMATION:
   ```
   sudo EASGProductCert --certInfo
   ```
   STEP RESULT:
   Avaya Aura® Web Gateway displays the product certificate details.

   For example:

```
[admin@amm-ova-test ~]$ EASGStatus
EASG is disabled
[admin@amm-ova-test ~]$ sudo /usr/sbin/EASGManage --enableEASG

By enabling Avaya Services Logins you are granting Avaya access to
your system.  This is required to maximize the performance and value
of your Avaya support entitlements, allowing Avaya to resolve product
issues in a timely manner.

The product must be registered using the Avaya Global Registration
Tool (GRT, see https://grt.avaya.com) to be eligible for Avaya remote
connectivity.  Please see the Avaya support site (https://support.avaya.com/
registration) for additional information for registering products and
establishing remote access and alarming.

Do you want to continue [yes/no]? yes

EASG Access is enabled. Performed by user ID: 'admin', on Oct 19 2016 - 12:28
[admin@amm-ova-test ~]$ EASGProductCert --certInfo
Subject:        CN=[redacted], OU=EASG, O=Avaya Inc.
Serial Number:  10005
Expiration:     Aug  6 04:00:00 2031 GMT
Trust Chain:
   1. O=Avaya, OU=IT, CN=AvayaITrootCA2
   2. DC=com, DC=avaya, DC=global, CN=AvayaITserverCA2
   3. O=Avaya Inc, OU=EASG, CN=EASG Intermediate CA
   4. CN=Product EASG Intermediate CA, OU=EASG, O=Avaya Inc.
   5. CN=[redacted].0, OU=EASG, O=Avaya Inc.
[admin@amm-ova-test ~]$
```

   If the certificate expires within 360, 180, 30, or 0 days, Avaya Aura® Web Gateway writes a certificate expiry notification to the *\/var/log/messages* log file.

Removing EASG

Use this procedure to remove EASG permanently. You can use the Avaya-provided OVA deployment process to reinstall EASG.

1. In the SSH console, run the following command to remove EASG:
   ADDITIONAL INFORMATION:
   ```
   sudo /opt/Avaya/permanentEASGRemoval.sh
   ```

Managing the Avaya Aura® Web Gateway server firewall

Use this procedure to reset the firewall settings back to the defaults, or to allow additional ports through the server firewall.

For detailed information about the ports that must be opened, go to http://support.avaya.com/security, scroll down, and click **Avaya Product Port Matrix Documents**. Navigate to the Avaya Aura® Web Gateway section and then click on the appropriate Port Matrix document for the release to open it.

1. Add the required ports to the firewall configuration file */opt/Avaya/CallSignallingAgent/<version>/CAS/<version>/os/security/firewall.conf*.
2. Run the Avaya Aura® Web Gateway configuration utility using the `app configure` command.
3. Select **Advanced Configuration › OS Security Utility › Run the firewall configuration script**.
   STEP RESULT:
   The firewall is configured automatically.

Viewing the firewall configuration

You can use the **firewall.pl** command to view the current Avaya Aura® Web Gateway firewall configuration.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator using an SSH connection.
2. Run the following command:
   ADDITIONAL INFORMATION:
   ```
   firewall.pl --print_ports
   ```

Backup and restore

This section describes how to perform backups and restore the backed up configuration files. You can back up existing configuration files, certificates from System Manager, and other certificates that you might have applied. Do not save backup files in the application installation directory.

> Note:
>
> - A backup file can only be used to restore data to the same software version that created it.
> - If you change the IP addresses or FQDN of an Avaya Aura® Web Gateway node, you cannot use backups created before you made these changes to restore the data. This limitation applies to both manual and automatic backups.

Backing up Avaya Aura® Web Gateway

Use this procedure to back up Avaya Aura® Web Gateway database and configuration files. This procedure applies to both standalone and cluster environments. In a cluster environment, you must back up files from all nodes in the cluster.

> Note:
> Avaya Aura® Device Services recommends that you start a Tmux session before running the backup command. If the SSH session terminates while the backup operation is in progress, you can re-connect to the Tmux session and complete the backup operation. For more information about the Tmux utility, see See "Using the Tmux utility" on page 43.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrative user.
2. To start a Tmux session, run the following command:
   ADDITIONAL INFORMATION:
   `tmux new-session -s <NAME>`

   In this command, `NAME` is a Tmux session name of your choice. For example:

   ```
   tmux new-session -s AAWG_backup_session
   ```
3. To create a backup in a *.tar.zip* file, run one of the following commands:
   - `app backup -t`
   - `app backup -t -P '<PASSWORD>'`

   ADDITIONAL INFORMATION:
   In the second command, `<PASSWORD>` is a password for protecting backup files. You must enclose the password with single quotation marks ('). If you do not enter a password or if the password does not comply with the password rules, Avaya Aura® Web Gateway prompts you to provide a password later. For information about supported characters, see See "Characters supported for Avaya Aura Web Gateway passwords" on page 162.

   In a cluster environment, you can run this command on any node. It creates a backup for all nodes in the cluster.

   For example, to set the `Password1234#` password for your backups, run the following command:

```
app backup -t -P 'Password1234#'
```

4. If you did not enter a password for backups, then enter it when prompted.
   ADDITIONAL INFORMATION:
   If the password does not comply with the password rules, Avaya Aura® Web Gateway prompts you to enter a new password.
5. Re-enter the password to confirm it.
6. Wait until the backup process is completed.
7. Ensure that the backup file with the name *<DateTimestamp>_<node_short-hostname>.tar.zip* is present in the administration directory, which is located at */home/admin*.
8. To avoid losing backed up files on the server, save a copy of them on a remote storage server.
   ADDITIONAL INFORMATION:
   You can add these files back to the server before restoring Avaya Aura® Web Gateway.
9. Store the password that you used for backup files in a safe place.
   ADDITIONAL INFORMATION:
   Avaya Aura® Web Gateway requires this password when you restore the backups.

Automatic backups

Avaya Aura® Web Gateway creates backups of configuration files and user data automatically on a weekly basis. Avaya Aura® Web Gateway uses the following default settings for automatic backups:

Setting	Value
Date and time	Each Sunday at 12:00 a.m.
Backup file location	*/var/log/Avaya/backup/*
Number of backups stored	Three rotating backups
Password	RAPtor@WELcomE

If required, you can update the default automatic backup settings using the Avaya Aura® Web Gateway web administration portal.

Important:

- Avaya Aura® Web Gateway recommends that you change the default password for automatic backups immediately after installing or upgrading Avaya Aura® Web Gateway.
- If STIG hardening is enabled, for automatic backups, you must configure a password 15-character long or more. The default password does not meet STIG requirements and will cause backup failure. You can review the current password rules using the **sys passwdrules show** command.
- In cluster deployments, if any Avaya Aura® Web Gateway node is down, then the automatic backup creation process fails.

If the backup creation process fails, Avaya Aura® Web Gateway generates an alarm on System Manager.

Configuring automatic backups

Use this procedure to modify the default automatic backup settings. By default, Avaya Aura® Web Gateway starts creating automatic backups each Sunday at 12:00 a.m.

The backup process requires significant system resources. Therefore, schedule automatic backups during the least busy periods to minimize the impact on Avaya Aura® Web Gateway performance.

> Important:
> Avaya Aura® Web Gateway recommends that you change the default password.

1. Log in to the Avaya Aura® Web Gateway administration portal.
2. Go to **Automatic Backup Configuration**.
3. In **Backup Time**, specify the time and date to start creating automatic backups.
4. Select the **Repeat every** check box and then specify the time interval between two consecutive automatic backup operations.
5. In **Max Backup**, specify how many backups you want to store on Avaya Aura® Web Gateway.
6. Change the password that is currently used for automatic backups as follows:
   a. Select the **Change Password** check box.
   b. In **Backup File Password**, type a password of your choice for backups.
   ADDITIONAL INFORMATION:
   The password must comply with the password complexity rules. You can view the rules by pointing the cursor to the **Backup File Password** field.

   > Important:
   > If STIG hardening is enabled, you must configure a password of 15 characters or more.

   c. In **Re-Enter Password**, type the password again.
7. Click **Save**.

Changing the location of automatic backups

By default, Avaya Aura® Web Gateway stores automatic backups in the */var/log/Avaya/backup/* directory. You can change the location of backup files by modifying the *tomcat.start.sh* file. You cannot change the location from the Avaya Aura® Web Gateway web administration portal.

Existing automatic backup files remain in the location where they were created. does not move these files to the new location.

> Note:

- Avaya recommends that you use the default location for automatic backups.
- After the Avaya Aura® Web Gateway upgrade, the location of automatic backups is changed back to the default location. You must repeat this procedure if you want to store automatic backups in another directory.

PREREQUISITE
Enable EASG functionality. This is required to obtain the sroot user privileges, which are required to run commands in this procedure.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator using an SSH connection.
2. To obtain the sroot user privileges, run the following command:
   ADDITIONAL INFORMATION:
   ```
   su - sroot
   ```
3. Run the following command to navigate to the directory that contains the *tomcat.start.sh* file:
   ADDITIONAL INFORMATION:
   ```
   cd /opt/Avaya/CallSignallingAgent/<version>/CAS/<version>/
   service.d/
   ```
4. Open the *tomcat.start.sh* in a text editor of your choice.
   ADDITIONAL INFORMATION:
   For example:
   ```
   vi ./tomcat.start.sh
   ```
5. In the `CATALINA_OPTS="$CATALINA_OPTS -Dcom.avaya.cas.autobackup.dir=<DIRECTORY_NAME>"` string, replace the existing *<DIRECTORY_NAME>* entry with the path to the directory where you want to store automatic backups.
   ADDITIONAL INFORMATION:
   For example, if you want to store automatic backups in the *home/admin/automatic_backups/* directory, modify the string as follows:

   `CATALINA_OPTS="$CATALINA_OPTS -Dcom.avaya.cas.autobackup.dir=/home/admin/automatic_backups/"`
6. Save the file.
7. Run the following command to restart Tomcat:
   ADDITIONAL INFORMATION:
   `svc tomcat restart`

Restoring options for standalone and cluster environments

Restoring Avaya Aura® Web Gateway in a standalone environment

Use this procedure to restore a standalone Avaya Aura® Web Gateway node.
PREREQUISITE

- Ensure that you have the password that was used to create backup files.
- If you changed the IP address or FQDN of the node, ensure that the required Avaya Aura® Web Gateway backup was created after these changes. You cannot restore the node using backups created before you changed the node IP address or FQDN.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Reinstall the same version of the Avaya Aura® Web Gateway that was backed up:
   a. Redeploy the OVA if the node needs to be re-imaged.

b. Run the following command:
ADDITIONAL INFORMATION:
```
app install
```
c. Select **Advanced Configuration** and set **Enable Cassandra DB initialization** to `n` (no).
d. To return to the previous menu, select **Return to Main Menu** and press `Enter`.
e. Select **Continue** and press `Enter`.
STEP RESULT:
The basic load of the Avaya Aura® Web Gateway, without System Manager and LDAP options, is installed.

3. To restore the backed-up configuration files, run one of the following commands:
   - ```
     app restore <full_path_to_backup_tar_zip_file>
     ```
   - ```
     app restore <full_path_to_backup_tar_zip_file> -P <PASSWORD>
     ```

   ADDITIONAL INFORMATION:
   In the second command, `<PASSWORD>` is the password you used to create the backup. If you do not enter a password, Avaya Aura® Web Gateway prompts you to enter it later.
4. If you did not enter the password for backups, then enter it when prompted.
   ADDITIONAL INFORMATION:
   If the password is wrong, Avaya Aura® Web Gateway prompts you to re-enter the password. After three failed attempts, Avaya Aura® Web Gateway aborts the restore process. In this case, you will need to run the `app restore` command again.
   STEP RESULT:
   The restore utility applies the configuration in the backup file and automatically restarts the service.

Restoring a node in an Avaya Aura® Web Gateway cluster

Use this procedure if you only need to restore a specific node in a cluster, and not the entire cluster.
PREREQUISITE

- Ensure that you have the password that was used to create backup files.
- If you changed the IP address or FQDN of the node, ensure that the required Avaya Aura® Web Gateway backup was created after these changes. You cannot restore the node using backups created before you changed the node IP address or FQDN.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Reinstall the same version of the Avaya Aura® Web Gateway that was backed up:
   a. Redeploy the OVA if the node needs to be re-imaged.
   b. Run the following command:
   ADDITIONAL INFORMATION:
   ```
   app install
   ```

c. Select **Advanced Configuration** and set **Enable Cassandra DB initialization** to n (no).
d. To return to the previous menu, select **Return to Main Menu** and press `Enter`.
e. Select **Continue** and press `Enter`.
STEP RESULT:
The basic load of the Avaya Aura® Web Gateway, without System Manager and LDAP options, is installed.

3. To restore the backed-up files, run one of the following commands:
   - ```
     app restore <full_path_to_backup_tar_zip_file>
     ```
   - ```
     app restore <full_path_to_backup_tar_zip_file> -P <PASSWORD>
     ```

   ADDITIONAL INFORMATION:
   In the second command, `<PASSWORD>` is the password you used to create the backup. If you do not enter a password, Avaya Aura® Web Gateway prompts you to enter it later.
4. If you did not enter the password for backups, then enter it when prompted.
   ADDITIONAL INFORMATION:
   If the password is wrong, Avaya Aura® Web Gateway prompts you to re-enter the password. After three failed attempts, Avaya Aura® Web Gateway aborts the restore process. In this case, you will need to run the `app restore` command again.
   STEP RESULT:
   The restore utility applies the configuration in the backup file and automatically restarts the service.
5. From another node in the cluster, set up the RSA public and private keys as described in the "Configuring RSA public and private keys for SSH connections in a cluster" procedure.
6. Run the following command to restart the Avaya Aura® Web Gateway service:
   ADDITIONAL INFORMATION:
   ```
   svc csa restart
   ```
7. Run the following Cassandra repair command:
   ADDITIONAL INFORMATION:
   ```
   /opt/Avaya/CallSignallingAgent/<version>/CAS/<version>/
   cassandra/cassandraRepair.sh -M
   ```

Restoring an entire Avaya Aura® Web Gateway cluster

Use this procedure to restore an entire cluster. Restoring a cluster is useful if a failure occurs that results in the loss of all nodes.

To restore an Avaya Aura® Web Gateway node, you must install the Avaya Aura® Web Gateway software first and then restore the configuration and data files from a previously made backup.
PREREQUISITE

- Ensure that you have the password that was used to create backup files.
- If you changed the IP address or FQDN of the node, ensure that the required Avaya Aura® Web Gateway backup was created after these changes. You cannot restore the node using backups created before you changed the node IP address or FQDN.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
2. Reinstall the same version of the Avaya Aura® Web Gateway that was backed up:
   a. Redeploy the OVA if the node needs to be re-imaged.
   b. Run the following command:
      ADDITIONAL INFORMATION:
      ```
      app install
      ```
   c. Select **Advanced Configuration** and set **Enable Cassandra DB initialization** to n (no).
   d. To return to the previous menu, select **Return to Main Menu** and press `Enter`.
   e. Select **Continue** and press `Enter`.
      STEP RESULT:
      The basic load of the Avaya Aura® Web Gateway, without System Manager and LDAP options, is installed.
3. To restore the backed-up data, do the following on the seed node first and then on all non-seed nodes, one node at a time:
   a. Run one of the following command to restore the backed-up data:
      ADDITIONAL INFORMATION:
      - ```
        app restore <full_path_to_backup_tar_zip_file>
        ```
      - ```
        app restore <full_path_to_backup_tar_zip_file> -
        P <PASSWORD>
        ```

      In these commands, `<full_path_to_backup_tar_zip_file>` is the absolute path to the *.tar.zip* backup file. In the second command, `<PASSWORD>` is the password you used to create the backup. If you do not enter a password, Avaya Aura® Web Gateway prompts you to enter it later.
   b. If you did not enter the password for backups, enter it when prompted.
      ADDITIONAL INFORMATION:
      If the password is wrong, Avaya Aura® Web Gateway prompts you to re-enter the password. After three failed attempts, Avaya Aura® Web Gateway aborts the restore process. In this case, you will need to run the `app restore` command again.

   ADDITIONAL INFORMATION:
   You must wait until the restore process is completed on the current node before restoring data on another node.
4. On the seed node, set up the RSA public and private keys as described in the "Configuring RSA public and private keys for SSH connections in a cluster" procedure.
5. Run the following command on the seed node first and then on other nodes to restart the Avaya Aura® Web Gateway:
   ADDITIONAL INFORMATION:
   ```
   svc csa restart
   ```
6. Run the following repair command from one of the nodes:
   ADDITIONAL INFORMATION:
   ```
   /opt/Avaya/CallSignallingAgent/<version>/CAS/<version>/
   cassandra/cassandraRepair.sh -M
   ```

Configuring RSA public and private keys for SSH connections in a cluster

After nodes are added to a cluster, you must configure the RSA public and private keys to enable internode SSH communications.

Use this procedure to configure the RSA public and private keys on the initial node for the entire cluster.
PREREQUISITE
Install all of the required nodes for the cluster.

1. Log in to the Linux shell on the initial Avaya Aura® Web Gateway node as an administrator.
2. Run the Avaya Aura® Web Gateway configuration utility using the `app configure` command.
3. Navigate to **Clustering Configuration › Cluster Utilities › Configure SSH RSA Public/ Private Keys**.
   STEP RESULT:
   Avaya Aura® Web Gateway displays the RSA Public and Private key configuration tool.
4. When the system displays the Add additional hosts to the list? (y/n) prompt, enter y (yes) if you are generating keys for the first time or if you need to generate keys for a new node in the cluster.
   ADDITIONAL INFORMATION:
   Otherwise, enter n (no).
5. If you chose to update the node list in the previous step, when Avaya Aura® Web Gateway prompts you, enter the IP address of the non-seed node in the cluster you want to generate keys for and then press `Enter`.
6. Repeat the previous step for all remaining non-seed nodes.
7. When Avaya Aura® Web Gateway prompts you to enter a user name for a node, enter the username for the Linux administrator account that you used to perform the Avaya Aura® Web Gateway installation.
8. If Avaya Aura® Web Gateway prompts you to replace the existing keys, enter y (yes).
9. When Avaya Aura® Web Gateway prompts you to enter a password, enter the password for the Linux administrator account that you used to perform the installation.
10. When the configuration is complete, press `Enter`.
11. Return to the main menu.
12. From the main menu, select **Continue** and then select **Yes** to apply the changes.

Exporting and importing Avaya Aura® Web Gateway configuration data

You can export the configuration data of an Avaya Aura® Web Gateway node to a *.properties* file. You can use the exported data to facilitate the Avaya Aura® Web Gateway installation process in migration or reinstall scenarios. The exported file contains all the parameters required for silent installation, and most of the parameters are already properly configured. Therefore, you can install Avaya Aura® Web Gateway in silent mode using this file instead of manually configuring the required settings during the interactive installation.

Configuration file structure

The exported *.properties* file contains the set of parameters in the following format:

```
<PARAMETER_NAME>=<PARAMETER VALUE>
```

For example:

```
CASSANDRA_VER=3.11.3
INCLUDE_OAUTH=
REST_FRONTEND_HOST=
USE_LDAP_FOR_AUTH=y
KEYSTORE_PW=<>
```

Most of the parameters required for silent installation contain the required values. For the parameters, such as KEYSTORE_PW, where values are set to "<>", you must provide values relevant to the server where you install Avaya Aura® Web Gateway. You do not need to modify parameters that have no value set, such as REST_FRONTEND_HOST.

Limitations

The exported configuration file only supports silent installation. You cannot use this file for interactive installations.

Use case example

The following is a simplified example of an Avaya Aura® Web Gateway node restoration using the exported configuration file:

1. Redeploy virtual machines.
2. In the exported configuration file, modify the parameters where values are set to "<>" according to your deployment settings.
3. Reinstall the Avaya Aura® Web Gateway application in silent mode using the exported configuration file.
4. Restore configuration and user data from backup files.

Exporting the configuration file using the web administration portal

Use this procedure to collect configuration data from an Avaya Aura® Web Gateway node using the web administration portal.
PREREQUISITE
In cluster deployments, if you plan to export the configuration file for another node, ensure that RSA keys are configured for SSH connections in the cluster. For more information, see "Configuring RSA public and private keys for SSH connections in a cluster" in *Deploying the Avaya Aura® Web Gateway*.

1. On the Avaya Aura® Web Gateway web administration portal, go to **Advanced › Application Management**.
2. In the Export Application Configuration area, click **Download** for the appropriate node.

Result
Avaya Aura® Web Gateway collects configuration data for the node and exports the file to your computer. The exported file name has the *_export.<NODE_IP_ADDRESS>-<DATE>-<TIME>.properties* format.

Exporting the configuration file using the CLI

Use this procedure to collect configuration data from an Avaya Aura® Web Gateway node using the CLI.
PREREQUISITE
In cluster deployments, if you plan to export the configuration file for another node, ensure that RSA keys are configured for SSH connections in the cluster. For more information, see "Configuring RSA public and private keys for SSH connections in a cluster" in *Deploying the Avaya Aura® Web Gateway*.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator using an SSH connection.
2. Run the following command:
   ADDITIONAL INFORMATION:
   `cdto misc`
3. Do one of the following:
   - To generate a configuration file for the current node, run the following command with sudo privileges:

     ```
     sudo ./exportConfigForSilentInstall.sh generate
     ```
   - To generate a configuration file for another node in a cluster, run the following command:

     ```
     sudo ./exportConfigForSilentInstall.sh generate -ip
     <NODE_IP_ADDRESS>
     ```

   ADDITIONAL INFORMATION:
   Avaya Aura® Web Gateway generates the configuration file and displays the full path to the file.
4. Download the file to your computer using a file transfer utility of your choice, such as SFTP, SCP, or WinSCP.

Preparing the exported configuration file for silent installation

Before using the exported configuration file for silent installation, you must provide valid values for parameters whose values equal to "<>".

You do not need to modify parameters that have no value set.
PREREQUISITE
Ensure that you have a configuration file for an appropriate Avaya Aura® Web Gateway node.

1. Open the configuration file in a text editor of your choice.

2. For each parameter whose value equals to "<>", provide a value appropriate for your deployment.
3. Save the file.

Avaya Aura® Web Gateway upgrade and migration operations

Depending on the currently installed release, the migration or upgrade path can flow through one or more prior releases. The following table shows the specific release upgrades along with the migration or upgrade path:

Upgrade or migration path number	From	To
1	3.8.0.x	3.8.1
2	3.8.1	3.9
3	3.9	3.9.1
4	3.9.1	3.11
5	3.11	10.1
6	3.11	10.2.x

Depending on the release you are currently using, you might need to perform several upgrade or migration procedures. For example, to upgrade Avaya Aura® Web Gateway Release 3.9 to 10.2.x, follow paths 3, 4, 5, and 6.

Avaya Aura® Web Gateway Release 3.11 and 10.2.x use different operating system versions. Therefore, you cannot upgrade from Release 3.11 to 10.2.x. You must perform a migration as described in this chapter. This requirement applies to all deployment types.

This document focuses on the migration process for Release 10.2.x. For information about upgrade procedures for previous releases, see the following documents:

- For Release 3.8.1, see Administering the Avaya Aura® Web Gateway.

  If you use Avaya Aura® Web Gateway Release 3.8.0.x or earlier, you must upgrade to Release 3.8.1 before performing the migration to Release 3.11.

- For Release 3.9, see Administering the Avaya Aura® Web Gateway.

  Avaya Aura® Web Gateway Release 3.11 and Release 3.8.1 use different operating system versions. Therefore, you cannot upgrade from Release 3.8.1 to 3.11. You must perform a migration as described in the specified document.

- For Release 3.9.1, see Administering the Avaya Aura® Web Gateway.
- For Release 3.11, see Administering the Avaya Aura® Web Gateway.

> Important:
>
> - Avaya Aura® Web Gateway does not support rollback with the migration process.
> - Perform the system layer update before upgrading Avaya Aura® Web Gateway.

Migrating Avaya Aura® Web Gateway

This procedure provides the high-level steps for migrating Avaya Aura® Web Gateway to Release 10.2.x. These migration steps apply to OVA-based and software-only deployment types.

This procedure describes cluster deployments. If you have a standalone Avaya Aura® Web Gateway deployment, ignore the references to other cluster nodes.

> Important:
>
> - You must use the original host name and IP address from the currently installed cluster when deploying OVAs.
> - You cannot roll back when performing the procedures.

1. Read and understand the migration limitations.
   ADDITIONAL INFORMATION:
   For more information, see See "Migration limitations" on page 191.
2. Read and understand the migration considerations for these deployment types.
   ADDITIONAL INFORMATION:
   For more information, see See "Migration considerations" on page 192.
3. Prepare for the migration:
   a. Obtain the latest Avaya Aura® Web Gateway Release 10.2.x installer and OVA.
   b. Create Avaya Aura® Web Gateway Release 3.11 backups and move them to an off-board location.
   c. Shut down Avaya Aura® Web Gateway and the virtual machines.

   ADDITIONAL INFORMATION:
   For more information, see See "Preparing for migration" on page 193.
4. Deploy new servers:
   a. Deploy new virtual machines for Avaya Aura® Web Gateway Release 10.2.x.
   b. Perform the operating system level changes, such as enabling FIPS or additional STIGs.
   c. Ensure that System Manager is running and is reachable from your virtual machines.

   ADDITIONAL INFORMATION:
   For more information, see See "Deploying new virtual machines" on page 196.
5. Complete the migration.
   a. Install the Avaya Aura® Web Gateway Release 10.2.x application in migration mode.
   b. Start Avaya Aura® Web Gateway.
   c. Perform post-installation procedures.

   ADDITIONAL INFORMATION:
   For more information, see the sections in See "Completing the migration" on page 199.

Migration limitations

The migration process has the following limitations:

- If FIPS is enabled on your Release 3.11 system, you must enable FIPS when migrating to Release 10.2.x. You cannot use backups from a FIPS system to deploy a non-FIPS system.

- If additional STIG hardening is enabled on your Release 3.11 system, you must enable additional STIG hardening when migrating to Release 10.2.x. You cannot use backups from a system where additional STIGs are enabled to deploy a system where additional STIGs are disabled.

Migration considerations

Starting from Release 10.2.x, Avaya Aura® Web Gateway does not support certain virtual environments. Read the following to understand if you need to update your virtual environment before migrating to Avaya Aura® Web Gateway Release 10.2.x:

- If you use Avaya Solutions Platform 120 or Avaya Solutions Platform 130, see See "Avaya Solutions Platform migration considerations" on page 192.
- If you use the VMware virtual environment, see See "VMware ESXi migration considerations" on page 192.
- If you use the AWS virtual environment, see See "Amazon Web Services migration considerations" on page 193.

Avaya Solutions Platform migration considerations

Avaya Solutions Platform 120

Avaya Aura® Web Gateway Release 10.2.x does not support Avaya Solutions Platform 120. Therefore, you must upgrade your Avaya Solutions Platform 120 to Avaya Solutions Platform 130 Release 5 before migrating to Avaya Aura® Web Gateway Release 10.2.x.

> Note:
> Avaya Solutions Platform 120 is synonymous with Avaya Converged Platform 120 and Appliance Virtualization Platform (AVP).

Avaya Solutions Platform 130

Avaya Solutions Platform 130 Release 4 uses the ESXi hypervisor version that Avaya Aura® Web Gateway Release 10.2.x does not support. Therefore, if you use Avaya Solutions Platform 130 Release 4, you must upgrade it to Avaya Solutions Platform 130 Release 5 before migrating to Avaya Aura® Web Gateway Release 10.2.x. For more information about upgrading Avaya Solutions Platform, see See "Upgrading Avaya Solutions Platform from Release 4.0 to 5.0" on page 201.

VMware ESXi migration considerations

Avaya Aura® Web Gateway Release 10.2.x does not support VMware ESXi versions 6.0 and 6.5. Therefore, if you use VMware ESXi version 6.0 or 6.5, you must upgrade it to version 6.7 or later before migrating Avaya Aura® Web Gateway to Release 10.2.x. For information about upgrading VMware ESXi, see the official VMware documentation.

Amazon Web Services migration considerations

Avaya Aura® Web Gateway Release 10.2.x does not support OVA-based Amazon Web Services deployments. Therefore, to continue using Avaya Aura® Web Gateway in the Amazon Web Services environment, migrate from the OVA-based to software-only deployment as follows:

1. Back up your OVA-based Avaya Aura® Web Gateway Release 3.11
2. Deploy new virtual machines in the software-only Amazon Web Services environment
3. Install Avaya Aura® Web Gateway Release 10.2.x using the Release 3.11 backups.

For more information about the migration process, see the migration procedures in this chapter.

Preparing for migration

Use this procedure to record server information and back up your Avaya Aura® Web Gateway Release 3.11 system. This information and backup files are required when you deploy new Avaya Aura® Web Gateway Release 10.2.x virtual machines.
PREREQUISITE

- Read and understand See "Migration limitations" on page 191.
- Configure an off-board backup server.
- If you use Avaya Solutions Platform 120 or Avaya Solutions Platform 130 Release 4 or below, you must upgrade the platform to ASP 130 Release 5. For more information about the upgrade process, see See "Upgrading Avaya Solutions Platform from Release 4.0 to 5.0" on page 201.

  > Note:
  > Avaya Solutions Platform 120 is synonymous with Avaya Converged Platform 120 and Appliance Virtualization Platform (AVP).

- If you use VMWare ESXi 6.0 or 6.5, you must upgrade it to VMware ESXi 6.7 or later.

1. Back up the cluster as described in See "Backing up Avaya Aura Web Gateway" on page 195.
   ADDITIONAL INFORMATION:

   > Important:
   > Do not rename backup files.

   In cluster deployments, perform steps See "2" on page 193 to See "5" on page 194 on the seed node first and then on all non-seed nodes.
2. Log in to Avaya Aura® Web Gateway as an administrator.
3. Run the following command to collect the current system logs:
   ADDITIONAL INFORMATION:

   ```
   app collectlogs collect
   ```

   STEP RESULT:
   Avaya Aura® Web Gateway creates a *.zip* file containing the logs in the */var/log/Avaya/collected-logs* directory.

Support teams can resolve issues that might occur on the new system by comparing the collected logs to the logs on the new system.

4. Do the following to record server information and then save the output in a safe location:
   a. To obtain the fully qualified host name, run the following command:
   ADDITIONAL INFORMATION:
   ```
   hostname -f
   ```
   b. To obtain the IP address and the network mask, run the following command:
   ADDITIONAL INFORMATION:
   ```
   ifconfig -a | grep inet | grep -v 127.0.0.1
   ```
   c. To obtain the IP address of the default gateway, run the following command:
   ADDITIONAL INFORMATION:
   ```
   netstat -nrv | grep '^0.0.0.0'
   ```
   d. To obtain the DNS search list and DNS server IP address, run the following command:
   ADDITIONAL INFORMATION:
   ```
   cat /etc/resolv.conf
   ```
   e. To obtain the NTP server IP address, run the following command:
   ADDITIONAL INFORMATION:
   ```
   cat /etc/ntp.conf | grep "^server"
   ```
   f. Log in as the administrator and run the following commands to display the user name and primary group for the administrative account:
   ADDITIONAL INFORMATION:
   ```
   id --user --name
   id --group --name
   ```
   g. If you want to use the same keystore password, obtain the existing keystore password by referencing notes from the original installation.
   ADDITIONAL INFORMATION:
   If you do not want to use the same keystore password, record a new keystore password.
   h. Record the current System Manager enrollment password by referencing notes from the original installation.
5. Copy the following files to an off-board storage location using a file transfer program, such as SFTP or SCP:
   - The *.zip* archive containing the logs.
   - Any other desired content from the */home* directory.
6. Repeat steps See "2" on page 193 to See "5" on page 194 for all remaining nodes in the cluster.
7. On each node, run the following command to stop Avaya Aura® Web Gateway and verify that services are stopped:
   ADDITIONAL INFORMATION:

```
svc csa stop
svc csa status
```

8. On each node, run the following commands to shut down the Avaya Aura® Web Gateway Release 3.11 virtual machine and turn off the power:
   ADDITIONAL INFORMATION:

   ```
   sudo shutdown -hP now
   ```

   If the virtual machine stays powered on, shut down the virtual machine from the virtual machine host interface.
9. For OVA-based deployments, for each virtual machine in the cluster, use the original installation notes to record the OVA profile used while deploying the original OVA.

AFTER COMPLETING THE TASK
Deploy Avaya Aura® Web Gateway Release 10.2.x virtual machines.

Backing up Avaya Aura® Web Gateway

Use this procedure to back up Avaya Aura® Web Gateway database and configuration files. This procedure applies to both standalone and cluster environments. In a cluster environment, you must back up files from all nodes in the cluster.

> Note:
> Avaya Aura® Web Gateway recommends that you run the Screen utility before running the backup command. If the SSH session terminates while the backup operation is in progress, you can re-connect to the Screen session and complete the backup operation.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrative user.
2. To start the Screen utility, run the following command:
   ADDITIONAL INFORMATION:
   `screen -S <SESSION_NAME>`

   In this command, `SESSION_NAME` is a Screen session name of your choice. For example:

   ```
   screen -S AAWG_backup_session
   ```

   If the SSH session is reset when you create a backup, you can re-connect to the Screen session and complete the backup operation.
3. To create a backup in a *.tar.zip* file, run one of the following commands:
   - `app backup -t`
   - `app backup -t -P '<PASSWORD>'`

   ADDITIONAL INFORMATION:
   In the second command, `<PASSWORD>` is a password for protecting backup files. You must enclose the password with single quotation marks ('). If you do not enter a password or if the password does not comply with the password rules, Avaya Aura® Web Gateway prompts you to provide a password later. For information about supported characters, see See "Characters supported for Avaya Aura Web Gateway passwords" on page 162.

   In a cluster environment, you can run this command on any node. It creates a backup for all nodes in the cluster.

For example, to set the `Password1234#` password for your backups, run the following command:

```
app backup -t -P 'Password1234#'
```

4. If you did not enter a password for backups, then enter it when prompted.
   ADDITIONAL INFORMATION:
   If the password does not comply with the password rules, Avaya Aura® Web Gateway prompts you to enter a new password.
5. Re-enter the password to confirm it.
6. Wait until the backup process is completed.
7. If the SSH session terminates while the backup operation is in progress, do the following to restore the session:
   a. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
   b. To obtain the ID of the Screen session, run the following command:
      ADDITIONAL INFORMATION:
      `screen -ls`

      Avaya Aura® Web Gateway displays the results in the following format:

      <SESSION_ID>.<SESSION_NAME>

      In this entry, <SESSION_ID> is the Screen session ID and <SESSION_NAME> is the Screen session name that you provided.
   c. To reattach to the appropriate session, run the following command:
      ADDITIONAL INFORMATION:
      `screen -rd <SESSION_ID>`
8. Ensure that the backup file with the name *<DateTimestamp>_<node_short-hostname>.tar.zip* is present in the administration directory, which is located at */home/admin*.
9. Copy the backup files to an external storage using a file transfer program, such as SFTP or SCP.
10. In a cluster environment, repeat step 8 and 9 on all remaining nodes in the cluster.
11. Store the password that you used for backup files in a safe place.
    ADDITIONAL INFORMATION:
    Avaya Aura® Web Gateway requires this password when you restore the backups.

AFTER COMPLETING THE TASK
Complete the remaining steps to prepare for migration.

Deploying new virtual machines

Use this procedure to deploy and prepare Avaya Aura® Web Gateway Release 10.2.x virtual machines for the migration. To improve efficiency, you can perform many steps simultaneously across all the nodes in the cluster.
PREREQUISITE

- Create system backups and gather the required server information by completing the migration preparation procedure. For more information, see See "Preparing for migration" on page 193.

- For OVA-based deployments, ensure that you have the latest Avaya Aura® Web Gateway Release 10.2.x installer and OVA.
- For software-only deployments, ensure that you have the latest Avaya Aura® Web Gateway Release 10.2.x software-only installation package.

1. Deploy Avaya Aura® Web Gateway Release 10.2.x virtual machines using the server configuration information gathered for the Release 3.11 system and then start the virtual machines.
   ADDITIONAL INFORMATION:
   For more information about the deployment procedures, see the "Initial setup for VMware and AWS deployments" chapter in *Deploying the Avaya Aura® Web Gateway* for your deployment type.
2. Attach the new virtual machine to the network recorded for the node in See "Preparing for migration" on page 193.
3. For OVA-based deployments, if you have an installer that is newer than the one staged in */opt/Avaya/* within the virtual machine, for each node in the cluster, copy the installer to the administrative user's home directory suing a file transfer program, such as SFTP or SCP.
4. If FIPS was enabled on your Release 3.11 system or if you want to use FIPS, perform See "Enabling FIPS mode" on page 197.
   ADDITIONAL INFORMATION:
   > Important:
   > You must enable FIPS if FIPS was enabled on your Release 3.11 system. You cannot use backups from a FIPS system to deploy a non-FIPS system. If FIPS was not enabled on your Release 3.11 system, you can enable FIPS as required.
5. If additional STIG hardening was enabled on your Release 3.11 or if you want to enable additional STIG hardening, perform See "Enabling additional STIG hardening" on page 160.
   ADDITIONAL INFORMATION:
   > Important:
   > You must enable additional STIG hardening if it was enabled on your Release 3.11 system. You cannot use backups from a system where additional STIGs are enabled to deploy a system where additional STIGs are disabled. If additional STIG hardening was not enabled on your Release 3.11 system, you can enable it as required.

AFTER COMPLETING THE TASK
Install the Avaya Aura® Web Gateway Release 10.2.x application as described in See "Reinstalling Avaya Aura Web Gateway" on page 199.

Enabling FIPS mode

FIPS is a cryptographic security standard. Use this procedure if your enterprise requires FIPS compliance.

FIPS mode is enabled at the operating system level before starting the Avaya Aura® Web Gateway installation. If FIPS is enabled for the operating system, then Avaya Aura® Web Gateway will be installed in FIPS mode. Otherwise, Avaya Aura® Web Gateway will be installed in non-FIPS mode. FIPS installation is only supported for new installations. You cannot upgrade a non-FIPS system to a FIPS system. If you want to enable FIPS on a non-FIPS system or disable FIPS on a FIPS system, you must uninstall the Avaya Aura® Web Gateway application first, change FIPS mode, and then install Avaya Aura® Web Gateway.

> Note:

- This procedure applies to OVA-based deployments. If you deploy Avaya Aura® Web Gateway as a software-only application, you must enable FIPS mode using the procedure for software-only deployments.
- If FIPS mode is enabled, you must use the Secure LDAP (LDAPS) protocol to configure LDAP.
- In cluster deployments, if FIPS mode is enabled, SSL encryption for internode communication between the database servers on the Avaya Aura® Web Gateway nodes is enabled by default.
- OAuth authorization is unavailable in FIPS mode.

1. Log in to the virtual machine with the Avaya Aura® Web Gateway OVA as an administrator.
2. Run the following command to enable FIPS mode:
   ADDITIONAL INFORMATION:
   ```
   sys secconfig --fips --enable
   ```
3. To review the FIPS status, run the following command:
   ADDITIONAL INFORMATION:
   ```
   sys secconfig --fips --query
   ```

Enabling additional STIG hardening

The Security Technical Implementation Guides (STIGs) are the requirements that, when implemented, enhance application security and monitoring capabilities. By default, Avaya Aura® Web Gateway enables essential STIG hardening options during the system layer installation or upgrade. These default settings are appropriate for most deployments. If your organization requires stricter STIG compliance, use this procedure to enable additional Linux STIG security hardening options.

Important:
You cannot enable additional STIG hardening in software-only deployments.

Note:
After you enable additional STIG hardening, ensure that the length of the automatic backup password is at least 15 characters.

1. Log in to the virtual machine with the Avaya Aura® Web Gateway OVA as an administrator.
2. Run the following command to enable additional STIG hardening options:
   ADDITIONAL INFORMATION:
   ```
   sys secconfig --stig --enable
   ```
3. When prompted, press c to apply new password rules.
4. To review the STIG hardening status, run the following command:
   ADDITIONAL INFORMATION:
   ```
   sys secconfig --stig --query
   ```

Checking the connection to System Manager

You must ensure that System Manager is reachable before installing the Avaya Aura® Web Gateway application.

1. Log in to the virtual machine CLI as an administrator using an SSH connection.
2. Run the following command:
   ADDITIONAL INFORMATION:

   ```
   ping -c 10 <SMGR_ADDRESS>
   ```

   In this command, `<SMGR_ADDRESS>` is the FQDN of System Manager.

   If System Manager is reachable, the system displays the following output for the command:

   10 packets transmitted, 10 received, 0% packet loss

AFTER COMPLETING THE TASK

Completing the migration

The following sections provide the tasks for completing the migration to Avaya Aura® Web Gateway Release 10.2.x.

Reinstalling Avaya Aura® Web Gateway

Use this procedure to install the Avaya Aura® Web Gateway Release 10.2.x application in standalone deployments or on a seed node in cluster deployments.

> Note:
> Avaya Aura® Web Gateway recommends that you start a Tmux session before running the installer. If the SSH session terminates while installation is in progress, you can re-connect to the Tmux session and complete the installation. For more information about the Tmux utility, see See "Using the Tmux utility" on page 43.

PREREQUISITE

- Deploy Release 10.2.x virtual machines as described in See "Deploying new virtual machines" on page 196.
- Ensure that the required Avaya Aura® Web Gateway backup files are available for the nodes that are restored. In cluster deployments, each node has its own specific backup file. When installing an Avaya Aura® Web Gateway node, you must use the correct backup file for that node.
- Ensure that System Manager is running and is reachable from your virtual machine. For more information, see See "Checking the connection to System Manager" on page 199.
- If FIPS was enabled on your Release 3.11 system, ensure that you know the System Manager enrollment password.

In cluster deployments, perform steps See "1" on page 200 to See "8" on page 200 on the seed node first and then on all remaining non-seed nodes.

1. Transfer the Avaya Aura® Web Gateway Release 3.11 backup file in *.tar.zip* format to the home directory of the administrator.
   ADDITIONAL INFORMATION:
   You can use a file transfer program of your choice, such as SFTP, SCP, or WinSCP.
2. In cluster deployments, ensure that you transfer each backup file to a node with the corresponding host name.
3. Log in to the Avaya Aura® Web Gateway node as an administrator using an SSH connection.
4. To start a Tmux session, run the following command:
   ADDITIONAL INFORMATION:

   ```
   tmux new-session -s <SESSION_NAME>
   ```

   In this command, `SESSION_NAME` is a tmux session name of your choice. For example:

   ```
   tmux new-session -s AAWG_migration_session
   ```

5. Do one of the following:
   - If you are using the binary installer bundled in the OVA, run the following command:

     ```
     app install -- --migrate <ABS_PATH_TO_BACKUP>
     ```

   - If you are using another binary installer, run the following command:

     ```
     sudo /opt/Avaya/csa-<version>.bin -- --
     migrate <ABS_PATH_TO_BACKUP>
     ```

   ADDITIONAL INFORMATION:
   In these commands, `<ABS_PATH_TO_BACKUP>` is the absolute path to the Release 3.11 backup file in *.tar.zip* format.

   > Important:
   > - Ensure that a double dash (`--`) separates the installer file name from the remainder of the command.
   > - Ensure that a double dash (`--`) prefixes the `migrate` argument.

6. When prompted, enter the password you used to create the backup.
7. Follow the system prompts to complete the migration procedure.
8. When the system prompts you to run *finishUpgrade.sh*, run the following command:
   ADDITIONAL INFORMATION:

   ```
   sudo /opt/Avaya/CallSignallingAgent/<release>/CAS/<release>/
   misc/finishUpgrade.sh
   ```

   In this command, `<release>` is the Avaya Aura® Web Gateway release number. For example: 10.2.x.x.x
9. In cluster deployments, perform steps See "1" on page 200 to See "8" on page 200 on the remaining non-seed nodes.

10. On each Avaya Aura® Web Gateway node, run the following command to start services:
    ADDITIONAL INFORMATION:

    ```
    svc csa start
    ```

AFTER COMPLETING THE TASK
Complete migration as described in See "Performing post-installation procedures" on page 201.

Performing post-installation procedures

After installing Avaya Aura® Web Gateway Release 10.2.x application, perform post-installation procedures to complete the migration process.

This procedure reference both cluster and standalone deployments. If you are working with a standalone Avaya Aura® Web Gateway system, ignore the steps related to other cluster nodes.
PREREQUISITE
Install Avaya Aura® Web Gateway Release 10.2.x application on all nodes.

1. In cluster deployments, to repair the Cassandra database, do the following:
    a. Log in to the seed node as an administrator using an SSH connection.
    b. Run the following commands:
       ADDITIONAL INFORMATION:

       ```
       cdto cas
       ./cassandra/cassandraRepair.sh -M
       ```

       The data replication process might take some time to complete.
2. In cluster deployments, on each non-seed node, run the following command to start services:
    ADDITIONAL INFORMATION:

    ```
    svc csa start
    ```

3. In cluster deployments, generate new RSA keys for SSH connections as described in See "Configuring RSA public and private keys for SSH connections in a cluster" on page 186.
4. Log in to the Avaya Aura® Web Gateway administration portal, navigate to the System Overview page and ensure that the connection status icons for all nodes are green.
5. If you used WebRTC media adaptation in the previous Avaya Aura® Web Gateway release and if you are using Avaya Meetings Server Release 9.1.9 or later, configure WebRTC media adaptation as described in See "Configuring WebRTC media adaptation" on page 64.

Upgrading Avaya Solutions Platform from Release 4.0 to 5.0

Use the following procedure to upgrade the Avaya Solutions Platform from Release 4.0 (ESXi 6.5, Update3) to Release 5.0 (ESXi 7.0, Update2).

1. Take the backup of your Avaya Aura® Device Services and keep it on remote servers.
2. Log in to the application through vSphere Web Client, select and right-click the Avaya Aura® Device Services application, and then click **Power › Power Off**.
3. Upgrade Avaya Solutions Platform from Release 4.0 to 5.0.

ADDITIONAL INFORMATION:
For information about upgrading Avaya Solutions Platform from Release 4.0 to 5.0, see "Avaya Solutions Platform 130 Series: Upgrading to ESXi 7.0 u2 from ESXi 6.5.x".
STEP RESULT:
If the Avaya Solutions Platform upgrade is successful, power on the Avaya Aura® Device Services application and ensure the applications are up and running.

If the Avaya Solutions Platform upgrade failed, then:

- Do the fresh deployment of Avaya Solutions Platform 130 Release 5.0. For information about installing Avaya Solutions Platform, see "Installing the Avaya Solutions Platform 130 Series".
- Deploy the application at the same version that was before the Avaya Solutions Platform upgrade. For information about deploying the application, see the application-specific deployment document.

4. If the Avaya Aura® Device Services applications are not up and running:
   a. Do the fresh deployment of the application at the same version that was before the Avaya Solutions Platform upgrade.
   b. Restore the backup taken at step 1.

Avaya Aura® Web Gateway upgrade checklist

Perform the tasks in this checklist to upgrade Avaya Aura® Web Gateway within Release 10.2.x. This checklist applies both to OVA-based and to software-only deployment types.

Important:
Do not use this checklist to upgrade from Release 3.11 to R10.2.x. To upgrade from Release 3.11 to R10.2.x, you must use the migration procedures as described in See "Migrating Avaya Aura Web Gateway" on page 191.

No.	Task	Description	✔
1	Back up Avaya Aura® Web Gateway.	See See "Backing up Avaya Aura Web Gateway" on page 179.	
2	Update the system layer, if required.	For OVA-based deployments, see the procedures in See "Checklist for updating the system layer" on page 203. For software-only deployments, see See "Updating the system layer for software-only deployments" on page 206. In a cluster environment, update the system layer on all cluster nodes before upgrading the application layer.	
3	Upgrade the application layer.	For upgrade steps, see See "Upgrading Avaya Aura Web Gateway to a new version or release" on page 207.	

System layer (operating system) updates for virtual machines deployed using Avaya-provided OVAs

Each virtual machine that is created by deploying the Avaya Aura® Web Gateway OVA file has a system layer (operating system). The system layer is updated with system layer updates provided by Avaya.

> Important:
> Do not apply updates obtained from sources other than Avaya to the system layer of Avaya Aura® Web Gateway virtual machines. Only use update artifacts provided by Avaya.

Checklist for updating the system layer

Use this checklist to update the system layer. This checklist apples to OVA-based deployments.

No.	Task	Notes	✔
1	Determine if the system layer update is applicable to the given virtual machine.	See See "Determining if a system update is applicable" on page 203. Skip the remaining steps if the update is not applicable.	
2	Download, extract, and stage the update.	See See "Downloading, extracting, and staging a system layer update" on page 204.	
3	Install the update during a maintenance window.	See See "Installing a staged system layer update" on page 205.	

Determining if a system update is applicable

Before installing a system update for a virtual machine, query the version of the currently installed system. Use the current version to determine if the system layer requires an update. The virtual machine might be installed using an OVA that was already built with the latest system layer version.

1. Log in to the virtual machine using the administrative user ID.
2. Query the version number of the system version by running the `sys versions` command.
   ADDITIONAL INFORMATION:
   > Note:
   > Ignore the patch level reported by the above command.

AFTER COMPLETING THE TASK

- If the above system version is already on the recommended system update, then no further action is required.
- If the system version is lower than the recommended system update version, then continue with the process to download and stage the update.

Downloading, extracting, and staging a system layer update

Before installing a system layer update, you must first download the update from the Avaya Support website, and then extract and stage the update on the system. The staging process places the update into a system area, which prepares the system for the installation of the update.

> Tip:
> Avaya recommends cleaning up the downloaded and extracted artifacts after staging. The staging operation copies the content to an internal system area. The downloaded and extracted content are no longer required.

1. Download the update from the Avaya Support website.
   ADDITIONAL INFORMATION:
   ucapp-system-3.4.3.0.7.tgz is an example of a system layer update artifact.
2. Transfer the update to the administrative account of the server to be updated, using standard file transfer methods, such as SFTP or SCP.
3. Log in to the administrative account of the server using SSH.
4. To extract the update, run the following command:
   ADDITIONAL INFORMATION:
   ```
   tar -zxf ucapp-system-<version>.tgz
   ```
   For example:
   ```
   tar -zxf ucapp-system-3.4.3.0.7.tgz
   ```
5. To stage the update, change to the required directory and perform the following staging command:
   ADDITIONAL INFORMATION:
   ```
   cd ucapp-system-<version>
   sudo ./update.sh --stage
   ```
   For example:
   ```
   cd ucapp-system-3.4.3.0.7
   sudo ./update.sh --stage
   ```
6. To free up disk space, clean up the downloaded and extracted files using the following commands:
   ADDITIONAL INFORMATION:
   ```
   cd..
   rm ucapp-system-<version>.tgz
   rm -rf ucapp-system-<version>
   ```
   For example:

```
rm ucapp-system-3.4.3.0.7.tgz
rm -rf ucapp-system-3.4.3.0.7
```

7. To verify that the update has been staged, query the status:
   ADDITIONAL INFORMATION:

   ```
   sysUpdate --status
   ```

   ADDITIONAL INFORMATION:
   The **sysUpdate** command is added to the system the first time a system update is staged. After staging, if the command is not recognized, you must exit the current session and establish a new session. Establishing a new session creates the **sysUpdate** command (alias) for the new session.
8. If a system update is staged in error, run the following command to delete this staged update:
   ADDITIONAL INFORMATION:

   ```
   sysUpdate --delete
   ```

   You cannot delete a staged update once the installation of the update has started.
9. For more information about the **sysUpdate** command, run one of the following commands:
   ADDITIONAL INFORMATION:

   ```
   sysUpdate --help
   sysUpdate --hhelp
   ```

   The **--help** option provides the command line syntax. The **--hhelp** option provides verbose help.

AFTER COMPLETING THE TASK
Install the staged update during a maintenance window.

Installing a staged system layer update

After a system update is staged, you can install it. The installation runs in the background in order to minimize the possibility of interference, such as the loss of an SSH session. The background installation process follows these steps:

- A login warning message is created so users logging into the system know that a system update is in progress.
- If the application is running, it is shut down.
- The update is installed onto the system.
- The server is rebooted.
- Post-reboot cleanup actions are performed.
- The application is started.
- The login warning message is removed.

> Important:

> Do not perform any system maintenance actions, such as starting, stopping, or upgrading the application, while the system update is in progress.

1. Log in to the administrative account using SSH.
2. Type `sysUpdate ––install` to start the installation.
   ADDITIONAL INFORMATION:
   Avaya Aura® Web Gateway reboots after the installation process completes.
3. Run one of the following commands to monitor the progress of the update:
   ADDITIONAL INFORMATION:
   ```
   sysUpdate --monitor
   sysUpdate --monitor less
   ```
   The **--monitor** option uses the Linux tail browser. The **-- monitor less** option uses the Linux less browser.
4. Run the following command to review the status of the update:
   ADDITIONAL INFORMATION:
   ```
   sysUpdate --status
   ```
5. Run the following command to obtain logs of the current and previous system layer update installations:
   ADDITIONAL INFORMATION:
   ```
   sysUpdate --logs
   ```
   This command gathers log files into an archive in ZIP format and places this archive in the current working directory.

AFTER COMPLETING THE TASK
If your organization requires stricter STIG compliance, re-enable additional STIG hardening after you finish installing the system layer update. For more information, see See "Enabling additional STIG hardening" on page 160.

Updating the system layer for software-only deployments

The system layer is low-level software that provides the required operational environment for the Avaya Aura® Web Gateway application. You must update the system layer before upgrading your Avaya Aura® Web Gateway application.

For software-only deployments, Avaya provides a system layer installer in a package, which also contains an Avaya Aura® Web Gateway application installer.

For information about installing system layer updates for OVA-based deployments, see See "System layer (operating system) updates for virtual machines deployed using Avaya-provided OVAs" on page 203.
PREREQUISITE
Download the latest *csa-swonly-<AAWG_VERSION>.tgz* software-only installation package from PLDS.

1. Upload the software-only installation package on the virtual machine to the */root* directory.
   ADDITIONAL INFORMATION:
   Use a file transfer program of your choice, such as SFTP, SCP or WinSCP.

2. Log in to the virtual machine as the root user.
3. To extract the content of the installation package, run the following command:
   ADDITIONAL INFORMATION:
   ```
   tar -xzf aads-swonly-<AADS_VERSION>.tgz
   ```
   For example: `tar -xzf aads-swonly-10.1.1.0.x.tgz`
   ADDITIONAL INFORMATION:
   ```
   tar -xzf csa-swonly-<AAWG_VERSION>.tgz
   ```
   For example: `tar -xzf csa-swonly-3.9.1.0.x.tgz`
   STEP RESULT:
   The */root/csa-swonly-AAWG_RELEASE/* directory contains the following files:
   - The system layer package in TGZ archive *ucapp-swonly-system-<VERSION>.tgz*.
   - The Avaya Aura® Web Gateway application binary *csa-<AAWG_RELEASE>.bin*.
4. Go to the directory with the extracted installation package:
   ADDITIONAL INFORMATION:
   ```
   cd aads-swonly-<AADS_VERSION>
   ```
   ADDITIONAL INFORMATION:
   ```
   cd csa-swonly-<AAWG_VERSION>
   ```
5. To extract the content of the archive with the system layer installation files, run the following command:
   ADDITIONAL INFORMATION:
   ```
   tar -xzf ucapp-swonly-system-<SYSTEM_LAYER_VERSION>.tgz
   ```
   For example: `tar - xzf ucapp-swonly-system-1.0.0.0.8.tgz`
6. Go to the directory with the extracted system layer package:
   ADDITIONAL INFORMATION:
   ```
   cd ucapp-swonly-system-<SYSTEM_LAYER_VERSION>
   ```
7. Run the following command to install the latest system layer:
   ADDITIONAL INFORMATION:
   ```
   ./swOnlyUpdate.sh --patch
   ```

Upgrading Avaya Aura® Web Gateway to a new version or release

Use this procedure to upgrade Avaya Aura® Web Gateway within the same major version. Avaya Aura® Web Gateway is automatically stopped for the upgrade and remains out of service until you restart it after the upgrade is completed.

> Important:
>
> - Do not use this procedure to upgrade from Release 3.11 to 10.2.x. To upgrade from Release 3.11 to 10.2.x, you must use the migration procedures as described in See "Migrating Avaya Aura Web Gateway" on page 191.
> - When upgrading an Avaya Aura® Web Gateway cluster, you can upgrade cluster nodes in any order. After the first node is upgraded, you can upgrade other nodes at the same time.
> - You can upgrade an Avaya Aura® Web Gateway cluster even if some nodes are turned off. You can upgrade these nodes later.

PREREQUISITE

- Download the required upgrade file from the Avaya support site. *csa-10.2.x.x.x.bin* is an example of an application layer upgrade file.
- Ensure that the Avaya Aura® Web Gateway application is installed. Otherwise, the `app upgrade` command will not work. For more information about Avaya Aura® Web Gateway installation, see *Deploying the Avaya Aura® Web Gateway*.
- Ensure that you know the enrollment password. Avaya Aura® Web Gateway does not store the enrollment password, so you must enter it during the upgrade.
- In a cluster environment, if you use Avaya Spaces Calling Gateway, enable Maintenance mode as described in See "Enabling Maintenance mode" on page 110.
- Upgrade the system layer (operating system):
    - For Avaya Aura® Web Gateway systems deployed using Avaya-provided OVAs, see the procedures in See "System layer (operating system) updates for virtual machines deployed using Avaya-provided OVAs" on page 203.

      > Important:
      > In a cluster environment, update the system layer on all nodes before upgrading Avaya Aura® Web Gateway to a new version.

    - For Avaya Aura® Web Gateway systems deployed as a software-only solution, see the procedures in See "Updating the system layer for software-only deployments" on page 206.

1. Transfer the upgrade file to the administrator home folder on the Avaya Aura® Web Gateway server using a file transfer tool of your choice.
2. Log in to the Avaya Aura® Web Gateway CLI as an administrator.
3. Set executable permissions on the upgrade file by using the following command:
   ADDITIONAL INFORMATION:
   ```
   chmod 770 <filename>
   ```
   For example:
   ```
   chmod 770 csa-10.2.x.x.x.bin
   ```
4. Run the following command to start the upgrade:
   ADDITIONAL INFORMATION:
   ```
   app upgrade <filename>
   ```
   For example:
   ```
   app upgrade csa-10.2.x.x.x.bin
   ```
5. Follow the prompts to complete the upgrade.
6. Restart services by running the `svc csa restart` command.
   ADDITIONAL INFORMATION:
   If you are upgrading a cluster, you can restart an upgraded node before the upgrade is completed on other nodes.

AFTER COMPLETING THE TASK

- If WebRTC media adaptation was enabled in the previous Avaya Aura® Web Gateway release and if you are using Avaya Meetings Server Release 9.1.9 or later, configure WebRTC media adaptation as described in See "Configuring WebRTC media adaptation" on page 64.
- If the push notifications feature was enabled in the previous Avaya Aura® Web Gateway release and if you use the Avaya Push Notification provider, configure the Avaya Push Notification provider with the "pnp.avaya.com" provider address. For more information, see See "Reconfiguring the Avaya Push Notification provider after upgrade or migration" on page 210.
- Optionally, configure AIDE scanning as described in See "Advanced Intrusion Detection Environment tool management" on page 148.
- In a cluster environment, if you use Avaya Spaces Calling Gateway, disable Maintenance mode as described in See "Disabling Maintenance mode" on page 111.

Rolling back to the earlier version

If you have performed at least one upgrade on your system, then you can roll back to the earlier installed version. Avaya Aura® Web Gateway is stopped while the rollback is in progress.

If you perform a rollback after changing Avaya Aura® Web Gateway IP addresses or FQDNs, Avaya Aura® Web Gateway will continue using these new IP addresses or FQDNs after the rollback. The rollback process does not revert IP address or FQDN changes.
PREREQUISITE
You must verify that it is possible to roll back a cluster by ensuring that all the nodes in the cluster have the same inactive version available. Do not attempt a cluster rollback if all nodes cannot rollback to the same inactive version. Run the following command to display the inactive version on each node:

```
cat /etc/ucapp.properties | grep UCAPP_CAS_INACTIVE
```

1. Open the Linux shell using your Linux administrator account credentials.
2. To roll back to the earlier version, run the `app rollback` command.
   ADDITIONAL INFORMATION:
   In a cluster, roll back each node one at a time. Roll back the seed node last.
3. After the rollback is completed, restart services by running the `svc csa restart` command.
   ADDITIONAL INFORMATION:
   If the node is part of a cluster, complete the rollback on the other nodes, one at a time, before restarting any of the cluster nodes.

Removing an inactive version

If you have performed at least one upgrade or rollback on your system, then the earlier version is retained on the system. Earlier versions are retained so that you can roll back when needed. Use this procedure if you would like to remove the inactive version to free disk space.

> Important:
> After you remove the inactive version from your system, you cannot perform any rollbacks.

1. Open the Linux shell using your Linux administrator account credentials.
2. To remove the inactive version from the system, run the `app removeinactive` command.

Reconfiguring the Avaya Push Notification provider after upgrade or migration

After you upgrade or migrate to Release 3.8, Avaya Aura® Web Gateway continues to use "apnp.avaya.com" as the default provider address for the Avaya Push Notification provider. Avaya recommends that you use the "pnp.avaya.com" domain instead. Since you cannot update or delete the default Avaya Push Notification provider, you can configure a new provider with the "pnp.avaya.com" provider address.

This procedure only applies if you are upgrading or migrating to Release 3.8. If you are performing a fresh installation, Avaya Aura® Web Gateway will use "pnp.avaya.com" as the default provider address for the Avaya Push Notification provider.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Push Notification Settings › Provider Settings**.
2. Click **Add** to add a new provider configuration.
3. In **Enter Company Domain**, enter the domain where your Avaya Aura® Web Gateway is deployed.
   ADDITIONAL INFORMATION:
   For example: `mycompany.com`
4. In **Push Notification Provider Name**, enter a name for the provider.
   ADDITIONAL INFORMATION:
   For example: `AvayaProviderUpdated`.

   This name is used in the Avaya Aura® Web Gateway administration portal for display purposes only.
5. In **Push Notification Provider Address**, enter `pnp.avaya.com`.
6. In **Push Notification Provider Port**, enter the port number for the push notification provider.
   ADDITIONAL INFORMATION:
   The default port is 443.
7. Click **Generate Key**.
   STEP RESULT:
   The Avaya Aura® Web Gateway generates a public and private key pair and an identifier. This data is required to authorize Avaya Aura® Web Gateway on the push notification provider. The Avaya Aura® Web Gateway also updates the following values:

   - **System Id**: A unique identifier for your system.
   - **Public Key**: A public key.
8. Click **Export**.
   ADDITIONAL INFORMATION:
   Avaya Aura® Web Gateway displays a pop-up window with the authorization information in JSON format. You must provide this data to your Avaya Cloud account.

   The following is an example of the authorization data:

```
{
    "systemId": "9bf8f4ab-99b1-452b-9b7f-
e75aacf31d19.mycompany.com",
    "description": "Avaya Aura Web Gateway Services",
    "publicKey": "-----BEGIN PUBLIC KEY-----
\nMFkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDQgAE9rtz4fuYhGm2JlvnI6lZmate8e
EX\na4wvmklSdHGYZHos7y8xNBNCEj9wc3klayOKHYIVIeL0ryVFgM16Ud5FDQ==
\n-----END PUBLIC KEY-----",
    "alg": "ES256"
}
```


Important:

- Do not close the Provider Settings page on the Avaya Aura® Web Gateway administration portal until you finish exporting authorization data to your Avaya Spaces account.
- Do not save the data you provided on the Provider Settings page of the Avaya Aura® Web Gateway administration portal until you export authorization data to your Avaya Cloud account. Otherwise, the push notification service might not work as expected.

9. Copy the authorization data, including the surrounding curly brackets, and save it in a file on your computer.
10. In a new browser tab, log in to your Avaya Cloud account and navigate to **Manage Companies**.
11. Select the required company and navigate to the **Apps** tab.
12. From **App**, select **Avaya Mobile Push Notification Service**.
13. In Data Configuration, select the **JSON** check box.
14. Replace the content in **Public Settings** with the content of the file that you created and then save your changes.
    ADDITIONAL INFORMATION:
    You must ensure that the authorization data you enter in **Public Settings** is a valid JSON string. If the data uses an invalid format, the Avaya Spaces account displays a warning message, but still allows you to save changes.

    The Avaya Spaces account stores authorization data internally. **Public Settings** only displays the authorization data that you entered last. You will not lose any previously entered authorization data if you overwrite existing content with the new authorization data.
15. Return to the Provider Settings page on the Avaya Aura® Web Gateway administration portal.
16. Click **Test** to verify that your system can connect and authenticate with the push notification provider.
17. Do one of the following:
    - If the verification completed successfully, click **Save**.
    - If the verification failed, fix the issue as described in See "Avaya Aura Web Gateway cannot connect to a push notification provider" on page 217 and then and re-run the connectivity test.

      If the problem persists, contact Avaya support personnel.

AFTER COMPLETING THE TASK

- Update the firewall rules for the "pnp.avaya.com" domain.
- Update push notification configurations for mobile applications.

Updating mobile application settings

After configuring a new Avaya Push Notification provider, you must select this provider for all iOS applications that currently use the default Avaya Push Notification provider.
PREREQUISITE
Configure a new Avaya Push Notification provider that uses "pnp.avaya.com" as the domain name. For more information, see See "Reconfiguring the Avaya Push Notification provider after upgrade or migration" on page 210.

1. On the Avaya Aura® Web Gateway administration portal, navigate to **Advanced › Push Notification Settings › Mobile Application Settings**.
2. In **Application Name**, select an application that uses the default Avaya Push Notification provider.
   ADDITIONAL INFORMATION:
   The name of the Avaya Push Notification Provider is "Avaya Provider".
3. Click **Edit**.
4. In **Push Notification Provider**, select the new Avaya Push Notification provider.
   ADDITIONAL INFORMATION:
   For example: **AvayaProviderUpdated**.
5. Click **Save**.
6. To verify that the Avaya Aura® Web Gateway can send notifications to the mobile application, click **Test**.
   STEP RESULT:
7. Do one of the following:
   - If the verification completed successfully, click **Save**.
   - If the verification failed, fix the issue as described in See "iOS application cannot connect to a push notification provider" on page 219 and then re-run the connection test.

     If the problem persists, contact Avaya support personnel.
8. Repeat the steps above for all mobile applications that use the default Avaya Push Notification provider.

Upgrading RHEL 8.4 to RHEL 8.10

To upgrade the Avaya Aura® Web Gateway OS from Red Hat Enterprise Linux (RHEL) 8.4 to RHEL 8.10, do the following procedure.

1. Upload the RHEL upgrade package to the virtual machine's admin directory.
   ADDITIONAL INFORMATION:
   Use a file transfer program of your choice, such as SFTP, SCP, or WinSCP.
2. Log in to the Avaya Aura® Web Gateway CLI
3. Run the following command to upgrade the OS:
   ADDITIONAL INFORMATION:
   av-upgrade-os AV-AAWG10.x-RHEL8.10-OSUpdate-001.tar.bz2

Upgrading VMware ESXi version for Avaya Aura® Web Gateway

If the ESXi upgrade is required for upgrading Avaya Aura® Web Gateway to Release 10.x, use the following procedure to upgrade ESXi to supported ESXi version.
PREREQUISITE

- You must do the backup of the AAWG Virtual Machine (VM).
- You must power off AAWG VM.

1. Log into **vSphere Client**
2. Log into **vCenter Server** using **vSphere Client** credentials.
   ADDITIONAL INFORMATION:
   The **vCenter Server** manages the ESXi host running the AAWG VM.
3. In the **vSphere Client**, navigate to the AAWG VM.
4. To power off the AAWG VM, do the following:
   a. Go to the **vSphere Client**
   b. Right-click on the name of the AAWG VM and select **Power › Power Off**.
5. Right-click on the name of the AAWG VM and select **Compatibility › Upgrade VM Compatibility**
   ADDITIONAL INFORMATION:
   The **Upgrade Virtual Machine Compatibility** window is displayed.
6. In the **Upgrade Virtual Machine Compatibility** window, select **ESXi 8.x** from the list.
   ADDITIONAL INFORMATION:
   A VMware shows the new VM hardware version that is compatible with the selected ESXi version.
7. To start the upgrade of VMware version, click **OK**.
   ADDITIONAL INFORMATION:
   The VM is upgraded to the selected compatibility level.
8. To power on the AAWG VM, do the following:
   a. Go to the **vSphere Client**
   b. Right-click on the name of the AAWG VM and select **Power › Power On**.

   ADDITIONAL INFORMATION:
   VM is started.

Troubleshooting

Avaya Aura® Web Gateway administration portal is not accessible when the primary node is not working

Condition

In a cluster, when the primary Avaya Aura® Web Gateway node is down, you cannot access the Avaya Aura® Web Gateway administration portal.

Solution

1. You can use the second node to access the Avaya Aura® Web Gateway administration portal.

Cannot log in to the Avaya Aura® Web Gateway web administration portal after changing the System Manager password

Condition

You cannot access the Avaya Aura® Web Gateway web administration portal after changing the System Manager administration password or migrating System Manager.

Cause

Avaya Aura® Web Gateway sends requests to System Manager approximately once per minute. These requests contain the System Manager password. When the System Manager password changes, Avaya Aura® Web Gateway continues to use the old password in requests. After multiple consecutive failed login attempts, System Manager locks out the administrative account used for Avaya Aura® Web Gateway authentication.

Solution

1. On the System Manager web administration portal, create a separate administrative account for Avaya Aura® Web Gateway.
   ADDITIONAL INFORMATION:
   If you use a separate administrative account for Avaya Aura® Web Gateway and this account gets locked, it will only affect Avaya Aura® Web Gateway and no other solution

components. For information about configuring administrative accounts on System Manager, see *Administering Avaya Aura® System Manager*.

> Note:
> If you already use a separate administrative account for Avaya Aura® Web Gateway, skip this step and continue with step 2.

2. To unlock the administrative account, on the System Manager web administration portal, disable the password lockout functionality.
   ADDITIONAL INFORMATION:
   For information about disabling the password lockout functionality, see *Administering Avaya Aura® System Manager*.
3. Log in to the Avaya Aura® Web Gateway web administration portal.
4. On the **General Network Settings › System Manager** page, type the current System Manager password.
5. Navigate to the **System Overview** page and wait until the status indicator for System Manager becomes green.
6. On the System Manager web administration portal, enable the password lockout functionality as appropriate.

System Manager does not show Avaya Aura® Web Gateway alarms

Condition

Alarms are generated on the Avaya Aura® Web Gateway, and user profiles are appropriately created and assigned, but System Manager does not show these alarms.

Solution

1. Enable EASG functionality.
   ADDITIONAL INFORMATION:
   You require EASG to obtain the sroot user privileges, which are required to run certain commands in this procedure.
2. Log in to the Avaya Aura® Web Gateway.
3. Go to the */var/net-snmp* directory.
4. Memorize the timestamp of the *snmpd.conf* file.
5. Log in to the System Manager web console and navigate to
   Home › Services › Inventory › Manage Serviceability Agents › Serviceability Agents.
6. From the agents list, select the Avaya Aura® Web Gateway node for which alarms are not displayed on System Manager.
7. On the Serviceability Agents page, select Avaya Aura® Web Gateway and click **Manage Profiles**.
8. On the next page, click **Commit**.
   STEP RESULT:
   The timestamp of the *snmpd.conf* file should be updated.
9. If the timestamp of the *snmpd.conf* file was not updated, perform the remaining steps.

ADDITIONAL INFORMATION:
If the timestamp was updated, then you do not need to do anything else.

10. Log in to System Manager as the root user using SSH.
11. Run the `locate recoverAgent.sh` command to obtain the full path to the `recoverAgent.sh` script.
ADDITIONAL INFORMATION:
For example:

```
>locate recoverAgent.sh
>/opt/Avaya/Mgmt/7.1.11/remoteSnmpConfig/utility/recoverAgent.sh
```

12. Run the `service jboss restart` command to restart JBoss on System Manager.
13. Run the following command to remove the Avaya Aura® Web Gateway entry from System Manager:
ADDITIONAL INFORMATION:

```
sh <path_to_recoverAgent.sh> <AAWG_IP_address>
```

Where, `<path_to_recoverAgent.sh>` is the full path to *recoverAgent.sh* and `<AAWG_IP_address>` is the IP address of the Avaya Aura® Web Gateway.

14. Log in to the Avaya Aura® Web Gateway.
15. Run the following command:
ADDITIONAL INFORMATION:

```
sudo -E $SPIRIT_HOME/scripts/utils/
reinitilizeSnmpdConfiguration.sh
```

> Note:
> The command might fail when restarting snmpd. This is the expected behavior.

```
Stopping existing snmpd service...
Stopping snmpd (via systemctl): [ OK ]
Restarting snmpd (via systemctl): Job for snmpd.service failed
because the control process exited with error code. See
"systemctl status snmpd.service" and "journalctl -xe" for
details.
 [FAILED]
Setting the reinitialized property to true
```

16. Run the following command:
ADDITIONAL INFORMATION:

```
sudo systemctl restart CSASpiritAgent.service
```

Clients cannot connect to the Avaya Aura® Web Gateway

Condition

Clients cannot connect to the Avaya Aura® Web Gateway if the Avaya Aura® Web Gateway does not use the default front-end port.

Cause

If the port number is not specified in the service URLs that are used to connect to the Avaya Aura® Web Gateway, clients continue to use the default 443 front-end port.

Solution

1. Include the actual Avaya Aura® Web Gateway front-end port number in all service URLs. For example: If you are using the `aawg_server.company.com` service URL to access the Avaya Aura® Web Gateway and the Avaya Aura® Web Gateway front-end port is 1000, set the service URL to `aawg_server.company.com:1000`.

Avaya Aura® Web Gateway cannot connect to a push notification provider

Condition

When you test the connectivity to the push notification server, it fails.

Solution

1. Ensure that the enterprise firewall is configured as described in See "Firewall configuration" on page 117.
2. Do one of the following:
    - If you are testing the connectivity to the Avaya Push Notification provider, ensure that the authorization data, which you entered in **Public Settings** on your Avaya Cloud account, is a valid JSON string with the correct Avaya Aura® Web Gateway data.
    - If you are testing the connectivity to a third-party push notification provider, ensure that you correctly entered the system ID and public key information on your push notification provider.
3. If the problem persists, contact Avaya support.

Avaya Aura® Web Gateway returns a TLS handshake error when testing the connectivity to a push notification server

Condition

When you test the connectivity to the push notification server, it fails with a TLS handshake error.

Cause

The issue might occur when your company uses an outgoing TLS inspector, such as ZScaler. The TLS inspector uses its own self-signed certificates to connect to the *pnp.avaya.com* Avaya Push Notification provider address.

Solution

1. From the Avaya Aura® Web Gateway CLI, run the following command to ensure that a TLS inspector is used:
   ADDITIONAL INFORMATION:

   ```
   openssl s_client -connect pnp.avaya.com:443
   ```

   If the command output does not display an Entrust certificate in the certificate chain, it means that the certificate is provided by a third-party.

   The following example shows a command output when Avaya Aura® Web Gateway uses built-in RHEL certificates for Avaya Push Notification service:

   ```
   Certificate chain
    0 s:C = US, ST = New Jersey, L = Morristown, O = "Avaya, Inc.",
   CN = pnp.avaya.com
      i:C = US, O = "Entrust, Inc.", OU = See www.entrust.net/
   legal-terms, OU = "(c) 2012 Entrust, Inc. - for authorized use
   only", CN = Entrust Certification Authority - L1K
    1 s:C = US, O = "Entrust, Inc.", OU = See www.entrust.net/
   legal-terms, OU = "(c) 2012 Entrust, Inc. - for authorized use
   only", CN = Entrust Certification Authority - L1K
      i:C = US, O = "Entrust, Inc.", OU = See www.entrust.net/
   legal-terms, OU = "(c) 2009 Entrust, Inc. - for authorized use
   only", CN = Entrust Root Certification Authority - G2
    2 s:C = US, O = "Entrust, Inc.", OU = See www.entrust.net/
   legal-terms, OU = "(c) 2009 Entrust, Inc. - for authorized use
   only", CN = Entrust Root Certification Authority - G2
      i:C = US, O = "Entrust, Inc.", OU = See www.entrust.net/
   legal-terms, OU = "(c) 2009 Entrust, Inc. - for authorized use
   only", CN = Entrust Root Certification Authority - G2
   ```

   apnp.avaya.com uses a different CA certificate, so if that FQDN is used, you should see the AvayaITserverCA2 certificate in the chain. For example:

```
openssl s_client -connect apnp.avaya.com:443
CONNECTED(00000005)
depth=2 O = Avaya, OU = IT, CN = AvayaITrootCA2
verify return:1
depth=1 DC = com, DC = avaya, DC = global, CN = AvayaITserverCA2
verify return:1
depth=0 C = US, ST = NJ, O = Avaya, CN = apnp.avaya.com
verify return:1
---
Certificate chain
 0 s:C = US, ST = NJ, O = Avaya, CN = apnp.avaya.com
   i:DC = com, DC = avaya, DC = global, CN = AvayaITserverCA2
 1 s:DC = com, DC = avaya, DC = global, CN = AvayaITserverCA2
   i:O = Avaya, OU = IT, CN = AvayaITrootCA2
 2 s:O = Avaya, OU = IT, CN = AvayaITrootCA2
   i:O = Avaya, OU = IT, CN = AvayaITrootCA2
```

2. If a third-party certificate is used, import the TLS inspector CA certificates into the Avaya Aura® Web Gateway truststore.
   ADDITIONAL INFORMATION:
   For more information, see See "Managing truststore certificates" on page 89.

iOS application cannot connect to a push notification provider

Condition

When you test the connectivity, the result indicates that the Avaya Aura® Web Gateway cannot send push notifications to the iOS application.

Cause

- Your system cannot connect with the push notification provider.
- Your mobile application is incorrectly configured on the push notification provider.

Solution

1. On the Provider Settings page of the Avaya Aura® Web Gateway administration portal, click **Test** to verify that your system can connect and authenticate with the push notification provider.
2. After successfully completing the previous step, if the problem persists, do the following:
   - For Avaya applications, such as Avaya Workplace Client for iOS, contact Avaya support.
   - For third-party iOS applications, verify that the application is correctly configured on the third-party provider.

Avaya Aura® Web Gateway displays a warning about SSH configuration during the upgrade process

Condition

After checking the configuration during an upgrade, Avaya Aura® Web Gateway displays a warning that the SSH protocol version is not defined in the *sshd_config* file. For example:

```
Checking SSH Configuration ..............................  [WARNING]
Avaya Aura Web Gateway Services uses EASG for Avaya Services access,
and requires certain sshd_config parameters.
Expected values are:
    Protocol 2
    PermitRootLogin no
    ChallengeResponseAuthentication yes
Actual values are:
    -not defined-
    PermitRootLogin no
    ChallengeResponseAuthentication yes
```

Solution

1. Ignore the warning.

Avaya Aura® Web Gateway does not display the default Avaya Push Notification provider or mobile application configuration

Condition

The Avaya Aura® Web Gateway administration portal does not display any of the following default configurations:

- Avaya Push Notification provider configuration. The configuration name is "Avaya Provider".
- Mobile application configuration. The configuration name is "Avaya Workplace".

You also cannot add your own push notification provider or mobile application configurations.

Solution

1. Log in to the Avaya Aura® Web Gateway as an administrator.
2. Run the `svc cassandra status` command and ensure that the Cassandra status is "Running".
3. If the Cassandra status is not "Running", run the `svc csa start` command.

4. Repeat the steps above for all remaining nodes in the cluster.
5. From any node in the cluster, run the following command:
   ADDITIONAL INFORMATION:
   ```
   /opt/Avaya/CallSignallingAgent/<version>/CAS/<version>/
   cassandra/cassandraRepair.sh -M
   ```

The location of automatic backups has changed after the upgrade

Condition

After the upgrade, Avaya Aura® Web Gateway uses the default location for automatic backup files instead of the location that you defined.

Cause

This is the expected behavior.

Solution

1. Configure the required location for automatic backups as described in See "Changing the location of automatic backups" on page 181.

Avaya Aura® Web Gateway installation or upgrade fails due to invalid System Manager certificates

Condition

Avaya Aura® Web Gateway installation or upgrade fails due to invalid System Manager certificates.

Cause

During the installation or upgrade process, Avaya Aura® Web Gateway connects to System Manager to obtain certificates required for installation. If the System Manager certificates contain unsupported critical extensions, expired, or the certificate start date is in the future, Avaya Aura® Web Gateway does not import these certificates and stops the installation or upgrade process.

Solution

1. Contact the System Manager administrator.

A user gets HTTP error 500 when performing an activate call request

Condition

A user can log in to their client, but cannot perform calls. The client HTTP logs contain an HTTP error 500 "incomplete telephone configuration for user" entry. The Avaya Aura® Web Gateway nginx access log contains an HTTP error 500 entry for that user.

Cause

The issue might be caused by incorrectly configured Avaya Aura® Device Services dynamic configuration settings on Avaya Aura® Device Services.

Solution

1. In your browser, type the following URL:
   ADDITIONAL INFORMATION:
   https://<AADS_FQDN>/acs/resources/configurations/user?username=<USERNAME>

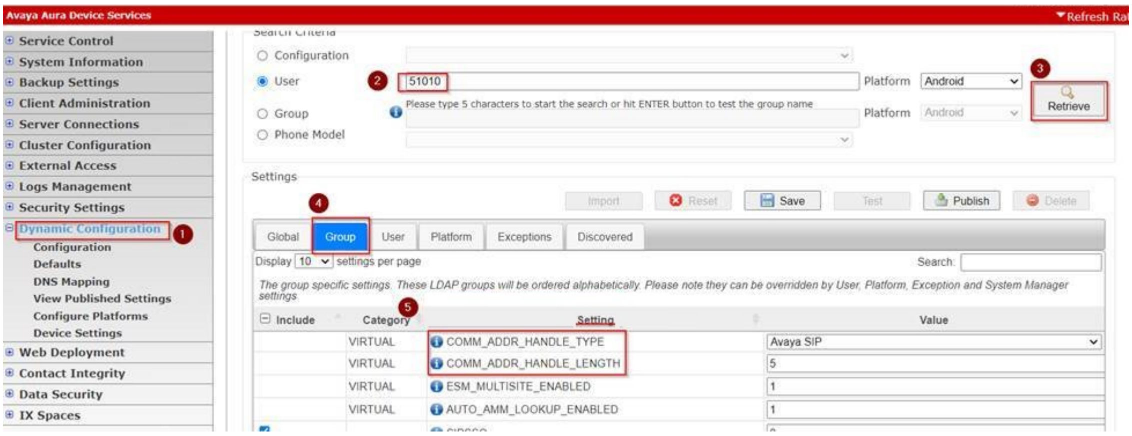
   In this URL, <*AADS_FQDN*> is the Avaya Aura® Device Services FQDN and <*USERNAME*> is the name of the user that gets HTTP error 500.
2. Enter the Avaya Aura® Device Services credentials.
   STEP RESULT:
   Avaya Aura® Device Services displays settings configured for the user.
3. Ensure that the SET SIP_CONTROLLER_LIST parameter contains one or more System Manager IP address and SIP ports.
4. If the SET SIP_CONTROLLER_LIST parameter does not exist or empty, ensure that:
   a. The SET SIPPROXYSRVR parameter exists and contains the System Manager IP address.
   b. The SET SIPSECURE parameter exists and is set to either
      0
      or
      1
      .
   c. The SET SIPPORT parameter exists and contains the System Manager port.
5. Ensure that the SET SIPUSERNAME parameter contains the correct user name.
6. Ensure that the SET SIPDOMAIN parameter contains the correct user domain.
7. If at least one of the parameters listed in steps See "3" on page 222 to See "6" on page 222 does not exist or has an incorrect value, do the following:
   a. Log in to the Avaya Aura® Device Services web administration portal.
   b. Navigate to **Dynamic Configuration › Configuration**.
   c. In the Search Criteria area, select **User** and type the user name.
   d. In **Platform**, select the appropriate platform.
   e. Click **Retrieve**.

f. In **Settings**, select **Group**.

g. Ensure that the following parameters are configured:
ADDITIONAL INFORMATION:

- COMM_ADDR_HANDLE_TYPE
- COMM_ADDR_HANDLE_LENGTH

ADDITIONAL INFORMATION:
The following is an example of the configured COMM_ADDR_HANDLE_TYPE and COMM_ADDR_HANDLE_LENGTH parameters:



8. If these settings are not configured, configure the settings as appropriate and publish them for the user group that the user belongs to.
ADDITIONAL INFORMATION:
For more information about editing and publishing dynamic configuration parameters on Avaya Aura® Device Services, see the " Administration of the Dynamic Configuration service" chapter in *Administering Avaya Aura® Device Services*.

RHEL single-user mode

Single-user mode is a maintenance mode available on Linux-based operating systems. In single-user mode, the operating system only starts certain services for troubleshooting and maintenance operations. In the single-user mode, you can perform operating system-level troubleshooting operations, such as repairing the file system or starting or stopping services. For more information about the use of the single-user mode, refer to the official RHEL documentation.

In single-user mode, network services are not available. Therefore, you cannot use Avaya Aura® Web Gateway functionality on a node that operates in single-user mode.

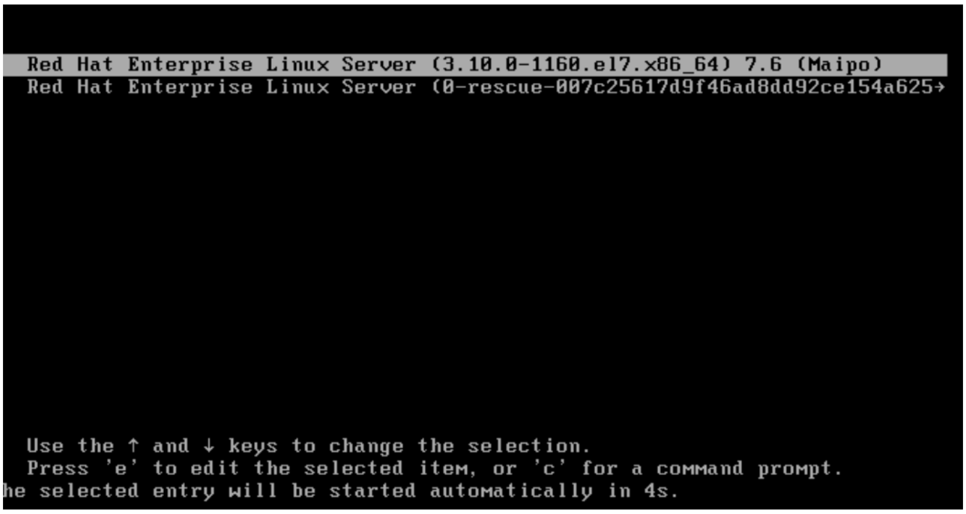
Avaya Aura® Web Gateway supports single-user mode only in a VMware-based environment for both OVA-based and software-only deployment types. You cannot enable single-user mode on because you cannot change the boot sequence of the operating system in Cloud environments.

Booting the RHEL into single-user mode

You can use single-user mode of the RHEL operating system to perform operating system-level troubleshooting and maintenance tasks. The Avaya Aura® Web Gateway functionality is unavailable in single-user mode.

You can only enable single-user mode in VMware-based environments.

In VMware OVA-based deployments, single-user mode is password-protected. The default password is the administrator password that you configure when installing Avaya Aura® Web Gateway. When you enable single-user mode for the first time, you must change the default password.

1. From vCenter, select your Avaya Aura® Web Gateway virtual machine.
2. On the Summary tab, click **Launch Web Console**.
3. Reboot your virtual machine.
4. When vCenter displays the GRUB boot menu, select the appropriate kernel version and then press e.
   ADDITIONAL INFORMATION:
   

5. When prompted to enter the user name, enter `root`.
6. When prompted to enter the password, do one of the following:
   - If you boot into single-user mode for the first time, type the administration password that you configured when installing Avaya Aura® Web Gateway, and then configure a password for single-user mode.

     Avaya Aura® Web Gateway does not enforce complexity rules for single-user mode passwords.
   - If you have already used single-user mode, type the password that you configured for single-user mode.
7. When Avaya Aura® Web Gateway displays the Setparams window, navigate to the line that starts with linux16 using the navigational keys.
8. At the end of the line, type the Space character (" ") and then type the following entry:
   ADDITIONAL INFORMATION:
   `rd.break console=tty1`
9. Press `Ctrl+X`.
   STEP RESULT:
   After the reboot, RHEL boots into the single-user mode.

Changing the password for single-user mode

You can change the password for single-user mode from the Avaya Aura® Web Gateway CLI. You do not need to log in to single-user mode to change the single-user mode password.

Avaya Aura® Web Gateway does not enforce any complexity rules for single-user mode passwords, therefore you can use any password that you want.

1. Log in to the Avaya Aura® Web Gateway CLI as an administrator using an SSH connection.
2. Run the following command:
   ADDITIONAL INFORMATION:
   `/opt/Avaya/bin/manageOSPassword.sh -p`
3. Follow the system prompts to change the single-user mode password.

Exiting the single-user mode

After performing maintenance tasks in single-user mode, switch back to normal Avaya Aura® Web Gateway operational mode.

1. When in the single-user mode, run the following command:
   ADDITIONAL INFORMATION:
   `reboot -f`
   STEP RESULT:
   After the reboot, Avaya Aura® Web Gateway starts in normal mode.

Resources

Documentation

The following table lists related documentation. All Avaya documentation is available at https://support.avaya.com. Many documents are also available at https://documentation.avaya.com/.

Title	Use this document to:	Audience
Overview and planning		
Avaya Aura® Core Solution Description	Understand the strategic, enterprise, and functional views of the Avaya Aura® architecture.	• Customers • Sales, services, and support personnel
Avaya Aura® Communication Manager Feature Description and Implementation	Understand the features of Avaya Aura® Communication Manager.	
What's New in Avaya Aura® Release 10.1	Understand the new and enhanced features of Avaya Aura® components.	• Contractors • Employees • Channel associates • Sales, services, and support personnel • Avaya Business Partners
Using		
Avaya Scopia® Desktop Client User Guide	Use the Avaya Scopia® desktop client. This document also provides usage information for the user portal.	End users
Deploying		
Deploying the Avaya Aura® Web Gateway	Install, configure, and administer the Avaya Aura® Web Gateway.	Implementation personnel
Deploying Avaya Aura® Device Services	Install, administer, configure, and maintain Avaya Aura® Device Services.	

Title	Use this document to:	Audience
Deploying Avaya Multimedia Messaging	Install, configure, and administer Avaya Multimedia Messaging.	
Deploying Avaya Meetings Server	Install, configure, and maintain the Avaya Meetings Server.	
Administering		
Administering Avaya Aura® Device Services	Use management tools, manage data and security, and perform periodic maintenance tasks in Avaya Aura® Device Services.	• System administrators • Implementation personnel • Services and support personnel
Administering Avaya Aura® Communication Manager	Administer Avaya Aura® Communication Manager on the following servers: • Avaya media gateways • Avaya S8XXX Servers	
Administering Network Connectivity on Avaya Aura® Communication Manager	Understand the network components of Avaya Aura® Communication Manager.	

Finding documents on the Avaya Support website

1. Go to https://support.avaya.com.
2. To log in, click **Login** at the top of the screen and then enter your login credentials when prompted.
3. Click **Support by Product › Documents**.
4. In **Enter your Product Here**, start typing the product name and then select the appropriate product from the list displayed.
5. In **Choose Release**, select the appropriate release number.
   ADDITIONAL INFORMATION:
   This field is not available if there is only one release for the product.
6. In , type keywords for your search.
7. From the **Content Type** list, select one or more content types.
   For example, if you only want to see user guides, click **User Guides** in the **Content Type** list.
8. Click to display the search results.

Avaya Documentation Center navigation

For many programs, the latest customer documentation is available on the Avaya Documentation Center website at https://documentation.avaya.com. Some functionality is only available when you log in to the Avaya Documentation Center. The available functionality depends on your role.

> Important:
> If the documentation you are looking for is not available on the Avaya Documentation Center, you can find it on the Avaya Support website.

While navigating through the Documentation Center, you can click the **Avaya Documentation Center** logo at the top of the screen to return to the home page anytime. On the Avaya Documentation Center, you can do the following:

- Click in the top menu bar to access other Avaya websites, including the Avaya Support website.
- Click ( ) in the top menu bar to change the display language and view localized documents.
- In the **Search** field, search for keywords and click **Filters** to filter by solution category, product, or user role.

  You can select multiple items in each filter category. For example, you can select a product and multiple user roles.
- Click in the top menu bar to access the complete library of documents. Use the filtering options to refine your results.
- After performing a search or accessing the library, you can sort content on the search results page. When you find the item you want to view, click it to open it.
- Use the table of contents in a document for navigation. You can also click **<** or **>** next to the document title to navigate to the previous topic or the next topic.
- Click ( ) to share a topic by email or copy the URL.
- Download a PDF of the current topic in a document, the topic and its subtopics, or the entire document.
- Print the section you are viewing.
- Add content to a collection by clicking ( ). You can add the topic and its subtopics or add the entire publication.
- View the topics in your collections. To access your collections, click your name in the top menu bar and then click .

  You can do the following:

  - Create, rename, and delete a collection.
  - Set a collection as the default or favorite collection.
  - Save a PDF of the selected content in a collection and download it to your computer.
  - Share content in a collection with others through email.
  - Receive collections that others have shared with you.
- Click ( ) to add a topic to your watchlist so you are notified when the content is updated or removed.
- View and manage your watchlist by clicking **Watchlist** from the top menu with your name.

  You can do the following:

  - Enable **Include in email notification** to receive email alerts.

    - Unwatch the selected content or all topics.

- Send feedback for a topic.

Training

Avaya Workplace Client solution courses and credentials, such as ACCS-7240, also include information about Avaya Aura® Web Gateway. You can access training courses at http://www.avaya-learning.com.

Viewing Avaya Mentor videos

Avaya Mentor videos provide technical content on how to install, configure, and troubleshoot Avaya products.
Videos are available on the Avaya Support website, listed under the video document type, and on the Avaya-run channel on YouTube.

- To find videos on the Avaya Support website, go to https://support.avaya.com/ and do one of the following:
    - In **Search**, type `Avaya Mentor Videos`, click **Clear All** and select **Video** in the **Content Type**.
    - In **Search**, type the product name. On the Search Results page, click **Clear All** and select **Video** in the **Content Type**.

      The **Video** content type is displayed only when videos are available for that product.

  STEP RESULT:
  In the right pane, the page displays a list of available videos.
- To find the Avaya Mentor videos on YouTube, go to www.youtube.com/AvayaMentor and do one of the following:
    - Enter a key word or key words in the **Search Channel** to search for a specific product or topic.
    - Scroll down Playlists, and click a topic name to see the list of videos available for the topic. For example, Contact Centers.

  ADDITIONAL INFORMATION:

  > Note:
  > Videos are not available for all products.

Support

Go to the Avaya Support website at https://support.avaya.com for the most up-to-date documentation, product notices, and knowledge articles. You can also search for release notes, downloads, and resolutions to issues. Use the online service request system to create a service request. Chat with live agents to get answers to questions, or request an agent to connect you to a support team if an issue requires additional expertise.

Using the Avaya InSite Knowledge Base

The Avaya InSite Knowledge Base is a web-based search engine that provides:

- Up-to-date troubleshooting procedures and technical tips
- Information about service packs
- Access to customer and technical documentation
- Information about training and certification programs
- Links to other pertinent information

If you are an authorized Avaya Partner or a current Avaya customer with a support contract, you can access the Knowledge Base without extra cost. You must have a login account and a valid Sold-To number.

Use the Avaya InSite Knowledge Base for any potential solutions to problems.

1. Go to http://www.avaya.com/support.
2. Log on to the Avaya website with a valid Avaya user ID and password.

   The system displays the Support page.
3. Click **Support by Product › Product-specific support**.
4. In **Enter Product Name**, enter the product, and press `Enter`.
5. Select the product from the list, and select a release.
6. Click the **Technical Solutions** tab to see articles.
7. Select relevant articles.

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